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The Foundations: Logic and Proofs

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The rules of logic specify the meaning of mathematical statements. For instance, these rules help us understand and reason with statements such as “There exists an integer that is not the sum of two squares” and “For every positive integer n , the sum of the positive integers not exceeding n is $n(n+1)/2$.” Logic is the basis of all mathematical reasoning, and of all automated reasoning. It has practical applications to the design of computing machines, to the specification of systems, to artificial intelligence, to computer programming, to programming languages, and to other areas of computer science, as well as to many other fields of study.

To understand mathematics, we must understand what makes up a correct mathematical argument, that is, a proof. Once we prove a mathematical statement is true, we call it a theorem. A collection of theorems on a topic organize what we know about this topic. To learn a mathematical topic, a person needs to actively construct mathematical arguments on this topic, and not just read exposition. Moreover, knowing the proof of a theorem often makes it possible to modify the result to fit new situations.

Everyone knows that proofs are important throughout mathematics, but many people find it surprising how important proofs are in computer science. In fact, proofs are used to verify that computer programs produce the correct output for all possible input values, to show that algorithms always produce the correct result, to establish the security of a system, and to create artificial intelligence. Furthermore, automated reasoning systems have been created to allow computers to construct their own proofs.

In this chapter, we will explain what makes up a correct mathematical argument and introduce tools to construct these arguments. We will develop an arsenal of different proof methods that will enable us to prove many different types of results. After introducing many different methods of proof, we will introduce several strategies for constructing proofs. We will introduce the notion of a conjecture and explain the process of developing mathematics by studying conjectures.

1.1 Propositional Logic

1.1.1 Introduction

The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments. Because a major goal of this book is to teach the reader how to understand and how to construct correct mathematical arguments, we begin our study of discrete mathematics with an introduction to logic.

Besides the importance of logic in understanding mathematical reasoning, logic has numerous applications to computer science. These rules are used in the design of computer circuits, the construction of computer programs, the verification of the correctness of programs, and in many other ways. Furthermore, software systems have been developed for constructing some, but not all, types of proofs automatically. We will discuss these applications of logic in this and later chapters.

1.1.2 Propositions

Our discussion begins with an introduction to the basic building blocks of logic—propositions. A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

EXAMPLE 1 All the following declarative sentences are propositions.

Extra Examples ➤

1. Washington, D.C., is the capital of the United States of America.
2. Toronto is the capital of Canada.
3. $1 + 1 = 2$.
4. $2 + 2 = 3$.

Propositions 1 and 3 are true, whereas 2 and 4 are false. ◀

Some sentences that are not propositions are given in Example 2.

EXAMPLE 2 Consider the following sentences.

1. What time is it?
2. Read this carefully.
3. $x + 1 = 2$.
4. $x + y = z$.

Sentences 1 and 2 are not propositions because they are not declarative sentences. Sentences 3 and 4 are not propositions because they are neither true nor false. Note that each of sentences 3 and 4 can be turned into a proposition if we assign values to the variables. We will also discuss other ways to turn sentences such as these into propositions in Section 1.4. ◀

We use letters to denote **propositional variables** (or **sentential variables**), that is, variables that represent propositions, just as letters are used to denote numerical variables. The conventional letters used for propositional variables are $p, q, r, s, \dots$. The **truth value** of a proposition

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Source: National Library of Medicine

ARISTOTLE (384 B.C.E.–322 B.C.E.) Aristotle was born in Stagirus (Stagira) in northern Greece. His father was the personal physician of the King of Macedonia. Because his father died when Aristotle was young, Aristotle could not follow the custom of following his father's profession. Aristotle became an orphan at a young age when his mother also died. His guardian who raised him taught him poetry, rhetoric, and Greek. At the age of 17, his guardian sent him to Athens to further his education. Aristotle joined Plato's Academy, where for 20 years he attended Plato's lectures, later presenting his own lectures on rhetoric. When Plato died in 347 B.C.E., Aristotle was not chosen to succeed him because his views differed too much from those of Plato. Instead, Aristotle joined the court of King Hermeas where he remained for three years, and married the niece of the King. When the Persians defeated Hermeas, Aristotle moved to Mytilene and, at the invitation of King Philip of Macedonia, he tutored Alexander, Philip's son, who later became Alexander the Great. Aristotle tutored Alexander for five years and after the death of King Philip, he returned to Athens and set up his own school, called the Lyceum.

Aristotle's followers were called the peripatetics, which means "to walk about," because Aristotle often walked around as he discussed philosophical questions. Aristotle taught at the Lyceum for 13 years where he lectured to his advanced students in the morning and gave popular lectures to a broad audience in the evening. When Alexander the Great died in 323 B.C.E., a backlash against anything related to Alexander led to trumped-up charges of impiety against Aristotle. Aristotle fled to Chalcis to avoid prosecution. He only lived one year in Chalcis, dying of a stomach ailment in 322 B.C.E.

Aristotle wrote three types of works: those written for a popular audience, compilations of scientific facts, and systematic treatises. The systematic treatises included works on logic, philosophy, psychology, physics, and natural history. Aristotle's writings were preserved by a student and were hidden in a vault where a wealthy book collector discovered them about 200 years later. They were taken to Rome, where they were studied by scholars and issued in new editions, preserving them for posterity.

is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition. Propositions that cannot be expressed in terms of simpler propositions are called **atomic propositions**.

The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**. It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.

Links 

We now turn our attention to methods for producing new propositions from those that we already have. These methods were discussed by the English mathematician George Boole in 1854 in his book *The Laws of Thought*. Many mathematical statements are constructed by combining one or more propositions. New propositions, called **compound propositions**, are formed from existing propositions using **logical operators**.

Definition 1

Let p be a proposition. The *negation of p* , denoted by $\neg p$ (also denoted by $\bar{p}$), is the statement
“It is not the case that p .”

The proposition $\neg p$ is read “not p .” The truth value of the negation of p , $\neg p$, is the opposite of the truth value of p .

Remark: The notation for the negation operator is not standardized. Although $\neg p$ and $\bar{p}$ are the most common notations used in mathematics to express the negation of p , other notations you might see are $\sim p$, $-p$, p' , Np , and $!p$.

EXAMPLE 3 Find the negation of the proposition

“Michael’s PC runs Linux”

Extra Examples 

and express this in simple English.

Solution: The negation is

“It is not the case that Michael’s PC runs Linux.”

This negation can be more simply expressed as

“Michael’s PC does not run Linux.” 

EXAMPLE 4 Find the negation of the proposition

“Vandana’s smartphone has at least 32 GB of memory”

and express this in simple English.

Solution: The negation is

“It is not the case that Vandana’s smartphone has at least 32 GB of memory.”

This negation can also be expressed as

“Vandana’s smartphone does not have at least 32 GB of memory”

or even more simply as

“Vandana’s smartphone has less than 32 GB of memory.” 

TABLE 1 The Truth Table for the Negation of a Proposition.

p	$\neg p$
T	F
F	T

Table 1 displays the **truth table** for the negation of a proposition p . This table has a row for each of the two possible truth values of p . Each row shows the truth value of $\neg p$ corresponding to the truth value of p for this row.

The negation of a proposition can also be considered the result of the operation of the **negation operator** on a proposition. The negation operator constructs a new proposition from a single existing proposition. We will now introduce the logical operators that are used to form new propositions from two or more existing propositions. These logical operators are also called **connectives**.

Definition 2

Let p and q be propositions. The *conjunction* of p and q , denoted by $p \wedge q$, is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Table 2 displays the truth table of $p \wedge q$. This table has a row for each of the four possible combinations of truth values of p and q . The four rows correspond to the pairs of truth values TT, TF, FT, and FF, where the first truth value in the pair is the truth value of p and the second truth value is the truth value of q .

Note that in logic the word “but” sometimes is used instead of “and” in a conjunction. For example, the statement “The sun is shining, but it is raining” is another way of saying “The sun is shining and it is raining.” (In natural language, there is a subtle difference in meaning between “and” and “but”; we will not be concerned with this nuance here.)

EXAMPLE 5

Find the conjunction of the propositions p and q where p is the proposition “Rebecca’s PC has more than 16 GB free hard disk space” and q is the proposition “The processor in Rebecca’s PC runs faster than 1 GHz.”

Solution: The conjunction of these propositions, $p \wedge q$, is the proposition “Rebecca’s PC has more than 16 GB free hard disk space, and the processor in Rebecca’s PC runs faster than 1 GHz.” This conjunction can be expressed more simply as “Rebecca’s PC has more than 16 GB free hard disk space, and its processor runs faster than 1 GHz.” For this conjunction to be true, both conditions given must be true. It is false when one or both of these conditions are false. ◀

Definition 3

Let p and q be propositions. The *disjunction* of p and q , denoted by $p \vee q$, is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Table 3 displays the truth table for $p \vee q$.

TABLE 2 The Truth Table for the Conjunction of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

TABLE 3 The Truth Table for the Disjunction of Two Propositions.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

The use of the connective *or* in a disjunction corresponds to one of the two ways the word *or* is used in English, namely, as an **inclusive or**. A disjunction is true when at least one of the two propositions is true. That is, $p \vee q$ is true when both p and q are true or when exactly one of p and q is true.

EXAMPLE 6 Translate the statement “Students who have taken calculus or introductory computer science can take this class” in a statement in propositional logic using the propositions p : “A student who has taken calculus can take this class” and q : “A student who has taken introductory computer science can take this class.”

Solution: We assume that this statement means that students who have taken both calculus and introductory computer science can take the class, as well as the students who have taken only one of the two subjects. Hence, this statement can be expressed as $p \vee q$, the inclusive or, or disjunction, of p and q . ◀

EXAMPLE 7 What is the disjunction of the propositions p and q , where p and q are the same propositions as in Example 5?

Extra Examples ▶

Solution: The disjunction of p and q , $p \vee q$, is the proposition

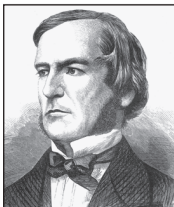
“Rebecca’s PC has at least 16 GB free hard disk space, or the processor in Rebecca’s PC runs faster than 1 GHz.”

This proposition is true when Rebecca’s PC has at least 16 GB free hard disk space, when the PC’s processor runs faster than 1 GHz, and when both conditions are true. It is false when both of these conditions are false, that is, when Rebecca’s PC has less than 16 GB free hard disk space and the processor in her PC runs at 1 GHz or slower. ◀

Besides its use in disjunctions, the connective *or* is also used to express an *exclusive or*. Unlike the disjunction of two propositions p and q , the exclusive or of these two propositions is true when exactly one of p and q is true; it is false when both p and q are true (and when both are false).

Definition 4 Let p and q be propositions. The *exclusive or* of p and q , denoted by $p \oplus q$ (or $p \text{ XOR } q$), is the proposition that is true when exactly one of p and q is true and is false otherwise.

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Source: Library of Congress
Washington, D.C. 20540
USA [LC-USZ62-61664]

GEORGE BOOLE (1815–1864) George Boole, the son of a cobbler, was born in Lincoln, England, in November 1815. Because of his family’s difficult financial situation, Boole struggled to educate himself while supporting his family. Nevertheless, he became one of the most important mathematicians of the 1800s. Although he considered a career as a clergyman, he decided instead to go into teaching, and soon afterward opened a school of his own. In his preparation for teaching mathematics, Boole—unsatisfied with textbooks of his day—decided to read the works of the great mathematicians. While reading papers of the great French mathematician Lagrange, Boole made discoveries in the calculus of variations, the branch of analysis dealing with finding curves and surfaces by optimizing certain parameters.

In 1848 Boole published *The Mathematical Analysis of Logic*, the first of his contributions to symbolic logic. In 1849 he was appointed professor of mathematics at Queen’s College in Cork, Ireland. In 1854 he published *The Laws of Thought*, his most famous work. In this book, Boole introduced what is now called *Boolean algebra* in his honor. Boole wrote textbooks on differential equations and on difference equations that were used in Great Britain until the end of the nineteenth century. Boole married in 1855; his wife was the niece of the professor of Greek at Queen’s College. In 1864 Boole died from pneumonia, which he contracted as a result of keeping a lecture engagement even though he was soaking wet from a rainstorm.

The truth table for the exclusive or of two propositions is displayed in Table 4.

EXAMPLE 8 Let p and q be the propositions that state “A student can have a salad with dinner” and “A student can have soup with dinner,” respectively. What is $p \oplus q$, the exclusive or of p and q ?

Solution: The exclusive or of p and q is the statement that is true when exactly one of p and q is true. That is, $p \oplus q$ is the statement “A student can have soup or salad, but not both, with dinner.” Note that this is often stated as “A student can have soup or a salad with dinner,” without explicitly stating that taking both is not permitted. ◀

EXAMPLE 9 Express the statement “I will use all my savings to travel to Europe or to buy an electric car” in propositional logic using the statement p : “I will use all my savings to travel to Europe” and the statement q : “I will use all my savings to buy an electric car.”

Solution: To translate this statement, we first note that the or in this statement must be an exclusive or because this student can either use all his or her savings to travel to Europe or use all these savings to buy an electric car, but cannot both go to Europe and buy an electric car. (This is clear because either option requires all his savings.) Hence, this statement can be expressed as $p \oplus q$. ◀

1.1.3 Conditional Statements

We will discuss several other important ways in which propositions can be combined.

Definition 5 Let p and q be propositions. The *conditional statement* $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the *hypothesis* (or *antecedent* or *premise*) and q is called the *conclusion* (or *consequence*).

Assessment ▶ The statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds. A conditional statement is also called an **implication**.

The truth table for the conditional statement $p \rightarrow q$ is shown in Table 5. Note that the statement $p \rightarrow q$ is true when both p and q are true and when p is false (no matter what truth value q has).

TABLE 4 The Truth Table for the Exclusive Or of Two Propositions.		
p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

TABLE 5 The Truth Table for the Conditional Statement $p \rightarrow q$.		
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Because conditional statements play such an essential role in mathematical reasoning, a variety of terminology is used to express $p \rightarrow q$. You will encounter most if not all of the following ways to express this conditional statement:

“if p , then q ”	“ p implies q ”
“if p , q ”	“ p only if q ”
“ p is sufficient for q ”	“a sufficient condition for q is p ”
“ q if p ”	“ q whenever p ”
“ q when p ”	“ q is necessary for p ”
“a necessary condition for p is q ”	“ q follows from p ”
“ q unless $\neg p$ ”	“ q provided that p ”

A useful way to understand the truth value of a conditional statement is to think of an obligation or a contract. For example, the pledge many politicians make when running for office is

“If I am elected, then I will lower taxes.”

If the politician is elected, voters would expect this politician to lower taxes. Furthermore, if the politician is not elected, then voters will not have any expectation that this person will lower taxes, although the person may have sufficient influence to cause those in power to lower taxes. It is only when the politician is elected but does not lower taxes that voters can say that the politician has broken the campaign pledge. This last scenario corresponds to the case when p is true but q is false in $p \rightarrow q$.

Similarly, consider a statement that a professor might make:

“If you get 100% on the final, then you will get an A.”

If you manage to get 100% on the final, then you would expect to receive an A. If you do not get 100%, you may or may not receive an A depending on other factors. However, if you do get 100%, but the professor does not give you an A, you will feel cheated.

Remark: Because some of the different ways to express the implication p implies q can be confusing, we will provide some extra guidance. To remember that “ p only if q ” expresses the same thing as “if p , then q ,” note that “ p only if q ” says that p cannot be true when q is not true. That is, the statement is false if p is true, but q is false. When p is false, q may be either true or false, because the statement says nothing about the truth value of q .

For example, suppose your professor tells you

“You can receive an A in the course only if your score on the final is at least 90%.”

Then, if you receive an A in the course, then you know that your score on the final is at least 90%. If you do not receive an A, you may or may not have scored at least 90% on the final. Be careful not to use “ q only if p ” to express $p \rightarrow q$ because this is incorrect. The word “only” plays an essential role here. To see this, note that the truth values of “ q only if p ” and $p \rightarrow q$ are different when p and q have different truth values. To see why “ q is necessary for p ” is equivalent to “if p , then q ,” observe that “ q is necessary for p ” means that p cannot be true unless q is true, or that if q is false, then p is false. This is the same as saying that if p is true, then q is true. To see why “ p is sufficient for q ” is equivalent to “if p , then q ,” note that “ p is sufficient for q ” means if p is true, it must be the case that q is also true. This is the same as saying that if p is true, then q is also true.

To remember that “ q unless $\neg p$ ” expresses the same conditional statement as “if p , then q ,” note that “ q unless $\neg p$ ” means that if $\neg p$ is false, then q must be true. That is, the statement “ q unless $\neg p$ ” is false when p is true but q is false, but it is true otherwise. Consequently, “ q unless $\neg p$ ” and $p \rightarrow q$ always have the same truth value.

You might have trouble understanding how “unless” is used in conditional statements unless you read this paragraph carefully.

We illustrate the translation between conditional statements and English statements in Example 10.

EXAMPLE 10

Extra
Examples

Let p be the statement “Maria learns discrete mathematics” and q the statement “Maria will find a good job.” Express the statement $p \rightarrow q$ as a statement in English.

Solution: From the definition of conditional statements, we see that when p is the statement “Maria learns discrete mathematics” and q is the statement “Maria will find a good job,” $p \rightarrow q$ represents the statement

“If Maria learns discrete mathematics, then she will find a good job.”

There are many other ways to express this conditional statement in English. Among the most natural of these are

“Maria will find a good job when she learns discrete mathematics.”

“For Maria to get a good job, it is sufficient for her to learn discrete mathematics.”

and

“Maria will find a good job unless she does not learn discrete mathematics.”

Note that the way we have defined conditional statements is more general than the meaning attached to such statements in the English language. For instance, the conditional statement in Example 10 and the statement

“If it is sunny, then we will go to the beach”

are statements used in normal language where there is a relationship between the hypothesis and the conclusion. Further, the first of these statements is true unless Maria learns discrete mathematics, but she does not get a good job, and the second is true unless it is indeed sunny, but we do not go to the beach. On the other hand, the statement

“If Juan has a smartphone, then $2 + 3 = 5$ ”

is true from the definition of a conditional statement, because its conclusion is true. (The truth value of the hypothesis does not matter then.) The conditional statement

“If Juan has a smartphone, then $2 + 3 = 6$ ”

is true if Juan does not have a smartphone, even though $2 + 3 = 6$ is false. We would not use these last two conditional statements in natural language (except perhaps in sarcasm), because there is no relationship between the hypothesis and the conclusion in either statement. In mathematical reasoning, we consider conditional statements of a more general sort than we use in English. The mathematical concept of a conditional statement is independent of a cause-and-effect relationship between hypothesis and conclusion. Our definition of a conditional statement specifies its truth values; it is not based on English usage. Propositional language is an artificial language; we only parallel English usage to make it easy to use and remember.

The if-then construction used in many programming languages is different from that used in logic. Most programming languages contain statements such as **if** p **then** S , where p is a proposition and S is a program segment (one or more statements to be executed). (Although this looks as if it might be a conditional statement, S is not a proposition, but rather is a set of executable instructions.) When execution of a program encounters such a statement, S is executed if p is true, but S is not executed if p is false, as illustrated in Example 11.

EXAMPLE 11 What is the value of the variable x after the statement

if $2 + 2 = 4$ **then** $x := x + 1$

if $x = 0$ before this statement is encountered? (The symbol $:=$ stands for assignment. The statement $x := x + 1$ means the assignment of the value of $x + 1$ to x .)

Solution: Because $2 + 2 = 4$ is true, the assignment statement $x := x + 1$ is executed. Hence, x has the value $0 + 1 = 1$ after this statement is encountered. 

CONVERSE, CONTRAPOSITIVE, AND INVERSE We can form some new conditional statements starting with a conditional statement $p \rightarrow q$. In particular, there are three related conditional statements that occur so often that they have special names. The proposition $q \rightarrow p$ is called the **converse** of $p \rightarrow q$. The **contrapositive** of $p \rightarrow q$ is the proposition $\neg q \rightarrow \neg p$. The proposition $\neg p \rightarrow \neg q$ is called the **inverse** of $p \rightarrow q$. We will see that of these three conditional statements formed from $p \rightarrow q$, only the contrapositive always has the same truth value as $p \rightarrow q$.

We first show that the contrapositive, $\neg q \rightarrow \neg p$, of a conditional statement $p \rightarrow q$ always has the same truth value as $p \rightarrow q$. To see this, note that the contrapositive is false only when $\neg p$ is false and $\neg q$ is true, that is, only when p is true and q is false. We now show that neither the converse, $q \rightarrow p$, nor the inverse, $\neg p \rightarrow \neg q$, has the same truth value as $p \rightarrow q$ for all possible truth values of p and q . Note that when p is true and q is false, the original conditional statement is false, but the converse and the inverse are both true.

When two compound propositions always have the same truth values, regardless of the truth values of its propositional variables, we call them **equivalent**. Hence, a conditional statement and its contrapositive are equivalent. The converse and the inverse of a conditional statement are also equivalent, as the reader can verify, but neither is equivalent to the original conditional statement. (We will study equivalent propositions in Section 1.3.) Take note that one of the most common logical errors is to assume that the converse or the inverse of a conditional statement is equivalent to this conditional statement.

We illustrate the use of conditional statements in Example 12.

EXAMPLE 12 Find the contrapositive, the converse, and the inverse of the conditional statement

**Extra
Examples** 

“The home team wins whenever it is raining.”

Solution: Because “ q whenever p ” is one of the ways to express the conditional statement $p \rightarrow q$, the original statement can be rewritten as

“If it is raining, then the home team wins.”

Consequently, the contrapositive of this conditional statement is

“If the home team does not win, then it is not raining.”

The converse is

“If the home team wins, then it is raining.”

The inverse is

“If it is not raining, then the home team does not win.”

Only the contrapositive is equivalent to the original statement. 

Remember that the contrapositive, but neither the converse or inverse, of a conditional statement is equivalent to it.

TABLE 6 The Truth Table for the Biconditional $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

BICONDITIONALS We now introduce another way to combine propositions that expresses that two propositions have the same truth value.

Definition 6

Let p and q be propositions. The *biconditional statement* $p \leftrightarrow q$ is the proposition “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called *bi-implications*.

The truth table for $p \leftrightarrow q$ is shown in Table 6. Note that the statement $p \leftrightarrow q$ is true when both the conditional statements $p \rightarrow q$ and $q \rightarrow p$ are true and is false otherwise. That is why we use the words “if and only if” to express this logical connective and why it is symbolically written by combining the symbols $\rightarrow$ and $\leftarrow$. There are some other common ways to express $p \leftrightarrow q$:

“ p is necessary and sufficient for q ”
 “if p then q , and conversely”
 “ p iff q .” “ p exactly when q .”

The last way of expressing the biconditional statement $p \leftrightarrow q$ uses the abbreviation “iff” for “if and only if.” Note that $p \leftrightarrow q$ has exactly the same truth value as $(p \rightarrow q) \wedge (q \rightarrow p)$.

EXAMPLE 13

Let p be the statement “You can take the flight,” and let q be the statement “You buy a ticket.” Then $p \leftrightarrow q$ is the statement

Extra Examples ➤

“You can take the flight if and only if you buy a ticket.”

This statement is true if p and q are either both true or both false, that is, if you buy a ticket and can take the flight or if you do not buy a ticket and you cannot take the flight. It is false when p and q have opposite truth values, that is, when you do not buy a ticket, but you can take the flight (such as when you get a free trip) and when you buy a ticket but you cannot take the flight (such as when the airline bumps you). ◀

IMPLICIT USE OF BICONDITIONALS You should be aware that biconditionals are not always explicit in natural language. In particular, the “if and only if” construction used in biconditionals is rarely used in common language. Instead, biconditionals are often expressed using an “if, then” or an “only if” construction. The other part of the “if and only if” is implicit. That is, the converse is implied, but not stated. For example, consider the statement in English “If you finish your meal, then you can have dessert.” What is really meant is “You can have dessert if and only if you finish your meal.” This last statement is logically equivalent to the two statements “If you finish your meal, then you can have dessert” and “You can have dessert only if you finish your meal.” Because of this imprecision in natural language, we need to make an assumption whether a conditional statement in natural language implicitly includes its converse. Because precision is essential in mathematics and in logic, we will always distinguish between the conditional statement $p \rightarrow q$ and the biconditional statement $p \leftrightarrow q$.

1.1.4 Truth Tables of Compound Propositions

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We have now introduced five important logical connectives—conjunction, disjunction, exclusive or, implication, and the biconditional operator—as well as negation. We can use these connectives to build up complicated compound propositions involving any number of propositional variables. We can use truth tables to determine the truth values of these compound propositions, as Example 14 illustrates. We use a separate column to find the truth value of each compound expression that occurs in the compound proposition as it is built up. The truth values of the compound proposition for each combination of truth values of the propositional variables in it is found in the final column of the table.

EXAMPLE 14 Construct the truth table of the compound proposition

$$(p \vee \neg q) \rightarrow (p \wedge q).$$

Solution: Because this truth table involves two propositional variables p and q , there are four rows in this truth table, one for each of the pairs of truth values TT, TF, FT, and FF. The first two columns are used for the truth values of p and q , respectively. In the third column we find the truth value of $\neg q$, needed to find the truth value of $p \vee \neg q$, found in the fourth column. The fifth column gives the truth value of $p \wedge q$. Finally, the truth value of $(p \vee \neg q) \rightarrow (p \wedge q)$ is found in the last column. The resulting truth table is shown in Table 7.

TABLE 7 The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

1.1.5 Precedence of Logical Operators

We can construct compound propositions using the negation operator and the logical operators defined so far. We will generally use parentheses to specify the order in which logical operators in a compound proposition are to be applied. For instance, $(p \vee q) \wedge (\neg r)$ is the conjunction of $p \vee q$ and $\neg r$. However, to reduce the number of parentheses, we specify that the negation operator is applied before all other logical operators. This means that $\neg p \wedge q$ is the conjunction of $\neg p$ and q , namely, $(\neg p) \wedge q$, not the negation of the conjunction of p and q , namely $\neg(p \wedge q)$. (It is generally the case that unary operators that involve only one object precede binary operators.)

Another general rule of precedence is that the conjunction operator takes precedence over the disjunction operator, so that $p \vee q \wedge r$ means $p \vee (q \wedge r)$ rather than $(p \vee q) \wedge r$ and $p \wedge q \vee r$ means $(p \wedge q) \vee r$ rather than $p \wedge (q \vee r)$. Because this rule may be difficult to remember, we will continue to use parentheses so that the order of the disjunction and conjunction operators is clear.

Finally, it is an accepted rule that the conditional and biconditional operators, $\rightarrow$ and $\leftrightarrow$, have lower precedence than the conjunction and disjunction operators, $\wedge$ and $\vee$. Consequently, $p \rightarrow q \vee r$ means $p \rightarrow (q \vee r)$ rather than $(p \rightarrow q) \vee r$ and $p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$ rather than $p \vee (q \rightarrow r)$. We will use parentheses when the order of the conditional operator and biconditional operator is at issue, although the conditional operator has precedence over the biconditional operator. Table 8 displays the precedence levels of the logical operators, $\neg$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$.

TABLE 8
Precedence of
Logical Operators.

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

1.1.6 Logic and Bit Operations

Truth Value	Bit
T	1
F	0

Links

Computers represent information using bits. A **bit** is a symbol with two possible values, namely, 0 (zero) and 1 (one). This meaning of the word bit comes from *binary digit*, because zeros and ones are the digits used in binary representations of numbers. The well-known statistician John Tukey introduced this terminology in 1946. A bit can be used to represent a truth value, because there are two truth values, namely, *true* and *false*. As is customarily done, we will use a 1 bit to represent true and a 0 bit to represent false. That is, 1 represents T (true), 0 represents F (false). A variable is called a **Boolean variable** if its value is either true or false. Consequently, a Boolean variable can be represented using a bit.

Computer **bit operations** correspond to the logical connectives. By replacing true by a one and false by a zero in the truth tables for the operators $\wedge$, $\vee$, and $\oplus$, the columns in Table 9 for the corresponding bit operations are obtained. We will also use the notation *OR*, *AND*, and *XOR* for the operators $\vee$, $\wedge$, and $\oplus$, as is done in various programming languages.

TABLE 9 Table for the Bit Operators <i>OR</i> , <i>AND</i> , and <i>XOR</i> .				
x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Information is often represented using bit strings, which are lists of zeros and ones. When this is done, operations on the bit strings can be used to manipulate this information.

Definition 7

A *bit string* is a sequence of zero or more bits. The *length* of this string is the number of bits in the string.

EXAMPLE 15 101010011 is a bit string of length nine.

Links



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The LIFE Picture
Collection/Getty Images

JOHN WILDER TUKEY (1915–2000) Tukey, born in New Bedford, Massachusetts, was an only child. His parents, both teachers, decided home schooling would best develop his potential. His formal education began at Brown University, where he studied mathematics and chemistry. He received a master’s degree in chemistry from Brown and continued his studies at Princeton University, changing his field of study from chemistry to mathematics. He received his Ph.D. from Princeton in 1939 for work in topology, when he was appointed an instructor in mathematics at Princeton. With the start of World War II, he joined the Fire Control Research Office, where he began working in statistics. Tukey found statistical research to his liking and impressed several leading statisticians with his skills. In 1945, at the conclusion of the war, Tukey returned to the mathematics department at Princeton as a professor of statistics, and he also took a position at AT&T Bell Laboratories. Tukey founded the Statistics Department at Princeton in 1966 and was its first chairman. Tukey made significant contributions to many areas of statistics, including the analysis of variance, the estimation of spectra of time series, inferences about the values of a set of parameters from a single experiment, and the philosophy of statistics. However, he is best known for his invention, with J. W. Cooley, of the fast Fourier transform. In addition to his contributions to statistics, Tukey was noted as a skilled wordsmith; he is credited with coining the terms *bit* and *software*.

Tukey contributed his insight and expertise by serving on the President’s Science Advisory Committee. He chaired several important committees dealing with the environment, education, and chemicals and health. He also served on committees working on nuclear disarmament. Tukey received many awards, including the National Medal of Science.

HISTORICAL NOTE There were several other suggested words for a binary digit, including *binit* and *bigit*, that never were widely accepted. The adoption of the word *bit* may be due to its meaning as a common English word. For an account of Tukey’s coining of the word *bit*, see the April 1984 issue of *Annals of the History of Computing*.

We can extend bit operations to bit strings. We define the **bitwise OR**, **bitwise AND**, and **bitwise XOR** of two strings of the same length to be the strings that have as their bits the *OR*, *AND*, and *XOR* of the corresponding bits in the two strings, respectively. We use the symbols $\vee$, $\wedge$, and $\oplus$ to represent the bitwise *OR*, bitwise *AND*, and bitwise *XOR* operations, respectively. We illustrate bitwise operations on bit strings with Example 16.

EXAMPLE 16 Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit strings 01 1011 0110 and 11 0001 1101. (Here, and throughout this book, bit strings will be split into blocks of four bits to make them easier to read.)

Solution: The bitwise *OR*, bitwise *AND*, and bitwise *XOR* of these strings are obtained by taking the *OR*, *AND*, and *XOR* of the corresponding bits, respectively. This gives us

01 1011 0110	
11 0001 1101	
11 1011 1111	bitwise <i>OR</i>
01 0001 0100	bitwise <i>AND</i>
10 1010 1011	bitwise <i>XOR</i>

Exercises

- Which of these sentences are propositions? What are the truth values of those that are propositions?
 - Boston is the capital of Massachusetts.
 - Miami is the capital of Florida.
 - $2 + 3 = 5$.
 - $5 + 7 = 10$.
 - $x + 2 = 11$.
 - Answer this question.
- Which of these are propositions? What are the truth values of those that are propositions?
 - Do not pass go.
 - What time is it?
 - There are no black flies in Maine.
 - $4 + x = 5$.
 - The moon is made of green cheese.
 - $2^n \geq 100$.
- What is the negation of each of these propositions?
 - Linda is younger than Sanjay.
 - Mei makes more money than Isabella.
 - Moshe is taller than Monica.
 - Abby is richer than Ricardo.
- What is the negation of each of these propositions?
 - Janice has more Facebook friends than Juan.
 - Quincy is smarter than Venkat.
 - Zelda drives more miles to school than Paola.
 - Briana sleeps longer than Gloria.
- What is the negation of each of these propositions?
 - Mei has an MP3 player.
 - There is no pollution in New Jersey.
 - $2 + 1 = 3$.
 - The summer in Maine is hot and sunny.
- What is the negation of each of these propositions?
 - Jennifer and Teja are friends.
 - There are 13 items in a baker's dozen.
 - Abby sent more than 100 text messages yesterday.
 - 121 is a perfect square.
- What is the negation of each of these propositions?
 - Steve has more than 100 GB free disk space on his laptop.
 - Zach blocks e-mails and texts from Jennifer.
 - $7 \cdot 11 \cdot 13 = 999$.
 - Diane rode her bicycle 100 miles on Sunday.
- Suppose that Smartphone A has 256 MB RAM and 32 GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Determine the truth value of each of these propositions.
 - Smartphone B has the most RAM of these three smartphones.
 - Smartphone C has more ROM or a higher resolution camera than Smartphone B.
 - Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A.
 - If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera.
 - Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone A.
- Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Determine the truth value of each of these propositions for the most recent fiscal year.

- a) Quixote Media had the largest annual revenue.
- b) Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.
- c) Acme Computer had the largest net profit or Quixote Media had the largest net profit.
- d) If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.
- e) Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.

10. Let p and q be the propositions p : I bought a lottery ticket this week. q : I won the million dollar jackpot.

Express each of these propositions as an English sentence.

- | | | |
|---------------------------|-------------------------------|--------------------------------|
| a) $\neg p$ | b) $p \vee q$ | c) $p \rightarrow q$ |
| d) $p \wedge q$ | e) $p \leftrightarrow q$ | f) $\neg p \rightarrow \neg q$ |
| g) $\neg p \wedge \neg q$ | h) $\neg p \vee (p \wedge q)$ | |

11. Let p and q be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore," respectively. Express each of these compound propositions as an English sentence.

- | | | |
|-------------------------------|------------------------------------|--------------------------------|
| a) $\neg q$ | b) $p \wedge q$ | c) $\neg p \vee q$ |
| d) $p \rightarrow \neg q$ | e) $\neg q \rightarrow p$ | f) $\neg p \rightarrow \neg q$ |
| g) $p \leftrightarrow \neg q$ | h) $\neg p \wedge (p \vee \neg q)$ | |

12. Let p and q be the propositions "The election is decided" and "The votes have been counted," respectively. Express each of these compound propositions as an English sentence.

- | | |
|--------------------------------|------------------------------------|
| a) $\neg p$ | b) $p \vee q$ |
| c) $\neg p \wedge q$ | d) $q \rightarrow p$ |
| e) $\neg q \rightarrow \neg p$ | f) $\neg p \rightarrow \neg q$ |
| g) $p \leftrightarrow q$ | h) $\neg q \vee (\neg p \wedge q)$ |

13. Let p and q be the propositions p : It is below freezing. q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- a) It is below freezing and snowing.
- b) It is below freezing but not snowing.
- c) It is not below freezing and it is not snowing.
- d) It is either snowing or below freezing (or both).
- e) If it is below freezing, it is also snowing.
- f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
- g) That it is below freezing is necessary and sufficient for it to be snowing.

14. Let p , q , and r be the propositions p : You have the flu. q : You miss the final examination. r : You pass the course.

Express each of these propositions as an English sentence.

- | | |
|---|--|
| a) $p \rightarrow q$ | b) $\neg q \leftrightarrow r$ |
| c) $q \rightarrow \neg r$ | d) $p \vee q \vee r$ |
| e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$ | f) $(p \wedge q) \vee (\neg q \wedge r)$ |

15. Let p and q be the propositions p : You drive over 65 miles per hour. q : You get a speeding ticket.

Write these propositions using p and q and logical connectives (including negations).

- a) You do not drive over 65 miles per hour.
- b) You drive over 65 miles per hour, but you do not get a speeding ticket.
- c) You will get a speeding ticket if you drive over 65 miles per hour.
- d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.
- e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.
- f) You get a speeding ticket, but you do not drive over 65 miles per hour.
- g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

16. Let p , q , and r be the propositions p : You get an A on the final exam. q : You do every exercise in this book. r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

- a) You get an A in this class, but you do not do every exercise in this book.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
- c) To get an A in this class, it is necessary for you to get an A on the final.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

17. Let p , q , and r be the propositions p : Grizzly bears have been seen in the area. q : Hiking is safe on the trail. r : Berries are ripe along the trail.

Write these propositions using p , q , and r and logical connectives (including negations).

- a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.
- b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.
- c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.
- d) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe.
- e) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area.

- f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.
18. Determine whether these biconditionals are true or false.
- $2 + 2 = 4$ if and only if $1 + 1 = 2$.
 - $1 + 1 = 2$ if and only if $2 + 3 = 4$.
 - $1 + 1 = 3$ if and only if monkeys can fly.
 - $0 > 1$ if and only if $2 > 1$.
19. Determine whether each of these conditional statements is true or false.
- If $1 + 1 = 2$, then $2 + 2 = 5$.
 - If $1 + 1 = 3$, then $2 + 2 = 4$.
 - If $1 + 1 = 3$, then $2 + 2 = 5$.
 - If monkeys can fly, then $1 + 1 = 3$.
20. Determine whether each of these conditional statements is true or false.
- If $1 + 1 = 3$, then unicorns exist.
 - If $1 + 1 = 3$, then dogs can fly.
 - If $1 + 1 = 2$, then dogs can fly.
 - If $2 + 2 = 4$, then $1 + 2 = 3$.
21. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.
- Coffee or tea comes with dinner.
 - A password must have at least three digits or be at least eight characters long.
 - The prerequisite for the course is a course in number theory or a course in cryptography.
 - You can pay using U.S. dollars or euros.
22. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.
- Experience with C++ or Java is required.
 - Lunch includes soup or salad.
 - To enter the country you need a passport or a voter registration card.
 - Publish or perish.
23. For each of these sentences, state what the sentence means if the logical connective or is an inclusive or (that is, a disjunction) versus an exclusive or. Which of these meanings of or do you think is intended?
- To take discrete mathematics, you must have taken calculus or a course in computer science.
 - When you buy a new car from Acme Motor Company, you get \$2000 back in cash or a 2% car loan.
 - Dinner for two includes two items from column A or three items from column B.
 - School is closed if more than two feet of snow falls or if the wind chill is below -100°F .
24. Write each of these statements in the form “if p , then q ” in English. [*Hint*: Refer to the list of common ways to express conditional statements provided in this section.]
- It is necessary to wash the boss’s car to get promoted.
 - Winds from the south imply a spring thaw.
 - A sufficient condition for the warranty to be good is that you bought the computer less than a year ago.
 - Willy gets caught whenever he cheats.
 - You can access the website only if you pay a subscription fee.
 - Getting elected follows from knowing the right people.
 - Carol gets seasick whenever she is on a boat.
25. Write each of these statements in the form “if p , then q ” in English. [*Hint*: Refer to the list of common ways to express conditional statements.]
- It snows whenever the wind blows from the northeast.
 - The apple trees will bloom if it stays warm for a week.
 - That the Pistons win the championship implies that they beat the Lakers.
 - It is necessary to walk eight miles to get to the top of Long’s Peak.
 - To get tenure as a professor, it is sufficient to be world famous.
 - If you drive more than 400 miles, you will need to buy gasoline.
 - Your guarantee is good only if you bought your CD player less than 90 days ago.
 - Jan will go swimming unless the water is too cold.
 - We will have a future, provided that people believe in science.
26. Write each of these statements in the form “if p , then q ” in English. [*Hint*: Refer to the list of common ways to express conditional statements provided in this section.]
- I will remember to send you the address only if you send me an e-mail message.
 - To be a citizen of this country, it is sufficient that you were born in the United States.
 - If you keep your textbook, it will be a useful reference in your future courses.
 - The Red Wings will win the Stanley Cup if their goalie plays well.
 - That you get the job implies that you had the best credentials.
 - The beach erodes whenever there is a storm.
 - It is necessary to have a valid password to log on to the server.
 - You will reach the summit unless you begin your climb too late.
 - You will get a free ice cream cone, provided that you are among the first 100 customers tomorrow.
27. Write each of these propositions in the form “ p if and only if q ” in English.
- If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.
 - For you to win the contest it is necessary and sufficient that you have the only winning ticket.
 - You get promoted only if you have connections, and you have connections only if you get promoted.
 - If you watch television your mind will decay, and conversely.
 - The trains run late on exactly those days when I take it.

28. Write each of these propositions in the form “ p if and only if q ” in English.
- For you to get an A in this course, it is necessary and sufficient that you learn how to solve discrete mathematics problems.
 - If you read the newspaper every day, you will be informed, and conversely.
 - It rains if it is a weekend day, and it is a weekend day if it rains.
 - You can see the wizard only if the wizard is not in, and the wizard is not in only if you can see him.
 - My airplane flight is late exactly when I have to catch a connecting flight.
29. State the converse, contrapositive, and inverse of each of these conditional statements.
- If it snows today, I will ski tomorrow.
 - I come to class whenever there is going to be a quiz.
 - A positive integer is a prime only if it has no divisors other than 1 and itself.
30. State the converse, contrapositive, and inverse of each of these conditional statements.
- If it snows tonight, then I will stay at home.
 - I go to the beach whenever it is a sunny summer day.
 - When I stay up late, it is necessary that I sleep until noon.
31. How many rows appear in a truth table for each of these compound propositions?
- $p \rightarrow \neg p$
 - $(p \vee \neg r) \wedge (q \vee \neg s)$
 - $q \vee p \vee \neg s \vee \neg r \vee \neg t \vee u$
 - $(p \wedge r \wedge t) \leftrightarrow (q \wedge t)$
32. How many rows appear in a truth table for each of these compound propositions?
- $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
 - $(p \vee \neg t) \wedge (p \vee \neg s)$
 - $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$
 - $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$
33. Construct a truth table for each of these compound propositions.
- $p \wedge \neg p$
 - $p \vee \neg p$
 - $(p \vee \neg q) \rightarrow q$
 - $(p \vee q) \rightarrow (p \wedge q)$
 - $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
 - $(p \rightarrow q) \rightarrow (q \rightarrow p)$
34. Construct a truth table for each of these compound propositions.
- $p \rightarrow \neg p$
 - $p \leftrightarrow \neg p$
 - $p \oplus (p \vee q)$
 - $(p \wedge q) \rightarrow (p \vee q)$
 - $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$
 - $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$
35. Construct a truth table for each of these compound propositions.
- $(p \vee q) \rightarrow (p \oplus q)$
 - $(p \oplus q) \rightarrow (p \wedge q)$
 - $(p \vee q) \oplus (p \wedge q)$
 - $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$
 - $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
 - $(p \oplus q) \rightarrow (p \oplus \neg q)$
36. Construct a truth table for each of these compound propositions.
- $p \oplus p$
 - $p \oplus \neg p$
 - $p \oplus \neg q$
 - $\neg p \oplus \neg q$
 - $(p \oplus q) \vee (p \oplus \neg q)$
 - $(p \oplus q) \wedge (p \oplus \neg q)$
37. Construct a truth table for each of these compound propositions.
- $p \rightarrow \neg q$
 - $\neg p \leftrightarrow q$
 - $(p \rightarrow q) \vee (\neg p \rightarrow q)$
 - $(p \rightarrow q) \wedge (\neg p \rightarrow q)$
 - $(p \leftrightarrow q) \vee (\neg p \leftrightarrow q)$
 - $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$
38. Construct a truth table for each of these compound propositions.
- $(p \vee q) \vee r$
 - $(p \vee q) \wedge r$
 - $(p \wedge q) \vee r$
 - $(p \wedge q) \wedge r$
 - $(p \vee q) \wedge \neg r$
 - $(p \wedge q) \vee \neg r$
39. Construct a truth table for each of these compound propositions.
- $p \rightarrow (\neg q \vee r)$
 - $\neg p \rightarrow (q \rightarrow r)$
 - $(p \rightarrow q) \vee (\neg p \rightarrow r)$
 - $(p \rightarrow q) \wedge (\neg p \rightarrow r)$
 - $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
 - $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$
40. Construct a truth table for $((p \rightarrow q) \rightarrow r) \rightarrow s$.
41. Construct a truth table for $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$.
42. Explain, without using a truth table, why $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when p, q , and r have the same truth value and it is false otherwise.
43. Explain, without using a truth table, why $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when at least one of p, q , and r is true and at least one is false, but is false when all three variables have the same truth value.
44. If $p_1, p_2, \dots, p_n$ are n propositions, explain why
- $$\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg p_i \vee \neg p_j)$$
- is true if and only if at most one of $p_1, p_2, \dots, p_n$ is true.
45. Use Exercise 44 to construct a compound proposition that is true if and only if exactly one of the propositions $p_1, p_2, \dots, p_n$ is true. [Hint: Combine the compound proposition in Exercise 44 and a compound proposition that is true if and only if at least one of $p_1, p_2, \dots, p_n$ is true.]
46. What is the value of x after each of these statements is encountered in a computer program, if $x = 1$ before the statement is reached?
- if $x + 2 = 3$ then $x := x + 1$
 - if $(x + 1 = 3)$ OR $(2x + 2 = 3)$ then $x := x + 1$
 - if $(2x + 3 = 5)$ AND $(3x + 4 = 7)$ then $x := x + 1$
 - if $(x + 1 = 2)$ XOR $(x + 2 = 3)$ then $x := x + 1$
 - if $x < 2$ then $x := x + 1$
47. Find the bitwise OR, bitwise AND, and bitwise XOR of each of these pairs of bit strings.
- 101 1110, 010 0001
 - 1111 0000, 1010 1010
 - 00 0111 0001, 10 0100 1000
 - 11 1111 1111, 00 0000 0000

48. Evaluate each of these expressions.

- a) $1\ 1000 \wedge (0\ 1011 \vee 1\ 1011)$
- b) $(0\ 1111 \wedge 1\ 0101) \vee 0\ 1000$
- c) $(0\ 1010 \oplus 1\ 1011) \oplus 0\ 1000$
- d) $(1\ 1011 \vee 0\ 1010) \wedge (1\ 0001 \vee 1\ 1011)$

Fuzzy logic is used in artificial intelligence. In fuzzy logic, a proposition has a truth value that is a number between 0 and 1, inclusive. A proposition with a truth value of 0 is false and one with a truth value of 1 is true. Truth values that are between 0 and 1 indicate varying degrees of truth. For instance, the truth value 0.8 can be assigned to the statement “Fred is happy,” because Fred is happy most of the time, and the truth value 0.4 can be assigned to the statement “John is happy,” because John is happy slightly less than half the time. Use these truth values to solve Exercises 49–51.

- 49. The truth value of the negation of a proposition in fuzzy logic is 1 minus the truth value of the proposition. What are the truth values of the statements “Fred is not happy” and “John is not happy”?
- 50. The truth value of the conjunction of two propositions in fuzzy logic is the minimum of the truth values of the two propositions. What are the truth values of the statements

“Fred and John are happy” and “Neither Fred nor John is happy”?

- 51. The truth value of the disjunction of two propositions in fuzzy logic is the maximum of the truth values of the two propositions. What are the truth values of the statements “Fred is happy, or John is happy” and “Fred is not happy, or John is not happy”?
- *52. Is the assertion “This statement is false” a proposition?
- *53. The n th statement in a list of 100 statements is “Exactly n of the statements in this list are false.”
 - a) What conclusions can you draw from these statements?
 - b) Answer part (a) if the n th statement is “At least n of the statements in this list are false.”
 - c) Answer part (b) assuming that the list contains 99 statements.
- 54. An ancient Sicilian legend says that the barber in a remote town who can be reached only by traveling a dangerous mountain road shaves those people, and only those people, who do not shave themselves. Can there be such a barber?

1.2 Applications of Propositional Logic

1.2.1 Introduction

Logic has many important applications to mathematics, computer science, and numerous other disciplines. Statements in mathematics and the sciences and in natural language often are imprecise or ambiguous. To make such statements precise, they can be translated into the language of logic. For example, logic is used in the specification of software and hardware, because these specifications need to be precise before development begins. Furthermore, propositional logic and its rules can be used to design computer circuits, to construct computer programs, to verify the correctness of programs, and to build expert systems. Logic can be used to analyze and solve many familiar puzzles. Software systems based on the rules of logic have been developed for constructing some, but not all, types of proofs automatically. We will discuss some of these applications of propositional logic in this section and in later chapters.

1.2.2 Translating English Sentences

There are many reasons to translate English sentences into expressions involving propositional variables and logical connectives. In particular, English (and every other human language) is often ambiguous. Translating sentences into compound statements (and other types of logical expressions, which we will introduce later in this chapter) removes the ambiguity. Note that this may involve making a set of reasonable assumptions based on the intended meaning of the sentence. Moreover, once we have translated sentences from English into logical expressions, we can analyze these logical expressions to determine their truth values, we can manipulate them, and we can use rules of inference (which are discussed in Section 1.6) to reason about them.

To illustrate the process of translating an English sentence into a logical expression, consider Examples 1 and 2.

EXAMPLE 1 How can this English sentence be translated into a logical expression?**Extra
Examples** 

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

Solution: There are many ways to translate this sentence into a logical expression. Although it is possible to represent the sentence by a single propositional variable, such as p , this would not be useful when analyzing its meaning or reasoning with it. Instead, we will use propositional variables to represent each sentence part and determine the appropriate logical connectives between them. In particular, we let a , c , and f represent “You can access the Internet from campus,” “You are a computer science major,” and “You are a freshman,” respectively. Noting that “only if” is one way a conditional statement can be expressed, this sentence can be represented as

$$a \rightarrow (c \vee \neg f).$$


EXAMPLE 2 How can this English sentence be translated into a logical expression?

“You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.”

Solution: Let q , r , and s represent “You can ride the roller coaster,” “You are under 4 feet tall,” and “You are older than 16 years old,” respectively. Then the sentence can be translated to

$$(r \wedge \neg s) \rightarrow \neg q.$$

There are other ways to represent the original sentence as a logical expression, but the one we have used should meet our needs. 

1.2.3 System Specifications

Translating sentences in natural language (such as English) into logical expressions is an essential part of specifying both hardware and software systems. System and software engineers take requirements in natural language and produce precise and unambiguous specifications that can be used as the basis for system development. Example 3 shows how compound propositions can be used in this process.

EXAMPLE 3 Express the specification “The automated reply cannot be sent when the file system is full” using logical connectives.**Extra
Examples** 

Solution: One way to translate this is to let p denote “The automated reply can be sent” and q denote “The file system is full.” Then $\neg p$ represents “It is not the case that the automated reply can be sent,” which can also be expressed as “The automated reply cannot be sent.” Consequently, our specification can be represented by the conditional statement $q \rightarrow \neg p$. 

System specifications should be **consistent**, that is, they should not contain conflicting requirements that could be used to derive a contradiction. When specifications are not consistent, there would be no way to develop a system that satisfies all specifications.

EXAMPLE 4 Determine whether these system specifications are consistent:

“The diagnostic message is stored in the buffer or it is retransmitted.”

“The diagnostic message is not stored in the buffer.”

“If the diagnostic message is stored in the buffer, then it is retransmitted.”

Solution: To determine whether these specifications are consistent, we first express them using logical expressions. Let p denote “The diagnostic message is stored in the buffer” and let q denote “The diagnostic message is retransmitted.” The specifications can then be written as $p \vee q$, $\neg p$, and $p \rightarrow q$. An assignment of truth values that makes all three specifications true must have p false to make $\neg p$ true. Because we want $p \vee q$ to be true but p must be false, q must be true. Because $p \rightarrow q$ is true when p is false and q is true, we conclude that these specifications are consistent, because they are all true when p is false and q is true. We could come to the same conclusion by use of a truth table to examine the four possible assignments of truth values to p and q . 

EXAMPLE 5 Do the system specifications in Example 4 remain consistent if the specification “The diagnostic message is not retransmitted” is added?

Solution: By the reasoning in Example 4, the three specifications from that example are true only in the case when p is false and q is true. However, this new specification is $\neg q$, which is false when q is true. Consequently, these four specifications are inconsistent. 

1.2.4 Boolean Searches



Logical connectives are used extensively in searches of large collections of information, such as indexes of Web pages. Because these searches employ techniques from propositional logic, they are called **Boolean searches**.

In Boolean searches, the connective *AND* is used to match records that contain both of two search terms, the connective *OR* is used to match one or both of two search terms, and the connective *NOT* (sometimes written as *AND NOT*) is used to exclude a particular search term. Careful planning of how logical connectives are used is often required when Boolean searches are used to locate information of potential interest. Example 6 illustrates how Boolean searches are carried out.

EXAMPLE 6 Web Page Searching Most Web search engines support Boolean searching techniques, which is useful for finding Web pages about particular subjects. For instance, using Boolean searching to find Web pages about universities in New Mexico, we can look for pages matching *NEW AND MEXICO AND UNIVERSITIES*. The results of this search will include those pages that contain the three words *NEW*, *MEXICO*, and *UNIVERSITIES*. This will include all of the pages of interest, together with others such as a page about new universities in Mexico. (Note that Google, and many other search engines, do require the use of “AND” because such search engines use all search terms by default.) Most search engines support the use of quotation marks to search for specific phrases. So, it may be more effective to search for pages matching “*NEW MEXICO*” *AND UNIVERSITIES*. 



Next, to find pages that deal with universities in New Mexico or Arizona, we can search for pages matching *(NEW AND MEXICO OR ARIZONA) AND UNIVERSITIES*. (*Note:* Here the *AND* operator takes precedence over the *OR* operator. Also, in Google, the terms used for this search would be *NEW MEXICO OR ARIZONA*.) The results of this search will include all pages that contain the word *UNIVERSITIES* and either both the words *NEW* and *MEXICO* or the word *ARIZONA*. Again, pages besides those of interest will be listed. Finally, to find Web pages that deal with universities in Mexico (and not New Mexico), we might first look for pages matching *MEXICO AND UNIVERSITIES*, but because the results of this search will include pages about universities in New Mexico, as well as universities in Mexico, it might be better to search for pages matching *(MEXICO AND UNIVERSITIES) NOT NEW*. The results of this search include pages that contain both the words *MEXICO* and *UNIVERSITIES* but do not contain the word *NEW*. (In Google, and many other search engines, the word “NOT” is

replaced by the symbol “-”. In Google, the terms used for this last search would be MEXICO UNIVERSITIES -NEW.)

1.2.5 Logic Puzzles

Links

Puzzles that can be solved using logical reasoning are known as **logic puzzles**. Solving logic puzzles is an excellent way to practice working with the rules of logic. Also, computer programs designed to carry out logical reasoning often use well-known logic puzzles to illustrate their capabilities. Many people enjoy solving logic puzzles, published in periodicals, books, and on the Web, as a recreational activity.

The next three examples present logic puzzles, in increasing level of difficulty. Many others can be found in the exercises. In Section 1.3 we will discuss the n -queens puzzle and the game of Sudoku.

EXAMPLE 7

As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each inscribed with the message “This trunk is empty,” and Trunk 3 is inscribed with the message “The treasure is in Trunk 2.” The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?

Solution: Let p_i be the proposition that the treasure is in Trunk i , for $i = 1, 2, 3$. To translate into propositional logic the Queen’s statement that exactly one of the inscriptions is true, we observe that the inscriptions on Trunk 1, Trunk 2, and Trunk 3, are $\neg p_1$, $\neg p_2$, and p_2 , respectively. So, her statement can be translated to

$$(\neg p_1 \wedge \neg(\neg p_2) \wedge \neg p_2) \vee (\neg(\neg p_1) \wedge \neg p_2 \wedge \neg p_2) \vee (\neg(\neg p_1) \wedge \neg(\neg p_2) \wedge p_2).$$

Using the rules for propositional logic, we see that this is equivalent to $(p_1 \wedge \neg p_2) \vee (p_1 \wedge p_2)$. By the distributive law, $(p_1 \wedge \neg p_2) \vee (p_1 \wedge p_2)$ is equivalent to $p_1 \wedge (\neg p_2 \vee p_2)$. But because $\neg p_2 \vee p_2$ must be true, this is then equivalent to $p_1 \wedge \mathbf{T}$, which is in turn equivalent to p_1 . So the treasure is in Trunk 1 (that is, p_1 is true), and p_2 and p_3 are false; and the inscription on Trunk 2 is the only true one. (Here, we have used the concept of propositional equivalence, which is discussed in Section 1.3.)

Next, we introduce a puzzle originally posed by Raymond Smullyan, a master of logic puzzles, who has published more than a dozen books containing challenging puzzles that involve logical reasoning.

EXAMPLE 8

In [Sm78] Smullyan posed many puzzles about an island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B . What are A and B if A says “ B is a knight” and B says “The two of us are opposite types”?

Extra Examples

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively, so that $\neg p$ and $\neg q$ are the statements that A is a knave and B is a knave, respectively.

We first consider the possibility that A is a knight; this is the statement that p is true. If A is a knight, then he is telling the truth when he says that B is a knight, so that q is true, and A and B are the same type. However, if B is a knight, then B ’s statement that A and B are of opposite

types, the statement $(p \wedge \neg q) \vee (\neg p \wedge q)$, would have to be true, which it is not, because A and B are both knights. Consequently, we can conclude that A is not a knight, that is, that p is false.

If A is a knave, then because everything a knave says is false, A 's statement that B is a knight, that is, that q is true, is a lie. This means that q is false and B is also a knave. Furthermore, if B is a knave, then B 's statement that A and B are opposite types is a lie, which is consistent with both A and B being knaves. We can conclude that both A and B are knaves. ▶

We pose more of Smullyan's puzzles about knights and knaves in Exercises 23–27. In Exercises 28–35 we introduce related puzzles where we have three types of people, knights and knaves as in this puzzle together with spies who can lie.

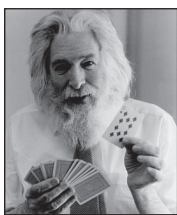
Next, we pose a puzzle known as the **muddy children puzzle** for the case of two children.

EXAMPLE 9 A father tells his two children, a boy and a girl, to play in their backyard without getting dirty. However, while playing, both children get mud on their foreheads. When the children stop playing, the father says “At least one of you has a muddy forehead,” and then asks the children to answer “Yes” or “No” to the question: “Do you know whether you have a muddy forehead?” The father asks this question twice. What will the children answer each time this question is asked, assuming that a child can see whether his or her sibling has a muddy forehead, but cannot see his or her own forehead? Assume that both children are honest and that the children answer each question simultaneously.

Solution: Let s be the statement that the son has a muddy forehead and let d be the statement that the daughter has a muddy forehead. When the father says that at least one of the two children has a muddy forehead, he is stating that the disjunction $s \vee d$ is true. Both children will answer “No” the first time the question is asked because each sees mud on the other child's forehead. That is, the son knows that d is true, but does not know whether s is true, and the daughter knows that s is true, but does not know whether d is true.

After the son has answered “No” to the first question, the daughter can determine that d must be true. This follows because when the first question is asked, the son knows that $s \vee d$ is true, but cannot determine whether s is true. Using this information, the daughter can conclude that d must be true, for if d were false, the son could have reasoned that because $s \vee d$ is true,

Links ▶



Courtesy of Indiana University Archives

RAYMOND SMULLYAN (1919–2017) Raymond Smullyan, the son of a businessman and a homemaker, was born in Far Rockaway, Queens, New York. He dropped out of high school because he wanted to study what he was really interested in and not standard high school material. After attending Pacific College and Reed College in Oregon, he earned an undergraduate degree in mathematics at the University of Chicago in 1955. He paid his college expenses by performing magic tricks at parties and clubs, using the stage name Five-Ace Merrill. He obtained a Ph.D. in logic in 1959 at Princeton, studying under Alonzo Church. After graduating from Princeton, he taught mathematics and logic at Dartmouth College, Princeton University, Yeshiva University, and the City University of New York. He joined the philosophy department at Indiana University in 1981, where he became an emeritus professor.

Smullyan wrote many books on recreational logic and mathematics, including *Satan, Cantor, and Infinity*; *What Is the Name of This Book?*; *The Lady or the Tiger?*; *Alice in Puzzleland*; *To Mock a Mockingbird*; *Forever Undecided*; and *The Riddle of Scheherazade: Amazing Logic Puzzles, Ancient and Modern*. Because his logic puzzles are challenging, entertaining, and thought-provoking, he was considered to be a modern-day Lewis Carroll. Smullyan also wrote several books about the application of deductive logic to chess, three collections of philosophical essays and aphorisms, and several advanced books on mathematical logic and set theory. He was particularly interested in self-reference and worked on extending some of Gödel's results that show that it is impossible to write a computer program that can solve all mathematical problems. He was also particularly interested in explaining ideas from mathematical logic to the public.

Smullyan was a talented musician and often played piano with his second wife, who was a concert-level pianist. Making telescopes was one of his hobbies and he was interested in optics and stereo photography. He said, “I’ve never had a conflict between teaching and research as some people do because when I’m teaching, I’m doing research.” Smullyan is the subject of a documentary short film entitled *This Film Needs No Title*.

then s must be true, and he would have answered “Yes” to the first question. The son can reason in a similar way to determine that s must be true. It follows that both children answer “Yes” the second time the question is asked. ◀

1.2.6 Logic Circuits

Propositional logic can be applied to the design of computer hardware. This was first observed in 1938 by Claude Shannon in his MIT master’s thesis. In Chapter 12 we will study this topic in depth. (See that chapter for a biography of Shannon.) We give a brief introduction to this application here.

A **logic circuit** (or **digital circuit**) receives input signals $p_1, p_2, \dots, p_n$, each a bit [either 0 (off) or 1 (on)], and produces output signals $s_1, s_2, \dots, s_n$, each a bit. In this section we will restrict our attention to logic circuits with a single output signal; in general, digital circuits may have multiple outputs.

Complicated digital circuits can be constructed from three basic circuits, called **gates**, shown in Figure 1. The **inverter**, or **NOT gate**, takes an input bit p , and produces as output $\neg p$. The **OR gate** takes two input signals p and q , each a bit, and produces as output the signal $p \vee q$. Finally, the **AND gate** takes two input signals p and q , each a bit, and produces as output the signal $p \wedge q$. We use combinations of these three basic gates to build more complicated circuits, such as that shown in Figure 2.

In Chapter 12 we will design some useful circuits.

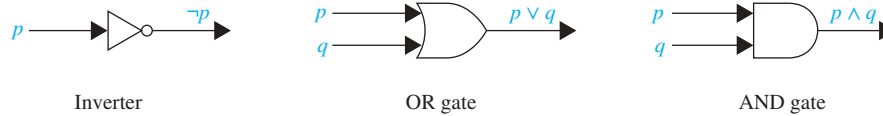


FIGURE 1 Basic logic gates.

Given a circuit built from the basic logic gates and the inputs to the circuit, we determine the output by tracing through the circuit, as Example 10 shows.

EXAMPLE 10 Determine the output for the combinatorial circuit in Figure 2.

Solution: In Figure 2 we display the output of each logic gate in the circuit. We see that the AND gate takes input of p and $\neg q$, the output of the inverter with input q , and produces $p \wedge \neg q$. Next, we note that the OR gate takes input $p \wedge \neg q$ and $\neg r$, the output of the inverter with input r , and produces the final output $(p \wedge \neg q) \vee \neg r$. ◀

Suppose that we have a formula for the output of a digital circuit in terms of negations, disjunctions, and conjunctions. Then, we can systematically build a digital circuit with the desired output, as illustrated in Example 11.

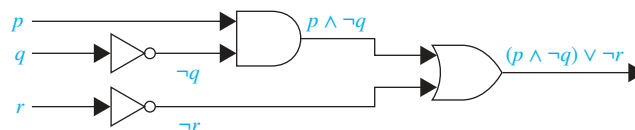


FIGURE 2 A combinatorial circuit.

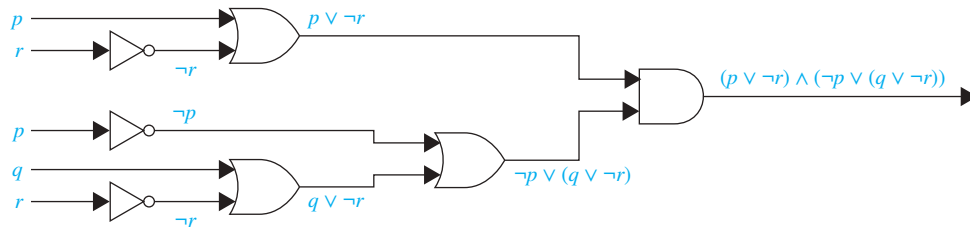


FIGURE 3 The circuit for $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$.

EXAMPLE 11 Build a digital circuit that produces the output $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$ when given input bits p , q , and r .

Solution: To construct the desired circuit, we build separate circuits for $p \vee \neg r$ and for $\neg p \vee (q \vee \neg r)$ and combine them using an AND gate. To construct a circuit for $p \vee \neg r$, we use an inverter to produce $\neg r$ from the input r . Then, we use an OR gate to combine p and $\neg r$. To build a circuit for $\neg p \vee (q \vee \neg r)$, we first use an inverter to obtain $\neg r$. Then we use an OR gate with inputs q and $\neg r$ to obtain $q \vee \neg r$. Finally, we use another inverter and an OR gate to get $\neg p \vee (q \vee \neg r)$ from the inputs p and $q \vee \neg r$.

To complete the construction, we employ a final AND gate, with inputs $p \vee \neg r$ and $\neg p \vee (q \vee \neg r)$. The resulting circuit is displayed in Figure 3. ◀

We will study logic circuits in great detail in Chapter 12 in the context of Boolean algebra, and with different notation.

Exercises

In Exercises 1–6, translate the given statement into propositional logic using the propositions provided.

1. You cannot edit a protected Wikipedia entry unless you are an administrator. Express your answer in terms of e : “You can edit a protected Wikipedia entry” and a : “You are an administrator.”
2. You can see the movie only if you are over 18 years old or you have the permission of a parent. Express your answer in terms of m : “You can see the movie,” e : “You are over 18 years old,” and p : “You have the permission of a parent.”
3. You can graduate only if you have completed the requirements of your major and you do not owe money to the university and you do not have an overdue library book. Express your answer in terms of g : “You can graduate,” m : “You owe money to the university,” r : “You have completed the requirements of your major,” and b : “You have an overdue library book.”
4. To use the wireless network in the airport you must pay the daily fee unless you are a subscriber to the service. Express your answer in terms of w : “You can use the wireless network in the airport,” d : “You pay the daily fee,” and s : “You are a subscriber to the service.”
5. You are eligible to be President of the U.S.A. only if you are at least 35 years old, were born in the U.S.A., or at the time of your birth both of your parents were citizens, and you have lived at least 14 years in the country. Express your answer in terms of e : “You are eligible to be President of the U.S.A.,” a : “You are at least 35 years old,” b : “You were born in the U.S.A.,” p : “At the time of your birth, both of your parents were citizens,” and r : “You have lived at least 14 years in the U.S.A.”
6. You can upgrade your operating system only if you have a 32-bit processor running at 1 GHz or faster, at least 1 GB RAM, and 16 GB free hard disk space, or a 64-bit processor running at 2 GHz or faster, at least 2 GB RAM, and at least 32 GB free hard disk space. Express your answer in terms of u : “You can upgrade your operating system,” b_{32} : “You have a 32-bit processor,” b_{64} : “You have a 64-bit processor,” g_1 : “Your processor runs at 1 GHz or faster,” g_2 : “Your processor runs at 2 GHz or faster,” r_1 : “Your processor has at least 1 GB RAM,” r_2 : “Your processor has at least 2 GB RAM,” h_{16} : “You have at least 16 GB free hard disk space,” and h_{32} : “You have at least 32 GB free hard disk space.”
7. Express these system specifications using the propositions p : “The message is scanned for viruses” and q : “The message was sent from an unknown system” together with logical connectives (including negations).
 - a) “The message is scanned for viruses whenever the message was sent from an unknown system.”

- b) "The message was sent from an unknown system but it was not scanned for viruses."
 - c) "It is necessary to scan the message for viruses whenever it was sent from an unknown system."
 - d) "When a message is not sent from an unknown system it is not scanned for viruses."
8. Express these system specifications using the propositions p : "The user enters a valid password," q : "Access is granted," and r : "The user has paid the subscription fee" and logical connectives (including negations).
- a) "The user has paid the subscription fee, but does not enter a valid password."
 - b) "Access is granted whenever the user has paid the subscription fee and enters a valid password."
 - c) "Access is denied if the user has not paid the subscription fee."
 - d) "If the user has not entered a valid password but has paid the subscription fee, then access is granted."
9. Are these system specifications consistent? "The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."
10. Are these system specifications consistent? "Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."
11. Are these system specifications consistent? "The router can send packets to the edge system only if it supports the new address space. For the router to support the new address space it is necessary that the latest software release be installed. The router can send packets to the edge system if the latest software release is installed. The router does not support the new address space."
12. Are these system specifications consistent? "If the file system is not locked, then new messages will be queued. If the file system is not locked, then the system is functioning normally, and conversely. If new messages are not queued, then they will be sent to the message buffer. If the file system is not locked, then new messages will be sent to the message buffer. New messages will not be sent to the message buffer."
13. What Boolean search would you use to look for Web pages about beaches in New Jersey? What if you wanted to find Web pages about beaches on the isle of Jersey (in the English Channel)?
14. What Boolean search would you use to look for Web pages about hiking in West Virginia? What if you wanted to find Web pages about hiking in Virginia, but not in West Virginia?
15. What Google search would you use to look for Web pages relating to Ethiopian restaurants in New York or New Jersey?
16. What Google search would you use to look for men's shoes or boots not designed for work?
17. Suppose that in Example 7, the inscriptions on Trunks 1, 2, and 3 are "The treasure is in Trunk 3," "The treasure is in Trunk 1," and "This trunk is empty." For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.
- a) "All the inscriptions are false."
 - b) "Exactly one of the inscriptions is true."
 - c) "Exactly two of the inscriptions are true."
 - d) "All three inscriptions are true."
18. Suppose that in Example 7 there are treasures in two of the three trunks. The inscriptions on Trunks 1, 2, and 3 are "This trunk is empty," "There is a treasure in Trunk 1," and "There is a treasure in Trunk 2." For each of these statements, determine whether the Queen who never lies could state this, and if so, which two trunks the treasures are in.
- a) "All the inscriptions are false."
 - b) "Exactly one of the inscriptions is true."
 - c) "Exactly two of the inscriptions are true."
 - d) "All three inscriptions are true."
- *19. Each inhabitant of a remote village always tells the truth or always lies. A villager will give only a "Yes" or a "No" response to a question a tourist asks. Suppose you are a tourist visiting this area and come to a fork in the road. One branch leads to the ruins you want to visit; the other branch leads deep into the jungle. A villager is standing at the fork in the road. What one question can you ask the villager to determine which branch to take?
20. An explorer is captured by a group of cannibals. There are two types of cannibals—those who always tell the truth and those who always lie. The cannibals will barbecue the explorer unless he can determine whether a particular cannibal always lies or always tells the truth. He is allowed to ask the cannibal exactly one question.
- a) Explain why the question "Are you a liar?" does not work.
 - b) Find a question that the explorer can use to determine whether the cannibal always lies or always tells the truth.
21. When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?" The first professor says "I do not know." The second professor then says "I do not know." Finally, the third professor says "No, not everyone wants coffee." The hostess comes back and gives coffee to the professors who want it. How did she figure out who wanted coffee?
22. When planning a party you want to know whom to invite. Among the people you would like to invite are three touchy friends. You know that if Jasmine attends, she will become unhappy if Samir is there, Samir will attend only if Kanti will be there, and Kanti will not attend unless Jasmine also does. Which combinations of these three friends can you invite so as not to make someone unhappy?

Exercises 23–27 relate to inhabitants of the island of knights and knaves created by Smullyan, where knights always tell the truth and knaves always lie. You encounter two people, *A* and *B*. Determine, if possible, what *A* and *B* are if they address you in the ways described. If you cannot determine what these two people are, can you draw any conclusions?

23. *A* says “At least one of us is a knave” and *B* says nothing.
24. *A* says “The two of us are both knights” and *B* says “*A* is a knave.”
25. *A* says “I am a knave or *B* is a knight” and *B* says nothing.
26. Both *A* and *B* say “I am a knight.”
27. *A* says “We are both knaves” and *B* says nothing.

Exercises 28–35 relate to inhabitants of an island on which there are three kinds of people: knights who always tell the truth, knaves who always lie, and spies (called normals by Smullyan [Sm78]) who can either lie or tell the truth. You encounter three people, *A*, *B*, and *C*. You know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of other two is. For each of these situations, if possible, determine whether there is a unique solution and determine who the knave, knight, and spy are. When there is no unique solution, list all possible solutions or state that there are no solutions.

28. *A* says “*C* is the knave,” *B* says “*A* is the knight,” and *C* says “I am the spy.”
29. *A* says “I am the knight,” *B* says “I am the knave,” and *C* says “*B* is the knight.”
30. *A* says “I am the knave,” *B* says “I am the knave,” and *C* says “I am the knave.”
31. *A* says “I am the knight,” *B* says “*A* is telling the truth,” and *C* says “I am the spy.”
32. *A* says “I am the knight,” *B* says “*A* is not the knave,” and *C* says “*B* is not the knave.”
33. *A* says “I am the knight,” *B* says “I am the knight,” and *C* says “I am the knight.”
34. *A* says “I am not the spy,” *B* says “I am not the spy,” and *C* says “*A* is the spy.”
35. *A* says “I am not the spy,” *B* says “I am not the spy,” and *C* says “I am not the spy.”

Exercises 36–42 are puzzles that can be solved by translating statements into logical expressions and reasoning from these expressions using truth tables.

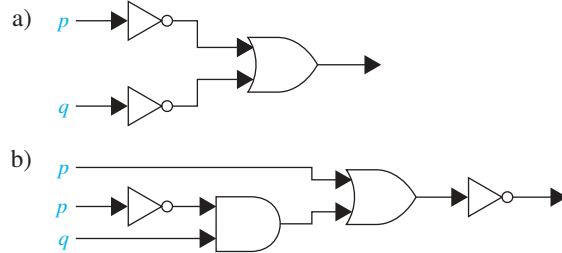
36. The police have three suspects for the murder of Mr. Cooper: Mr. Smith, Mr. Jones, and Mr. Williams. Smith, Jones, and Williams each declare that they did not kill Cooper. Smith also states that Cooper was a friend of Jones and that Williams disliked him. Jones also states that he did not know Cooper and that he was out of town the day Cooper was killed. Williams also states that he saw both Smith and Jones with Cooper the day of the killing and that either Smith or Jones must have killed him. Can you determine who the murderer was if
 - a) one of the three men is guilty, the two innocent men are telling the truth, but the statements of the guilty man may or may not be true?
 - b) innocent men do not lie?

37. Steve would like to determine the relative salaries of three coworkers using two facts. First, he knows that if Fred is not the highest paid of the three, then Janice is. Second, he knows that if Janice is not the lowest paid, then Maggie is paid the most. Is it possible to determine the relative salaries of Fred, Maggie, and Janice from what Steve knows? If so, who is paid the most and who the least? Explain your reasoning.
38. Five friends have access to a chat room. Is it possible to determine who is chatting if the following information is known? Either Kevin or Heather, or both, are chatting. Either Randy or Vijay, but not both, are chatting. If Abby is chatting, so is Randy. Vijay and Kevin are either both chatting or neither is. If Heather is chatting, then so are Abby and Kevin. Explain your reasoning.
39. A detective has interviewed four witnesses to a crime. From the stories of the witnesses the detective has concluded that if the butler is telling the truth then so is the cook; the cook and the gardener cannot both be telling the truth; the gardener and the handyman are not both lying; and if the handyman is telling the truth then the cook is lying. For each of the four witnesses, can the detective determine whether that person is telling the truth or lying? Explain your reasoning.
40. Four friends have been identified as suspects for an unauthorized access into a computer system. They have made statements to the investigating authorities. Alice said, “Carlos did it.” John said, “I did not do it.” Carlos said, “Diana did it.” Diana said, “Carlos lied when he said that I did it.”
 - a) If the authorities also know that exactly one of the four suspects is telling the truth, who did it? Explain your reasoning.
 - b) If the authorities also know that exactly one is lying, who did it? Explain your reasoning.
41. Suppose there are signs on the doors to two rooms. The sign on the first door reads “In this room there is a lady, and in the other one there is a tiger”; and the sign on the second door reads “In one of these rooms, there is a lady, and in one of them there is a tiger.” Suppose that you know that one of these signs is true and the other is false. Behind which door is the lady?
- *42. Solve this famous logic puzzle, attributed to Albert Einstein, and known as the **zebra puzzle**. Five men with different nationalities and with different jobs live in consecutive houses on a street. These houses are painted different colors. The men have different pets and have different favorite drinks. Determine who owns a zebra and whose favorite drink is mineral water (which is one of the favorite drinks) given these clues: The Englishman lives in the red house. The Spaniard owns a dog. The Japanese man is a painter. The Italian drinks tea. The Norwegian lives in the first house on the left. The green house is immediately to the right of the white one. The photographer breeds snails. The diplomat lives in the yellow house. Milk is drunk in the middle house. The owner of the green house drinks coffee. The Norwegian’s house is next to the

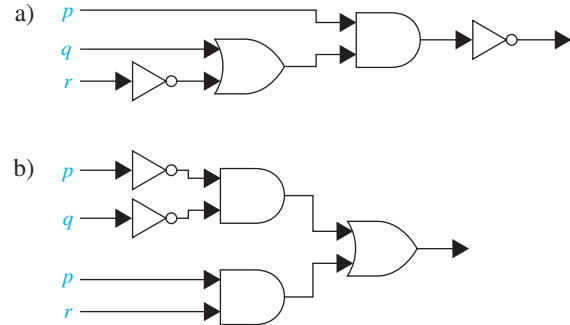


blue one. The violinist drinks orange juice. The fox is in a house next to that of the physician. The horse is in a house next to that of the diplomat. [Hint: Make a table where the rows represent the men and columns represent the color of their houses, their jobs, their pets, and their favorite drinks and use logical reasoning to determine the correct entries in the table.]

43. Freedonia has 50 senators. Each senator is either honest or corrupt. Suppose you know that at least one of the Freedonian senators is honest and that, given any two Freedonian senators, at least one is corrupt. Based on these facts, can you determine how many Freedonian senators are honest and how many are corrupt? If so, what is the answer?
44. Find the output of each of these combinatorial circuits.



45. Find the output of each of these combinatorial circuits.



46. Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $(p \wedge \neg r) \vee (\neg q \wedge r)$ from input bits p , q , and r .
47. Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $((\neg p \vee \neg r) \wedge \neg q) \vee (\neg p \wedge (q \vee r))$ from input bits p , q , and r .

1.3 Propositional Equivalences

1.3.1 Introduction

An important type of step used in a mathematical argument is the replacement of a statement with another statement with the same truth value. Because of this, methods that produce propositions with the same truth value as a given compound proposition are used extensively in the construction of mathematical arguments. Note that we will use the term “compound proposition” to refer to an expression formed from propositional variables using logical operators, such as $p \wedge q$.

We begin our discussion with a classification of compound propositions according to their possible truth values.

Definition 1

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a *tautology*. A compound proposition that is always false is called a *contradiction*. A compound proposition that is neither a tautology nor a contradiction is called a *contingency*.

Tautologies and contradictions are often important in mathematical reasoning. Example 1 illustrates these types of compound propositions.

EXAMPLE 1 We can construct examples of tautologies and contradictions using just one propositional variable. Consider the truth tables of $p \vee \neg p$ and $p \wedge \neg p$, shown in Table 1. Because $p \vee \neg p$ is always true, it is a tautology. Because $p \wedge \neg p$ is always false, it is a contradiction. ◀

TABLE 1 Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

1.3.2 Logical Equivalences



Compound propositions that have the same truth values in all possible cases are called **logically equivalent**. We can also define this notion as follows.

Definition 2

The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

Remark: The symbol $\equiv$ is not a logical connective, and $p \equiv q$ is not a compound proposition but rather is the statement that $p \leftrightarrow q$ is a tautology. The symbol $\Leftrightarrow$ is sometimes used instead of $\equiv$ to denote logical equivalence.



One way to determine whether two compound propositions are equivalent is to use a truth table. In particular, the compound propositions p and q are equivalent if and only if the columns giving their truth values agree. Example 2 illustrates this method to establish an extremely important and useful logical equivalence, namely, that of $\neg(p \vee q)$ with $\neg p \wedge \neg q$. This logical equivalence is one of the two **De Morgan laws**, shown in Table 2, named after the English mathematician Augustus De Morgan, of the mid-nineteenth century.

TABLE 2 De Morgan's Laws.

$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

EXAMPLE 2 Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

Solution: The truth tables for these compound propositions are displayed in Table 3. Because the truth values of the compound propositions $\neg(p \vee q)$ and $\neg p \wedge \neg q$ agree for all possible combinations of the truth values of p and q , it follows that $\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$ is a tautology and that these compound propositions are logically equivalent. 

TABLE 3 Truth Tables for $\neg(p \vee q)$ and $\neg p \wedge \neg q$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

The next example establishes an extremely important equivalence. It allows us to replace conditional statements with negations and disjunctions.

EXAMPLE 3 Show that $p \rightarrow q$ and $\neg p \vee q$ are logically equivalent. (This is known as the **conditional-disjunction equivalence**.)

Solution: We construct the truth table for these compound propositions in Table 4. Because the truth values of $\neg p \vee q$ and $p \rightarrow q$ agree, they are logically equivalent. ◀

TABLE 4 Truth Tables for $\neg p \vee q$ and $p \rightarrow q$.				
p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

We will now establish a logical equivalence of two compound propositions involving three different propositional variables p , q , and r . To use a truth table to establish such a logical equivalence, we need eight rows, one for each possible combination of truth values of these three variables. We symbolically represent these combinations by listing the truth values of p , q , and r , respectively. These eight combinations of truth values are TTT, TTF, TFT, TFF, FTT, FTF, FFT, and FFF; we use this order when we display the rows of the truth table. Note that we need to double the number of rows in the truth tables we use to show that compound propositions are equivalent for each additional propositional variable, so that 16 rows are needed to establish the logical equivalence of two compound propositions involving four propositional variables, and so on. In general, 2^n rows are required if a compound proposition involves n propositional variables. Because of the rapid growth of 2^n , more efficient ways are needed to establish logical equivalences, such as by using ones we already know. This technique will be discussed later.

EXAMPLE 4 Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. This is the *distributive law* of disjunction over conjunction.

Solution: We construct the truth table for these compound propositions in Table 5. Because the truth values of $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ agree, these compound propositions are logically equivalent. ◀

TABLE 5 A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.							
p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

TABLE 6 Logical Equivalences.	
<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

The identities in Table 6 are a special case of Boolean algebra identities found in Table 5 of Section 12.1. See Table 1 in Section 2.2 for analogous set identities.

Table 6 contains some important equivalences. In these equivalences, **T** denotes the compound proposition that is always true and **F** denotes the compound proposition that is always false. We also display some useful equivalences for compound propositions involving conditional statements and biconditional statements in Tables 7 and 8, respectively. The reader is asked to verify the equivalences in Tables 6–8 in the exercises.

TABLE 7 Logical Equivalences Involving Conditional Statements.
$p \rightarrow q \equiv \neg p \vee q$ $p \rightarrow q \equiv \neg q \rightarrow \neg p$ $p \vee q \equiv \neg p \rightarrow q$ $p \wedge q \equiv \neg(p \rightarrow \neg q)$ $\neg(p \rightarrow q) \equiv p \wedge \neg q$ $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$ $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$ $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

TABLE 8 Logical Equivalences Involving Biconditional Statements.
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Be careful not to apply logical identities, such as associative laws, distributive laws, or De Morgan's laws, to expressions that have a mix of conjunctions and disjunctions when the identities only apply when all these operators are the same.

The associative law for disjunction shows that the expression $p \vee q \vee r$ is well defined, in the sense that it does not matter whether we first take the disjunction of p with q and then the disjunction of $p \vee q$ with r , or if we first take the disjunction of q and r and then take the disjunction of p with $q \vee r$. Similarly, the expression $p \wedge q \wedge r$ is well defined. By extending this reasoning, it follows that $p_1 \vee p_2 \vee \cdots \vee p_n$ and $p_1 \wedge p_2 \wedge \cdots \wedge p_n$ are well defined whenever $p_1, p_2, \dots, p_n$ are propositions.

Furthermore, note that De Morgan's laws extend to

$$\neg(p_1 \vee p_2 \vee \cdots \vee p_n) \equiv (\neg p_1 \wedge \neg p_2 \wedge \cdots \wedge \neg p_n)$$

and

$$\neg(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \equiv (\neg p_1 \vee \neg p_2 \vee \cdots \vee \neg p_n).$$

We will sometimes use the notation $\bigvee_{j=1}^n p_j$ for $p_1 \vee p_2 \vee \cdots \vee p_n$ and $\bigwedge_{j=1}^n p_j$ for $p_1 \wedge p_2 \wedge \cdots \wedge p_n$. Using this notation, the extended version of De Morgan's laws can be written concisely as $\neg(\bigvee_{j=1}^n p_j) \equiv \bigwedge_{j=1}^n \neg p_j$ and $\neg(\bigwedge_{j=1}^n p_j) \equiv \bigvee_{j=1}^n \neg p_j$. (Methods for proving these identities will be given in Section 5.1.)

A truth table with 2^n rows is needed to prove the equivalence of two compound propositions in n variables. (Note that the number of rows doubles for each additional propositional variable added. See Chapter 6 for details about solving counting problems such as this.) Because 2^n grows extremely rapidly as n increases (see Section 3.2), the use of truth tables to establish equivalences becomes impractical as the number of variables grows. It is quicker to use other methods, such as employing logical equivalences that we already know. How that can be done is discussed later in this section.

1.3.3 Using De Morgan's Laws

When using De Morgan's laws, remember to change the logical connective after you negate.

The two logical equivalences known as De Morgan's laws are particularly important. They tell us how to negate conjunctions and how to negate disjunctions. In particular, the equivalence $\neg(p \vee q) \equiv \neg p \wedge \neg q$ tells us that the negation of a disjunction is formed by taking the conjunction of the negations of the component propositions. Similarly, the equivalence $\neg(p \wedge q) \equiv \neg p \vee \neg q$ tells us that the negation of a conjunction is formed by taking the disjunction of the negations of the component propositions. Example 5 illustrates the use of De Morgan's laws.

EXAMPLE 5 Use De Morgan's laws to express the negations of "Miguel has a cellphone and he has a laptop computer" and "Heather will go to the concert or Steve will go to the concert."

Assessment  **Solution:** Let p be "Miguel has a cellphone" and q be "Miguel has a laptop computer." Then "Miguel has a cellphone and he has a laptop computer" can be represented by $p \wedge q$. By the first of De Morgan's laws, $\neg(p \wedge q)$ is equivalent to $\neg p \vee \neg q$. Consequently, we can express the negation of our original statement as "Miguel does not have a cellphone or he does not have a laptop computer."

Let r be "Heather will go to the concert" and s be "Steve will go to the concert." Then "Heather will go to the concert or Steve will go to the concert" can be represented by $r \vee s$. By the second of De Morgan's laws, $\neg(r \vee s)$ is equivalent to $\neg r \wedge \neg s$. Consequently, we can express the negation of our original statement as "Heather will not go to the concert and Steve will not go to the concert." 

1.3.4 Constructing New Logical Equivalences

The logical equivalences in Table 6, as well as any others that have been established (such as those shown in Tables 7 and 8), can be used to construct additional logical equivalences. The reason for this is that a proposition in a compound proposition can be replaced by a compound proposition that is logically equivalent to it without changing the truth value of the original compound proposition. This technique is illustrated in Examples 6–8, where we also use the fact that if p and q are logically equivalent and q and r are logically equivalent, then p and r are logically equivalent (see Exercise 60).

EXAMPLE 6 Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

Extra Examples ▶

Solution: We could use a truth table to show that these compound propositions are equivalent (similar to what we did in Example 4). Indeed, it would not be hard to do so. However, we want to illustrate how to use logical identities that we already know to establish new logical identities, something that is of practical importance for establishing equivalences of compound propositions with a large number of variables. So, we will establish this equivalence by developing a series of logical equivalences, using one of the equivalences in Table 6 at a time, starting with $\neg(p \rightarrow q)$ and ending with $p \wedge \neg q$. We have the following equivalences.

$$\begin{aligned}\neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{by the conditional-disjunction equivalence (Example 3)} \\ &\equiv \neg(\neg p) \wedge \neg q && \text{by the second De Morgan law} \\ &\equiv p \wedge \neg q && \text{by the double negation law}\end{aligned}$$

EXAMPLE 7 Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Solution: We will use one of the equivalences in Table 6 at a time, starting with $\neg(p \vee (\neg p \wedge q))$ and ending with $\neg p \wedge \neg q$. (Note: we could also easily establish this equivalence using a truth table.) We have the following equivalences.

Links ▶



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AUGUSTUS DE MORGAN (1806–1871) Augustus De Morgan was born in India, where his father was a colonel in the Indian army. De Morgan's family moved to England when he was 7 months old. He attended private schools, where in his early teens he developed a strong interest in mathematics. De Morgan studied at Trinity College, Cambridge, graduating in 1827. Although he considered medicine or law, he decided on mathematics for his career. He won a position at University College, London, in 1828, but resigned after the college dismissed a fellow professor without giving reasons. However, he resumed this position in 1836 when his successor died, remaining until 1866.

De Morgan was a noted teacher who stressed principles over techniques. His students included many famous mathematicians, including Augusta Ada, Countess of Lovelace, who was Charles Babbage's collaborator in his work on computing machines (see page 32 for biographical notes on Augusta Ada). (De Morgan cautioned the countess against studying too much mathematics, because it might interfere with her childbearing abilities!)

De Morgan was an extremely prolific writer, publishing more than 1000 articles in more than 15 periodicals. De Morgan also wrote textbooks on many subjects, including logic, probability, calculus, and algebra. In 1838 he presented what was perhaps the first clear explanation of an important proof technique known as *mathematical induction* (discussed in Section 5.1 of this text), a term he coined. In the 1840s De Morgan made fundamental contributions to the development of symbolic logic. He invented notations that helped him prove propositional equivalences, such as the laws that are named after him. In 1842 De Morgan presented what is considered to be the first precise definition of a limit and developed new tests for convergence of infinite series. De Morgan was also interested in the history of mathematics and wrote biographies of Newton and Halley.

In 1837 De Morgan married Sophia Frend, who wrote his biography in 1882. De Morgan's research, writing, and teaching left little time for his family or social life. Nevertheless, he was noted for his kindness, humor, and wide range of knowledge.

$$\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the second De Morgan law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the first De Morgan law} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{by the double negation law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the second distributive law} \\
&\equiv \mathbf{F} \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv \mathbf{F} \\
&\equiv (\neg p \wedge \neg q) \vee \mathbf{F} && \text{by the commutative law for disjunction} \\
&\equiv \neg p \wedge \neg q && \text{by the identity law for } \mathbf{F}
\end{aligned}$$

Consequently $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent. ◀

EXAMPLE 8 Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Solution: To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to **T**. (Note: This could also be done using a truth table.)

$$\begin{aligned}
(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{by Example 3} \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{by the first De Morgan law} \\
&\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{by the associative and commutative} \\
&&& \text{laws for disjunction} \\
&\equiv \mathbf{T} \vee \mathbf{T} && \text{by Example 1 and the commutative} \\
&&& \text{law for disjunction} \\
&\equiv \mathbf{T} && \text{by the domination law}
\end{aligned}$$

Links ▶



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AUGUSTA ADA, COUNTESS OF LOVELACE (1815–1852) Augusta Ada was the only child from the marriage of the flamboyant and notorious poet Lord Byron and Lady Byron, Annabella Millbanke, who separated when Ada was 1 month old, because of Lord Byron's scandalous affair with his half sister. The Lord Byron had quite a reputation, being described by one of his lovers as “mad, bad, and dangerous to know.” Lady Byron was noted for her intellect and had a passion for mathematics; she was called by Lord Byron “The Princess of Parallelograms.” Augusta was raised by her mother, who encouraged her intellectual talents especially in music and mathematics, to counter what Lady Byron considered dangerous poetic tendencies. At this time, women were not allowed to attend universities and could not join learned societies. Nevertheless, Augusta pursued her mathematical studies independently and with mathematicians, including William Frend. She was also encouraged by another female mathematician, Mary Somerville, and in 1834 at a dinner party hosted by Mary Somerville, she learned about Charles Babbage's ideas for a calculating machine, called the Analytic Engine. In 1838 Augusta Ada married Lord King, later elevated to Earl of Lovelace. Together they had three children.

Augusta Ada continued her mathematical studies after her marriage. Charles Babbage had continued work on his Analytic Engine and lectured on this in Europe. In 1842 Babbage asked Augusta Ada to translate an article in French describing Babbage's invention. When Babbage saw her translation, he suggested she add her own notes, and the resulting work was three times the length of the original. The most complete accounts of the Analytic Engine are found in Augusta Ada's notes. In her notes, she compared the working of the Analytic Engine to that of the Jacquard loom, with Babbage's punch cards analogous to the cards used to create patterns on the loom. Furthermore, she recognized the promise of the machine as a general purpose computer much better than Babbage did. She stated that the “engine is the material expression of any indefinite function of any degree of generality and complexity.” Her notes on the Analytic Engine anticipate many future developments, including computer-generated music. Augusta Ada published her writings under her initials A.A.L., concealing her identity as a woman as did many women at a time when women were not considered to be the intellectual equals of men. After 1845 she and Babbage worked toward the development of a system to predict horse races. Unfortunately, their system did not work well, leaving Augusta Ada heavily in debt at the time of her death at an unfortunately young age from uterine cancer.

In 1953 Augusta Ada's notes on the Analytic Engine were republished more than 100 years after they were written, and after they had been long forgotten. In his work in the 1950s on the capacity of computers to think (and his influential Turing test for determining whether a machine is intelligent), Alan Turing responded to Augusta Ada's statement that “The Analytic Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform.” This “dialogue” between Turing and Augusta Ada is still the subject of controversy. Because of her fundamental contributions to computing, the programming language Ada is named in honor of the Countess of Lovelace.

1.3.5 Satisfiability

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true (that is, when it is a tautology or a contingency). When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**. Note that a compound proposition is unsatisfiable if and only if its negation is true for all assignments of truth values to the variables, that is, if and only if its negation is a tautology.

When we find a particular assignment of truth values that makes a compound proposition true, we have shown that it is satisfiable; such an assignment is called a **solution** of this particular satisfiability problem. However, to show that a compound proposition is unsatisfiable, we need to show that *every* assignment of truth values to its variables makes it false. Although we can always use a truth table to determine whether a compound proposition is satisfiable, it is often more efficient not to, as Example 9 demonstrates.

EXAMPLE 9 Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable.

Solution: Instead of using a truth table to solve this problem, we will reason about truth values. Note that $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when the three variables p , q , and r have the same truth value (see Exercise 42 of Section 1.1). Hence, it is satisfiable as there is at least one assignment of truth values for p , q , and r that makes it true. Similarly, note that $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is true when at least one of p , q , and r is true and at least one is false (see Exercise 43 of Section 1.1). Hence, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable, as there is at least one assignment of truth values for p , q , and r that makes it true.

Finally, note that for $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ to be true, $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ and $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ must both be true. For the first to be true, the three variables must have the same truth values, and for the second to be true, at least one of the three variables must be true and at least one must be false. However, these conditions are contradictory. From these observations we conclude that no assignment of truth values to p , q , and r makes $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ true. Hence, it is unsatisfiable. 

1.3.6 Applications of Satisfiability

Many problems, in diverse areas such as robotics, software testing, artificial intelligence planning, computer-aided design, machine vision, integrated circuit design, scheduling, computer networking, and genetics, can be modeled in terms of propositional satisfiability. Although most applications are quite complex and beyond the scope of this book, we can illustrate how two puzzles can be modeled as satisfiability problems.

EXAMPLE 10 The n -Queens Problem The n -queens problem asks for a placement of n queens on an $n \times n$ chessboard so that no queen can attack another queen. This means that no two queens can be placed in the same row, in the same column, or on the same diagonal. We display a solution to the eight-queens problem in Figure 1. (The eight-queens problem dates back to 1848 when it was proposed by Max Bezzel and was completely solved by Franz Nauck in 1850. We will return to the n -queens problem in Section 11.4.)

To model the n -queens problem as a satisfiability problem, we introduce n^2 variables, $p(i, j)$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$. For a given placement of n queens on the chessboard, $p(i, j)$ is true when there is a queen on the square in the i th row and j th column, and is false

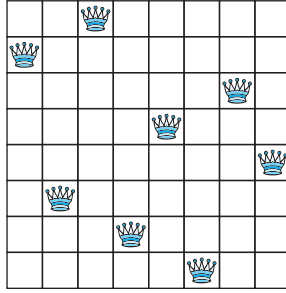


FIGURE 1

otherwise. Note that squares (i, j) and (i', j') are on the same diagonal if either $i + i' = j + j'$ or $i - i' = j - j'$. In the chessboard in Figure 1, $p(6, 2)$ and $p(2, 1)$ are true, while $p(3, 4)$ and $p(5, 4)$ are false.

For no two of the n queens to be in the same row, there must be one queen in each row. We can show that there is one queen in each row by verifying that every row contains at least one queen and that every row contains at most one queen. We first note that $\bigvee_{j=1}^n p(i, j)$ asserts that row i contains at least one queen, and

$$Q_1 = \bigwedge_{i=1}^n \bigvee_{j=1}^n p(i, j)$$

asserts that every row contains at least one queen.

For every row to include at most one queen, it must be the case that $p(i, j)$ and $p(k, j)$ are not both true for integers j and k with $1 \leq j < k \leq n$. Observe that $\neg p(i, j) \vee \neg p(k, j)$ asserts that at least one of $\neg p(i, j)$ and $\neg p(k, j)$ is true, which means that at least one of $p(i, j)$ and $p(k, j)$ is false. So, to check that there is at most one queen in each row, we assert

$$Q_2 = \bigwedge_{i=1}^n \bigwedge_{j=1}^{n-1} \bigwedge_{k=j+1}^n (\neg p(i, j) \vee \neg p(k, j)).$$

To assert that no column contains more than one queen, we assert that

$$Q_3 = \bigwedge_{j=1}^n \bigwedge_{i=1}^{n-1} \bigwedge_{k=i+1}^n (\neg p(i, j) \vee \neg p(k, j)).$$

(This assertion, together with the previous assertion that every row contains a queen, implies that every column contains a queen.)

To assert that no diagonal contains two queens, we assert

$$Q_4 = \bigwedge_{i=2}^n \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(i-1, n-j)} (\neg p(i, j) \vee \neg p(i-k, k+j))$$

and

$$Q_5 = \bigwedge_{i=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(n-i, n-j)} (\neg p(i, j) \vee \neg p(i+k, j+k)).$$

The innermost conjunction in Q_4 and in Q_5 for a pair (i, j) runs through the positions on a diagonal that begin at (i, j) and runs rightward along this diagonal. The upper limits on these innermost conjunctions identify the last cell in the board on each diagonal.

Putting all this together, we find that the solutions of the n -queens problem are given by the assignments of truth values to the variables $p(i, j)$, $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$ that make

$$Q = Q_1 \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5$$

true.

Using this and other approaches, the number of ways n queens can be placed on a chessboard so that no queen can attack another has been computed for $n \leq 27$. When $n = 8$ there are 92 such placements, while for $n = 16$ this number grows to 14,772,512. (See the OEIS discussed in Section 2.4 for details.)

EXAMPLE 11 **Sudoku** Sudoku puzzles are constructed using a 9×9 grid made up of nine 3×3 subgrids, known as **blocks**, as shown in Figure 2. For each puzzle, some of the 81 cells, called **givens**, are assigned one of the numbers 1, 2, $\dots$, 9, and the other cells are blank. The puzzle is solved by assigning a number to each blank cell so that every row, every column, and every one of the nine 3×3 blocks contains each of the nine possible numbers. Note that instead of using a 9×9 grid, Sudoku puzzles can be based on $n^2 \times n^2$ grids, for any positive integer n , with the $n^2 \times n^2$ grid made up of n^2 $n \times n$ subgrids.

Links

The popularity of Sudoku dates back to the 1980s when it was introduced in Japan. It took 20 years for Sudoku to spread to rest of the world, but by 2005, Sudoku puzzles were a worldwide craze. The name Sudoku is short for the Japanese *suuji wa dokushin ni kagiru*, which means “the digits must remain single.” The modern game of Sudoku was apparently designed in the late 1970s by an American puzzle designer. The basic ideas of Sudoku date back even further; puzzles printed in French newspapers in the 1890s were quite similar, but not identical, to modern Sudoku.

Sudoku puzzles designed for entertainment have two additional important properties. First, they have exactly one solution. Second, they can be solved using reasoning alone, that is, without resorting to searching all possible assignments of numbers to the cells. As a Sudoku puzzle is solved, entries in blank cells are successively determined by already known values. For instance, in the grid in Figure 2, the number 4 must appear in exactly one cell in the second row. How can we determine in which of the seven blank cells it must appear? First, we observe that 4 cannot appear in one of the first three cells or in one of the last three cells of this row, because it already appears in another cell in the block each of these cells is in. We can also see that 4

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

FIGURE 2 A 9×9 Sudoku puzzle.

cannot appear in the fifth cell in this row, as it already appears in the fifth column in the fourth row. This means that 4 must appear in the sixth cell of the second row.

Many strategies based on logic and mathematics have been devised for solving Sudoku puzzles (see [Da10], for example). Here, we discuss one of the ways that have been developed for solving Sudoku puzzles with the aid of a computer, which depends on modeling the puzzle as a propositional satisfiability problem. Using the model we describe, particular Sudoku puzzles can be solved using software developed to solve satisfiability problems. Currently, Sudoku puzzles can be solved in less than 10 milliseconds this way. It should be noted that there are many other approaches for solving Sudoku puzzles via computers using other techniques.

To encode a Sudoku puzzle, let $p(i, j, n)$ denote the proposition that is true when the number n is in the cell in the i th row and j th column. There are $9 \times 9 \times 9 = 729$ such propositions, as i , j , and n all range from 1 to 9. For example, for the puzzle in Figure 2, the number 6 is given as the value in the fifth row and first column. Hence, we see that $p(5, 1, 6)$ is true, but $p(5, j, 6)$ is false for $j = 2, 3, \dots, 9$.

Given a particular Sudoku puzzle, we begin by encoding each of the given values. Then, we construct compound propositions that assert that every row contains every number, every column contains every number, every 3×3 block contains every number, and each cell contains no more than one number. It follows, as the reader should verify, that the Sudoku puzzle is solved by finding an assignment of truth values to the 729 propositions $p(i, j, n)$ with i , j , and n each ranging from 1 to 9 that makes the conjunction of all these compound propositions true. After listing these assertions, we will explain how to construct the assertion that every row contains every integer from 1 to 9. We will leave the construction of the other assertions that every column contains every number and each of the nine 3×3 blocks contains every number to the exercises.

- ▶ For each cell with a given value, we assert $p(i, j, n)$ when the cell in row i and column j has the given value n .
- ▶ We assert that every row contains every number:

$$\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$$

- ▶ We assert that every column contains every number:

$$\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$$

- ▶ We assert that each of the nine 3×3 blocks contains every number:

$$\bigwedge_{r=0}^2 \bigwedge_{s=0}^2 \bigwedge_{n=1}^9 \bigvee_{i=1}^3 \bigvee_{j=1}^3 p(3r + i, 3s + j, n)$$

- ▶ To assert that no cell contains more than one number, we take the conjunction over all values of n, n', i , and j , where each variable ranges from 1 to 9 and $n \neq n'$ of $p(i, j, n) \rightarrow \neg p(i, j, n')$.

We now explain how to construct the assertion that every row contains every number. First, to assert that row i contains the number n , we form $\bigvee_{j=1}^9 p(i, j, n)$. To assert that row i contains all n numbers, we form the conjunction of these disjunctions over all nine possible values of n , giving us $\bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$. Finally, to assert that every row contains



It is tricky setting up the two inner indices so that all nine cells in each square block are examined.

every number, we take the conjunction of $\bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$ over all nine rows. This gives us $\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$. (Exercises 71 and 72 ask for explanations of the assertions that every column contains every number and that each of the nine 3×3 blocks contains every number.)

Given a particular Sudoku puzzle, to solve this puzzle we can find a solution to the satisfiability problems that asks for a set of truth values for the 729 variables $p(i, j, n)$ that makes the conjunction of all the listed assertions true. ▶

1.3.7 Solving Satisfiability Problems

A truth table can be used to determine whether a compound proposition is satisfiable, or equivalently, whether its negation is a tautology (see Exercise 64). This can be done by hand for a compound proposition with a small number of variables, but when the number of variables grows, this becomes impractical. For instance, there are $2^{20} = 1,048,576$ rows in the truth table for a compound proposition with 20 variables. Thus, you need a computer to help you determine, in this way, whether a compound proposition in 20 variables is satisfiable.

When many applications are modeled, questions concerning the satisfiability of compound propositions with hundreds, thousands, or millions of variables arise. Note, for example, that when there are 1000 variables, checking every one of the 2^{1000} (a number with more than 300 decimal digits) possible combinations of truth values of the variables in a compound proposition cannot be done by a computer in even trillions of years. No procedure is known that a computer can follow to determine in a reasonable amount of time whether an arbitrary compound proposition in such a large number of variables is satisfiable. However, progress has been made developing methods for solving the satisfiability problem for the particular types of compound propositions that arise in practical applications, such as for the solution of Sudoku puzzles. Many computer programs have been developed for solving satisfiability problems which have practical use. In our discussion of the subject of algorithms in Chapter 3, we will discuss this question further. In particular, we will explain the important role the propositional satisfiability problem plays in the study of the complexity of algorithms.

Links ▶

Links ▶



Courtesy of Harvard University Portrait Collection, Department of Philosophy

HENRY MAURICE SHEFFER (1883–1964) Henry Maurice Sheffer, born to Jewish parents in the western Ukraine, emigrated to the United States in 1892 with his parents and six siblings. He studied at the Boston Latin School before entering Harvard, where he completed his undergraduate degree in 1905, his master's in 1907, and his Ph.D. in philosophy in 1908. After holding a postdoctoral position at Harvard, Henry traveled to Europe on a fellowship. Upon returning to the United States, he became an academic nomad, spending one year each at the University of Washington, Cornell, the University of Minnesota, the University of Missouri, and City College in New York. In 1916 he returned to Harvard as a faculty member in the philosophy department. He remained at Harvard until his retirement in 1952.

Sheffer introduced what is now known as the Sheffer stroke in 1913; it became well known only after its use in the 1925 edition of Whitehead and Russell's *Principia Mathematica*. In this same edition Russell wrote that Sheffer had invented a powerful method that could be used to simplify the *Principia*. Because of this comment, Sheffer was something of a mystery man to logicians, especially because Sheffer, who published

little in his career, never published the details of this method, only describing it in mimeographed notes and in a brief published abstract.

Sheffer was a dedicated teacher of mathematical logic. He liked his classes to be small and did not like auditors. When strangers appeared in his classroom, Sheffer would order them to leave, even his colleagues or distinguished guests visiting Harvard. Sheffer was barely five feet tall; he was noted for his wit and vigor, as well as for his nervousness and irritability. Although widely liked, he was quite lonely. He is noted for a quip he spoke at his retirement: "Old professors never die, they just become emeriti." Sheffer is also credited with coining the term "Boolean algebra" (the subject of Chapter 12 of this text). Sheffer was briefly married and lived most of his later life in small rooms at a hotel packed with his logic books and vast files of slips of paper he used to jot down his ideas. Unfortunately, Sheffer suffered from severe depression during the last two decades of his life.

Exercises

1. Use truth tables to verify these equivalences.
 - a) $p \wedge \mathbf{T} \equiv p$
 - b) $p \vee \mathbf{F} \equiv p$
 - c) $p \wedge \mathbf{F} \equiv \mathbf{F}$
 - d) $p \vee \mathbf{T} \equiv \mathbf{T}$
 - e) $p \vee p \equiv p$
 - f) $p \wedge p \equiv p$
2. Show that $\neg(\neg p)$ and p are logically equivalent.
3. Use truth tables to verify the commutative laws
 - a) $p \vee q \equiv q \vee p$.
 - b) $p \wedge q \equiv q \wedge p$.
4. Use truth tables to verify the associative laws
 - a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
 - b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
5. Use a truth table to verify the distributive law
 $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
6. Use a truth table to verify the first De Morgan law
 $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
7. Use De Morgan's laws to find the negation of each of the following statements.
 - a) Jan is rich and happy.
 - b) Carlos will bicycle or run tomorrow.
 - c) Mei walks or takes the bus to class.
 - d) Ibrahim is smart and hard working.
8. Use De Morgan's laws to find the negation of each of the following statements.
 - a) Kwame will take a job in industry or go to graduate school.
 - b) Yoshiko knows Java and calculus.
 - c) James is young and strong.
 - d) Rita will move to Oregon or Washington.
9. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.
 - a) $p \rightarrow \neg q$
 - b) $(p \rightarrow q) \rightarrow r$
 - c) $(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q)$
10. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.
 - a) $\neg p \rightarrow \neg q$
 - b) $(p \vee q) \rightarrow \neg p$
 - c) $(p \rightarrow \neg q) \rightarrow (\neg p \rightarrow q)$
-  11. Show that each of these conditional statements is a tautology by using truth tables.
 - a) $(p \wedge q) \rightarrow p$
 - b) $p \rightarrow (p \vee q)$
 - c) $\neg p \rightarrow (p \rightarrow q)$
 - d) $(p \wedge q) \rightarrow (p \rightarrow q)$
 - e) $\neg(p \rightarrow q) \rightarrow p$
 - f) $\neg(p \rightarrow q) \rightarrow \neg q$
-  12. Show that each of these conditional statements is a tautology by using truth tables.
 - a) $[\neg p \wedge (p \vee q)] \rightarrow q$
 - b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
 - c) $[p \wedge (p \rightarrow q)] \rightarrow q$
 - d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$
13. Show that each conditional statement in Exercise 11 is a tautology using the fact that a conditional statement is false exactly when the hypothesis is true and the conclusion is false. (Do not use truth tables.)
14. Show that each conditional statement in Exercise 12 is a tautology using the fact that a conditional statement is false exactly when the hypothesis is true and the conclusion is false. (Do not use truth tables.)
15. Show that each conditional statement in Exercise 11 is a tautology by applying a chain of logical identities as in Example 8. (Do not use truth tables.)
16. Show that each conditional statement in Exercise 12 is a tautology by applying a chain of logical identities as in Example 8. (Do not use truth tables.)
17. Use truth tables to verify the absorption laws.
 - a) $p \vee (p \wedge q) \equiv p$
 - b) $p \wedge (p \vee q) \equiv p$
18. Determine whether $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$ is a tautology.
-  19. Determine whether $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a tautology.
 Each of Exercises 20–32 asks you to show that two compound propositions are logically equivalent. To do this, either show that both sides are true, or that both sides are false, for exactly the same combinations of truth values of the propositional variables in these expressions (whichever is easier).
20. Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent.
21. Show that $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent.
22. Show that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.
23. Show that $\neg p \leftrightarrow q$ and $p \leftrightarrow \neg q$ are logically equivalent.
24. Show that $\neg(p \oplus q)$ and $p \leftrightarrow q$ are logically equivalent.
25. Show that $\neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent.
26. Show that $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$ are logically equivalent.
27. Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent.
28. Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent.
29. Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent.
30. Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent.
31. Show that $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent.
32. Show that $p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$ are logically equivalent.
33. Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology.
-  34. Show that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.
35. Show that $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent.
36. Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

37. Show that $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$ are not logically equivalent.

The **dual** of a compound proposition that contains only the logical operators $\vee$, $\wedge$, and $\neg$ is the compound proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each **T** by **F**, and each **F** by **T**. The dual of s is denoted by s^* .

38. Find the dual of each of these compound propositions.

- a) $p \vee \neg q$ b) $p \wedge (q \vee (r \wedge \mathbf{T}))$
c) $(p \wedge \neg q) \vee (q \wedge \mathbf{F})$

39. Find the dual of each of these compound propositions.

- a) $p \wedge \neg q \wedge \neg r$ b) $(p \wedge q \wedge r) \vee s$
c) $(p \vee \mathbf{F}) \wedge (q \vee \mathbf{T})$

40. When does $s^* = s$, where s is a compound proposition?

41. Show that $(s^*)^* = s$ when s is a compound proposition.

42. Show that the logical equivalences in Table 6, except for the double negation law, come in pairs, where each pair contains compound propositions that are duals of each other.

- **43. Why are the duals of two equivalent compound propositions also equivalent, where these compound propositions contain only the operators $\wedge$, $\vee$, and $\neg$?

44. Find a compound proposition involving the propositional variables p , q , and r that is true when p and q are true and r is false, but is false otherwise. [Hint: Use a conjunction of each propositional variable or its negation.]

45. Find a compound proposition involving the propositional variables p , q , and r that is true when exactly two of p , q , and r are true and is false otherwise. [Hint: Form a disjunction of conjunctions. Include a conjunction for each combination of values for which the compound proposition is true. Each conjunction should include each of the three propositional variables or its negations.]

-  46. Suppose that a truth table in n propositional variables is specified. Show that a compound proposition with this truth table can be formed by taking the disjunction of conjunctions of the variables or their negations, with one conjunction included for each combination of values for which the compound proposition is true. The resulting compound proposition is said to be in **disjunctive normal form**.

A collection of logical operators is called **functionally complete** if every compound proposition is logically equivalent to a compound proposition involving only these logical operators.

47. Show that $\neg$, $\wedge$, and $\vee$ form a functionally complete collection of logical operators. [Hint: Use the fact that every compound proposition is logically equivalent to one in disjunctive normal form, as shown in Exercise 46.]

- *48. Show that $\neg$ and $\wedge$ form a functionally complete collection of logical operators. [Hint: First use a De Morgan law to show that $p \vee q$ is logically equivalent to $\neg(\neg p \wedge \neg q)$.]

- *49. Show that $\neg$ and $\vee$ form a functionally complete collection of logical operators.

We now present a group of exercises that involve the logical operators *NAND* and *NOR*. The proposition p *NAND* q is true

when either p or q , or both, are false; and it is false when both p and q are true. The proposition p *NOR* q is true when both p and q are false, and it is false otherwise. The propositions p *NAND* q and p *NOR* q are denoted by $p \mid q$ and $p \downarrow q$, respectively. (The operators $\mid$ and $\downarrow$ are called the **Sheffer stroke** and the **Peirce arrow** after H. M. Sheffer and C. S. Peirce, respectively.)

50. Construct a truth table for the logical operator *NAND*.

51. Show that $p \mid q$ is logically equivalent to $\neg(p \wedge q)$.

52. Construct a truth table for the logical operator *NOR*.

53. Show that $p \downarrow q$ is logically equivalent to $\neg(p \vee q)$.

54. In this exercise we will show that $\{\downarrow\}$ is a functionally complete collection of logical operators.

- a) Show that $p \downarrow p$ is logically equivalent to $\neg p$.

- b) Show that $(p \downarrow q) \downarrow (p \downarrow q)$ is logically equivalent to $p \vee q$.

- c) Conclude from parts (a) and (b), and Exercise 49, that $\{\downarrow\}$ is a functionally complete collection of logical operators.

- *55. Find a compound proposition logically equivalent to $p \rightarrow q$ using only the logical operator $\downarrow$.

56. Show that $\{\mid\}$ is a functionally complete collection of logical operators.

57. Show that $p \mid q$ and $q \mid p$ are equivalent.

58. Show that $p \mid (q \mid r)$ and $(p \mid q) \mid r$ are not equivalent, so that the logical operator $\mid$ is not associative.

- *59. How many different truth tables of compound propositions are there that involve the propositional variables p and q ?

60. Show that if p , q , and r are compound propositions such that p and q are logically equivalent and q and r are logically equivalent, then p and r are logically equivalent.

61. The following sentence is taken from the specification of a telephone system: "If the directory database is opened, then the monitor is put in a closed state, if the system is not in its initial state." This specification is hard to understand because it involves two conditional statements. Find an equivalent, easier-to-understand specification that involves disjunctions and negations but not conditional statements.

62. How many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, and $\neg q \vee \neg r$ can be made simultaneously true by an assignment of truth values to p , q , and r ?

63. How many of the disjunctions $p \vee \neg q \vee s$, $\neg p \vee \neg r \vee s$, $\neg p \vee \neg r \vee \neg s$, $\neg p \vee q \vee \neg s$, $q \vee r \vee \neg s$, $q \vee \neg r \vee \neg s$, $\neg p \vee \neg q \vee \neg s$, $p \vee r \vee s$, and $p \vee r \vee \neg s$ can be made simultaneously true by an assignment of truth values to p , q , r , and s ?

64. Show that the negation of an unsatisfiable compound proposition is a tautology and the negation of a compound proposition that is a tautology is unsatisfiable.

65. Determine whether each of these compound propositions is satisfiable.

- a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$

- b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$

- c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

66. Determine whether each of these compound propositions is satisfiable.

- a) $(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (p \vee \neg r \vee \neg s) \wedge (\neg p \vee \neg q \vee \neg s) \wedge (p \vee q \vee \neg s)$
 b) $(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg s) \wedge (\neg p \vee \neg r \vee \neg s) \wedge (p \vee q \vee \neg r) \wedge (p \vee \neg r \vee \neg s)$
 c) $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg s) \wedge (q \vee \neg r \vee s) \wedge (\neg p \vee r \vee s) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee s) \wedge (\neg p \vee \neg r \vee \neg s)$

67. Find the compound proposition Q constructed in Example 10 for the n -queens problem, and use it to find all the ways that n queens can be placed on an $n \times n$ chessboard, so that no queen can attack another when n is

- a) 2. b) 3. c) 4.

68. Starting with the compound proposition Q found in Example 10, construct a compound proposition that can be

used to find all solutions of the n -queens problem where the queen in the first column is in an odd-numbered row.

69. Show how the solution of a given 4×4 Sudoku puzzle can be found by solving a satisfiability problem.

70. Construct a compound proposition that asserts that every cell of a 9×9 Sudoku puzzle contains at least one number.

71. Explain the steps in the construction of the compound proposition given in the text that asserts that every column of a 9×9 Sudoku puzzle contains every number.

*72. Explain the steps in the construction of the compound proposition given in the text that asserts that each of the nine 3×3 blocks of a 9×9 Sudoku puzzle contains every number.

1.4 Predicates and Quantifiers

1.4.1 Introduction

Propositional logic, studied in Sections 1.1–1.3, cannot adequately express the meaning of all statements in mathematics and in natural language. For example, suppose that we know that

“Every computer connected to the university network is functioning properly.”

No rules of propositional logic allow us to conclude the truth of the statement

“MATH3 is functioning properly,”

where MATH3 is one of the computers connected to the university network. Likewise, we cannot use the rules of propositional logic to conclude from the statement

“CS2 is under attack by an intruder,”

where CS2 is a computer on the university network, to conclude the truth of

“There is a computer on the university network that is under attack by an intruder.”

In this section we will introduce a more powerful type of logic called **predicate logic**. We will see how predicate logic can be used to express the meaning of a wide range of statements in mathematics and computer science in ways that permit us to reason and explore relationships between objects. To understand predicate logic, we first need to introduce the concept of a predicate. Afterward, we will introduce the notion of quantifiers, which enable us to reason with statements that assert that a certain property holds for all objects of a certain type and with statements that assert the existence of an object with a particular property.

1.4.2 Predicates

Statements involving variables, such as

“ $x > 3$,” “ $x = y + 3$,” “ $x + y = z$,”

and

“Computer x is under attack by an intruder,”

and

“Computer x is functioning properly,”

are often found in mathematical assertions, in computer programs, and in system specifications. These statements are neither true nor false when the values of the variables are not specified. In this section, we will discuss the ways that propositions can be produced from such statements.

The statement “ x is greater than 3” has two parts. The first part, the variable x , is the subject of the statement. The second part—the **predicate**, “is greater than 3”—refers to a property that the subject of the statement can have. We can denote the statement “ x is greater than 3” by $P(x)$, where P denotes the predicate “is greater than 3” and x is the variable. The statement $P(x)$ is also said to be the value of the **propositional function** P at x . Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value. Consider Examples 1 and 2.

EXAMPLE 1 Let $P(x)$ denote the statement “ $x > 3$.” What are the truth values of $P(4)$ and $P(2)$?

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement “ $x > 3$.” Hence, $P(4)$, which is the statement “ $4 > 3$,” is true. However, $P(2)$, which is the statement “ $2 > 3$,” is false. 

EXAMPLE 2 Let $A(x)$ denote the statement “Computer x is under attack by an intruder.” Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are truth values of $A(\text{CS1})$, $A(\text{CS2})$, and $A(\text{MATH1})$?

Solution: We obtain the statement $A(\text{CS1})$ by setting $x = \text{CS1}$ in the statement “Computer x is under attack by an intruder.” Because CS1 is not on the list of computers currently under attack, we conclude that $A(\text{CS1})$ is false. Similarly, because CS2 and MATH1 are on the list of computers under attack, we know that $A(\text{CS2})$ and $A(\text{MATH1})$ are true. 

We can also have statements that involve more than one variable. For instance, consider the statement “ $x = y + 3$.” We can denote this statement by $Q(x, y)$, where x and y are variables and Q is the predicate. When values are assigned to the variables x and y , the statement $Q(x, y)$ has a truth value.

EXAMPLE 3 Let $Q(x, y)$ denote the statement “ $x = y + 3$.” What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Extra Examples 

Solution: To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement “ $1 = 2 + 3$,” which is false. The statement $Q(3, 0)$ is the proposition “ $3 = 0 + 3$,” which is true. 

EXAMPLE 4 Let $A(c, n)$ denote the statement “Computer c is connected to network n ,” where c is a variable representing a computer and n is a variable representing a network. Suppose that the computer MATH1 is connected to network CAMPUS2, but not to network CAMPUS1. What are the values of $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$?

Solution: Because MATH1 is not connected to the CAMPUS1 network, we see that $A(\text{MATH1}, \text{CAMPUS1})$ is false. However, because MATH1 is connected to the CAMPUS2 network, we see that $A(\text{MATH1}, \text{CAMPUS2})$ is true. ◀

Similarly, we can let $R(x, y, z)$ denote the statement “ $x + y = z$.” When values are assigned to the variables x, y , and z , this statement has a truth value.

EXAMPLE 5 What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution: The proposition $R(1, 2, 3)$ is obtained by setting $x = 1$, $y = 2$, and $z = 3$ in the statement $R(x, y, z)$. We see that $R(1, 2, 3)$ is the statement “ $1 + 2 = 3$,” which is true. Also note that $R(0, 0, 1)$, which is the statement “ $0 + 0 = 1$,” is false. ◀

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by

$$P(x_1, x_2, \dots, x_n).$$

A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the **propositional function** P at the n -tuple $(x_1, x_2, \dots, x_n)$, and P is also called an **n -place predicate** or an **n -ary predicate**.

Propositional functions occur in computer programs, as Example 6 demonstrates.

EXAMPLE 6 Consider the statement

if $x > 0$ **then** $x := x + 1$.

Links ▶



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CHARLES SANDERS PEIRCE (1839–1914) Many consider Charles Peirce, born in Cambridge, Massachusetts, to be the most original and versatile American intellect. He made important contributions to an amazing number of disciplines, including mathematics, astronomy, chemistry, geodesy, metrology, engineering, psychology, philology, the history of science, and economics. Peirce was also an inventor, a lifelong student of medicine, a book reviewer, a dramatist and an actor, a short story writer, a phenomenologist, a logician, and a metaphysician. He is noted as the preeminent system-building philosopher competent and productive in logic, mathematics, and a wide range of sciences. He was encouraged by his father, Benjamin Peirce, a professor of mathematics and natural philosophy at Harvard, to pursue a career in science. Instead, he decided to study logic and scientific methodology. Peirce attended Harvard (1855–1859) and received a Harvard master of arts degree (1862) and an advanced degree in chemistry from the Lawrence Scientific School (1863).

In 1861, Peirce became an aide in the U.S. Coast Survey, with the goal of better understanding scientific methodology. His service for the Survey exempted him from military service during the Civil War. While working for the Survey, Peirce did astronomical and geodesic work. He made fundamental contributions to the design of pendulums and to map projections, applying new mathematical developments in the theory of elliptic functions. He was the first person to use the wavelength of light as a unit of measurement. Peirce rose to the position of Assistant for the Survey, a position he held until forced to resign in 1891 when he disagreed with the direction taken by the Survey’s new administration.

While making his living from work in the physical sciences, Peirce developed a hierarchy of sciences, with mathematics at the top rung, in which the methods of one science could be adapted for use by those sciences under it in the hierarchy. During this time, he also founded the American philosophical theory of pragmatism.

The only academic position Peirce ever held was lecturer in logic at Johns Hopkins University in Baltimore (1879–1884). His mathematical work during this time included contributions to logic, set theory, abstract algebra, and the philosophy of mathematics. His work is still relevant today, with recent applications to artificial intelligence. Peirce believed that the study of mathematics could develop the mind’s powers of imagination, abstraction, and generalization. His diverse activities after retiring from the Survey included writing for periodicals, contributing to scholarly dictionaries, translating scientific papers, guest lecturing, and textbook writing. Unfortunately, his income from these pursuits was insufficient to protect him and his second wife from abject poverty. He was supported in his later years by a fund created by his many admirers and administered by the philosopher William James, his lifelong friend. Although Peirce wrote and published voluminously in a vast range of subjects, he left more than 100,000 pages of unpublished manuscripts. Because of the difficulty of studying his unpublished writings, scholars have only recently started to understand some of his varied contributions. A group of people is devoted to making his work available over the Internet to bring a better appreciation of Peirce’s accomplishments to the world.

When this statement is encountered in a program, the value of the variable x at that point in the execution of the program is inserted into $P(x)$, which is “ $x > 0$.” If $P(x)$ is true for this value of x , the assignment statement $x := x + 1$ is executed, so the value of x is increased by 1. If $P(x)$ is false for this value of x , the assignment statement is not executed, so the value of x is not changed. 

PRECONDITIONS AND POSTCONDITIONS Predicates are also used to establish the correctness of computer programs, that is, to show that computer programs always produce the desired output when given valid input. (Note that unless the correctness of a computer program is established, no amount of testing can show that it produces the desired output for all input values, unless every input value is tested.) The statements that describe valid input are known as **preconditions** and the conditions that the output should satisfy when the program has run are known as **postconditions**. As Example 7 illustrates, we use predicates to describe both preconditions and postconditions. We will study this process in greater detail in Section 5.5.

EXAMPLE 7 Consider the following program, designed to interchange the values of two variables x and y .

```
temp := x
x := y
y := temp
```

Find predicates that we can use as the precondition and the postcondition to verify the correctness of this program. Then explain how to use them to verify that for all valid input the program does what is intended.

Solution: For the precondition, we need to express that x and y have particular values before we run the program. So, for this precondition we can use the predicate $P(x, y)$, where $P(x, y)$ is the statement “ $x = a$ and $y = b$,” where a and b are the values of x and y before we run the program. Because we want to verify that the program swaps the values of x and y for all input values, for the postcondition we can use $Q(x, y)$, where $Q(x, y)$ is the statement “ $x = b$ and $y = a$.”

To verify that the program always does what it is supposed to do, suppose that the precondition $P(x, y)$ holds. That is, we suppose that the statement “ $x = a$ and $y = b$ ” is true. This means that $x = a$ and $y = b$. The first step of the program, $temp := x$, assigns the value of x to the variable $temp$, so after this step we know that $x = a$, $temp = a$, and $y = b$. After the second step of the program, $x := y$, we know that $x = b$, $temp = a$, and $y = b$. Finally, after the third step, we know that $x = b$, $temp = a$, and $y = a$. Consequently, after this program is run, the postcondition $Q(x, y)$ holds, that is, the statement “ $x = b$ and $y = a$ ” is true. 

1.4.3 Quantifiers

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value. However, there is another important way, called **quantification**, to create a proposition from a propositional function. Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words *all*, *some*, *many*, *none*, and *few* are used in quantifications. We will focus on two types of quantification here: universal quantification, which tells us that a predicate is true for every element under consideration, and existential quantification, which tells us that there is one or more element under consideration for which the predicate is true. The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

Assessment 

Assessment

THE UNIVERSAL QUANTIFIER Many mathematical statements assert that a property is true for all values of a variable in a particular domain, called the **domain of discourse** (or the **universe of discourse**), often just referred to as the **domain**. Such a statement is expressed using universal quantification. The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain. Note that the domain specifies the possible values of the variable x . The meaning of the universal quantification of $P(x)$ changes when we change the domain. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not defined.

Definition 1

The *universal quantification* of $P(x)$ is the statement

“ $P(x)$ for all values of x in the domain.”

The notation $\forall xP(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier**. We read $\forall xP(x)$ as “for all $xP(x)$ ” or “for every $xP(x)$.” An element for which $P(x)$ is false is called a **counterexample** to $\forall xP(x)$.

The meaning of the universal quantifier is summarized in the first row of Table 1. We illustrate the use of the universal quantifier in Examples 8–12 and 15.

EXAMPLE 8

Let $P(x)$ be the statement “ $x + 1 > x$.” What is the truth value of the quantification $\forall xP(x)$, where the domain consists of all real numbers?

Extra Examples

Solution: Because $P(x)$ is true for all real numbers x , the quantification

$$\forall xP(x)$$

is true.

Remark: Generally, an implicit assumption is made that all domains of discourse for quantifiers are nonempty. Note that if the domain is empty, then $\forall xP(x)$ is true for any propositional function $P(x)$ because there are no elements x in the domain for which $P(x)$ is false.

Besides “for all” and “for every,” universal quantification can be expressed in many other ways, including “all of,” “for each,” “given any,” “for arbitrary,” “for each,” and “for any.”

Remark: It is best to avoid using “for any x ” because it is often ambiguous as to whether “any” means “every” or “some.” In some cases, “any” is unambiguous, such as when it is used in negatives: “There is not any reason to avoid studying.”

A statement $\forall xP(x)$ is false, where $P(x)$ is a propositional function, if and only if $P(x)$ is not always true when x is in the domain. One way to show that $P(x)$ is not always true when x is in the domain is to find a counterexample to the statement $\forall xP(x)$. Note that a single counterexample is all we need to establish that $\forall xP(x)$ is false. Example 9 illustrates how counterexamples are used.

TABLE 1 Quantifiers.

Statement	When True?	When False?
$\forall xP(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists xP(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

Remember that the truth value of $\forall xP(x)$ depends on the domain!

EXAMPLE 9 Let $Q(x)$ be the statement “ $x < 2$.” What is the truth value of the quantification $\forall xQ(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall xQ(x)$. Thus,

$$\forall xQ(x)$$

is false. 

EXAMPLE 10 Suppose that $P(x)$ is “ $x^2 > 0$.” To show that the statement $\forall xP(x)$ is false where the universe of discourse consists of all integers, we give a counterexample. We see that $x = 0$ is a counterexample because $x^2 = 0$ when $x = 0$, so that x^2 is not greater than 0 when $x = 0$. 

Looking for counterexamples to universally quantified statements is an important activity in the study of mathematics, as we will see in subsequent sections of this book.

EXAMPLE 11 What does the statement $\forall xN(x)$ mean if $N(x)$ is “Computer x is connected to the network” and the domain consists of all computers on campus?

Solution: The statement $\forall xN(x)$ means that for every computer x on campus, that computer x is connected to the network. This statement can be expressed in English as “Every computer on campus is connected to the network.” 

As we have pointed out, specifying the domain is mandatory when quantifiers are used. The truth value of a quantified statement often depends on which elements are in this domain, as Example 12 shows.

EXAMPLE 12 What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

Solution: The universal quantification $\forall x(x^2 \geq x)$, where the domain consists of all real numbers, is false. For example, $(\frac{1}{2})^2 \not\geq \frac{1}{2}$. Note that $x^2 \geq x$ if and only if $x^2 - x = x(x - 1) \geq 0$. Consequently, $x^2 \geq x$ if and only if $x \leq 0$ or $x \geq 1$. It follows that $\forall x(x^2 \geq x)$ is false if the domain consists of all real numbers (because the inequality is false for all real numbers x with $0 < x < 1$). However, if the domain consists of the integers, $\forall x(x^2 \geq x)$ is true, because there are no integers x with $0 < x < 1$. 

THE EXISTENTIAL QUANTIFIER Many mathematical statements assert that there is an element with a certain property. Such statements are expressed using existential quantification. With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

Definition 2 The *existential quantification* of $P(x)$ is the proposition

“There exists an element x in the domain such that $P(x)$.”

We use the notation $\exists xP(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

A domain must always be specified when a statement $\exists xP(x)$ is used. Furthermore, the meaning of $\exists xP(x)$ changes when the domain changes. Without specifying the domain, the statement $\exists xP(x)$ has no meaning.

Besides the phrase “there exists,” we can also express existential quantification in many other ways, such as by using the words “for some,” “for at least one,” or “there is.” The existential quantification $\exists xP(x)$ is read as

“There is an x such that $P(x)$,”

“There is at least one x such that $P(x)$,”

or

“For some $xP(x)$.”

The meaning of the existential quantifier is summarized in the second row of Table 1. We illustrate the use of the existential quantifier in Examples 13, 14, and 16.

EXAMPLE 13 Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists xP(x)$, where the domain consists of all real numbers?

Extra Examples ➤

Solution: Because “ $x > 3$ ” is sometimes true—for instance, when $x = 4$ —the existential quantification of $P(x)$, which is $\exists xP(x)$, is true. ◀

Observe that the statement $\exists xP(x)$ is false if and only if there is no element x in the domain for which $P(x)$ is true. That is, $\exists xP(x)$ is false if and only if $P(x)$ is false for every element of the domain. We illustrate this observation in Example 14.

EXAMPLE 14 Let $Q(x)$ denote the statement “ $x = x + 1$.” What is the truth value of the quantification $\exists xQ(x)$, where the domain consists of all real numbers?

Solution: Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists xQ(x)$, is false. ◀

Remember that the truth value of $\exists xP(x)$ depends on the domain!

Remark: Generally, an implicit assumption is made that all domains of discourse for quantifiers are nonempty. If the domain is empty, then $\exists xQ(x)$ is false whenever $Q(x)$ is a propositional function because when the domain is empty, there can be no element x in the domain for which $Q(x)$ is true.

THE UNIQUENESS QUANTIFIER We have now introduced universal and existential quantifiers. These are the most important quantifiers in mathematics and computer science. However, there is no limitation on the number of different quantifiers we can define, such as “there are exactly two,” “there are no more than three,” “there are at least 100,” and so on. Of these other quantifiers, the one that is most often seen is the **uniqueness quantifier**, denoted by $\exists!$ or $\exists_1$. The notation $\exists!xP(x)$ [or $\exists_1xP(x)$] states “There exists a unique x such that $P(x)$ is true.” (Other phrases for uniqueness quantification include “there is exactly one” and “there is one and only one.”) For instance, $\exists!x(x - 1 = 0)$, where the domain is the set of real numbers, states that there is a unique real number x such that $x - 1 = 0$. This is a true statement, as $x = 1$ is the unique real number such that $x - 1 = 0$. Observe that we can use quantifiers and propositional logic to express uniqueness (see Exercise 52 in Section 1.5), so the uniqueness quantifier can be avoided. Generally, it is best to stick with existential and universal quantifiers so that rules of inference for these quantifiers can be used.

1.4.4 QUANTIFIERS OVER FINITE DOMAINS

When the domain of a quantifier is finite, that is, when all its elements can be listed, quantified statements can be expressed using propositional logic. In particular, when the elements of the domain are $x_1, x_2, \dots, x_n$, where n is a positive integer, the universal quantification $\forall xP(x)$ is the same as the conjunction

$$P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n),$$

because this conjunction is true if and only if $P(x_1), P(x_2), \dots, P(x_n)$ are all true.

EXAMPLE 15 What is the truth value of $\forall xP(x)$, where $P(x)$ is the statement “ $x^2 < 10$ ” and the domain consists of the positive integers not exceeding 4?

Solution: The statement $\forall xP(x)$ is the same as the conjunction

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4),$$

because the domain consists of the integers 1, 2, 3, and 4. Because $P(4)$, which is the statement “ $4^2 < 10$,” is false, it follows that $\forall xP(x)$ is false. ◀

Similarly, when the elements of the domain are $x_1, x_2, \dots, x_n$, where n is a positive integer, the existential quantification $\exists xP(x)$ is the same as the disjunction

$$P(x_1) \vee P(x_2) \vee \dots \vee P(x_n),$$

because this disjunction is true if and only if at least one of $P(x_1), P(x_2), \dots, P(x_n)$ is true.

EXAMPLE 16 What is the truth value of $\exists xP(x)$, where $P(x)$ is the statement “ $x^2 > 10$ ” and the universe of discourse consists of the positive integers not exceeding 4?

Solution: Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists xP(x)$ is the same as the disjunction

$$P(1) \vee P(2) \vee P(3) \vee P(4).$$

Because $P(4)$, which is the statement “ $4^2 > 10$,” is true, it follows that $\exists xP(x)$ is true. ◀

CONNECTIONS BETWEEN QUANTIFICATION AND LOOPING It is sometimes helpful to think in terms of looping and searching when determining the truth value of a quantification. Suppose that there are n objects in the domain for the variable x . To determine whether $\forall xP(x)$ is true, we can loop through all n values of x to see whether $P(x)$ is always true. If we encounter a value x for which $P(x)$ is false, then we have shown that $\forall xP(x)$ is false. Otherwise, $\forall xP(x)$ is true. To see whether $\exists xP(x)$ is true, we loop through the n values of x searching for a value for which $P(x)$ is true. If we find one, then $\exists xP(x)$ is true. If we never find such an x , then we have determined that $\exists xP(x)$ is false. (Note that this searching procedure does not apply if there are infinitely many values in the domain. However, it is still a useful way of thinking about the truth values of quantifications.)

1.4.5 Quantifiers with Restricted Domains

An abbreviated notation is often used to restrict the domain of a quantifier. In this notation, a condition a variable must satisfy is included after the quantifier. This is illustrated in Example 17. We will also describe other forms of this notation involving set membership in Section 2.1.

EXAMPLE 17 What do the statements $\forall x < 0 (x^2 > 0)$, $\forall y \neq 0 (y^3 \neq 0)$, and $\exists z > 0 (z^2 = 2)$ mean, where the domain in each case consists of the real numbers?

Solution: The statement $\forall x < 0 (x^2 > 0)$ states that for every real number x with $x < 0$, $x^2 > 0$. That is, it states “The square of a negative real number is positive.” This statement is the same as $\forall x(x < 0 \rightarrow x^2 > 0)$.

The statement $\forall y \neq 0 (y^3 \neq 0)$ states that for every real number y with $y \neq 0$, we have $y^3 \neq 0$. That is, it states “The cube of every nonzero real number is nonzero.” This statement is equivalent to $\forall y(y \neq 0 \rightarrow y^3 \neq 0)$.

Finally, the statement $\exists z > 0 (z^2 = 2)$ states that there exists a real number z with $z > 0$ such that $z^2 = 2$. That is, it states “There is a positive square root of 2.” This statement is equivalent to $\exists z(z > 0 \wedge z^2 = 2)$. 

Note that the restriction of a universal quantification is the same as the universal quantification of a conditional statement. For instance, $\forall x < 0 (x^2 > 0)$ is another way of expressing $\forall x(x < 0 \rightarrow x^2 > 0)$. On the other hand, the restriction of an existential quantification is the same as the existential quantification of a conjunction. For instance, $\exists z > 0 (z^2 = 2)$ is another way of expressing $\exists z(z > 0 \wedge z^2 = 2)$.

1.4.6 Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus. For example, $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$. In other words, it means $(\forall x P(x)) \vee Q(x)$ rather than $\forall x (P(x) \vee Q(x))$.

1.4.7 Binding Variables

When a quantifier is used on the variable x , we say that this occurrence of the variable is **bound**. An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be **free**. All the variables that occur in a propositional function must be bound or set equal to a particular value to turn it into a proposition. This can be done using a combination of universal quantifiers, existential quantifiers, and value assignments.

The part of a logical expression to which a quantifier is applied is called the **scope** of this quantifier. Consequently, a variable is free if it is outside the scope of all quantifiers in the formula that specify this variable.

EXAMPLE 18 In the statement $\exists x(x + y = 1)$, the variable x is bound by the existential quantification $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable. This illustrates that in the statement $\exists x(x + y = 1)$, x is bound, but y is free.

In the statement $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$, all variables are bound. The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$, because $\exists x$ is applied only to $P(x) \wedge Q(x)$ and not to the rest of the statement. Similarly, the scope of the second quantifier, $\forall x$, is the expression $R(x)$. That is, the existential quantifier binds the variable x in $P(x) \wedge Q(x)$ and the universal quantifier $\forall x$ binds the variable x in $R(x)$. Observe that we could have written our statement using two different variables x and y , as $\exists x(P(x) \wedge Q(x)) \vee \forall y R(y)$, because the scopes of the two quantifiers do not overlap. The reader should be aware that in common usage, the same

letter is often used to represent variables bound by different quantifiers with scopes that do not overlap. ◀

1.4.8 Logical Equivalences Involving Quantifiers

In Section 1.3 we introduced the notion of logical equivalences of compound propositions. We can extend this notion to expressions involving predicates and quantifiers.

Definition 3

Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value no matter which predicates are substituted into these statements and which domain of discourse is used for the variables in these propositional functions. We use the notation $S \equiv T$ to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.

Example 19 illustrates how to show that two statements involving predicates and quantifiers are logically equivalent.

EXAMPLE 19

Show that $\forall x(P(x) \wedge Q(x))$ and $\forall xP(x) \wedge \forall xQ(x)$ are logically equivalent (where the same domain is used throughout). This logical equivalence shows that we can distribute a universal quantifier over a conjunction. Furthermore, we can also distribute an existential quantifier over a disjunction. However, we cannot distribute a universal quantifier over a disjunction, nor can we distribute an existential quantifier over a conjunction. (See Exercises 52 and 53.)

Solution: To show that these statements are logically equivalent, we must show that they always take the same truth value, no matter what the predicates P and Q are, and no matter which domain of discourse is used. Suppose we have particular predicates P and Q , with a common domain. We can show that $\forall x(P(x) \wedge Q(x))$ and $\forall xP(x) \wedge \forall xQ(x)$ are logically equivalent by doing two things. First, we show that if $\forall x(P(x) \wedge Q(x))$ is true, then $\forall xP(x) \wedge \forall xQ(x)$ is true. Second, we show that if $\forall xP(x) \wedge \forall xQ(x)$ is true, then $\forall x(P(x) \wedge Q(x))$ is true.

So, suppose that $\forall x(P(x) \wedge Q(x))$ is true. This means that if a is in the domain, then $P(a) \wedge Q(a)$ is true. Hence, $P(a)$ is true and $Q(a)$ is true. Because $P(a)$ is true and $Q(a)$ is true for every element a in the domain, we can conclude that $\forall xP(x)$ and $\forall xQ(x)$ are both true. This means that $\forall xP(x) \wedge \forall xQ(x)$ is true.

Next, suppose that $\forall xP(x) \wedge \forall xQ(x)$ is true. It follows that $\forall xP(x)$ is true and $\forall xQ(x)$ is true. Hence, if a is in the domain, then $P(a)$ is true and $Q(a)$ is true [because $P(x)$ and $Q(x)$ are both true for all elements in the domain, there is no conflict using the same value of a here]. It follows that for all a , $P(a) \wedge Q(a)$ is true. It follows that $\forall x(P(x) \wedge Q(x))$ is true. We can now conclude that

$$\forall x(P(x) \wedge Q(x)) \equiv \forall xP(x) \wedge \forall xQ(x).$$

1.4.9 Negating Quantified Expressions

We will often want to consider the negation of a quantified expression. For instance, consider the negation of the statement

“Every student in your class has taken a course in calculus.”

This statement is a universal quantification, namely,

$$\forall xP(x),$$

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where $P(x)$ is the statement “ x has taken a course in calculus” and the domain consists of the students in your class. The negation of this statement is “It is not the case that every student in your class has taken a course in calculus.” This is equivalent to “There is a student in your class who has not taken a course in calculus.” And this is simply the existential quantification of the negation of the original propositional function, namely,

$$\exists x \neg P(x).$$

This example illustrates the following logical equivalence:

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

To show that $\neg \forall x P(x)$ and $\exists x \neg P(x)$ are logically equivalent no matter what the propositional function $P(x)$ is and what the domain is, first note that $\neg \forall x P(x)$ is true if and only if $\forall x P(x)$ is false. Next, note that $\forall x P(x)$ is false if and only if there is an element x in the domain for which $P(x)$ is false. This holds if and only if there is an element x in the domain for which $\neg P(x)$ is true. Finally, note that there is an element x in the domain for which $\neg P(x)$ is true if and only if $\exists x \neg P(x)$ is true. Putting these steps together, we can conclude that $\neg \forall x P(x)$ is true if and only if $\exists x \neg P(x)$ is true. It follows that $\neg \forall x P(x)$ and $\exists x \neg P(x)$ are logically equivalent.

Suppose we wish to negate an existential quantification. For instance, consider the proposition “There is a student in this class who has taken a course in calculus.” This is the existential quantification

$$\exists x Q(x),$$

where $Q(x)$ is the statement “ x has taken a course in calculus.” The negation of this statement is the proposition “It is not the case that there is a student in this class who has taken a course in calculus.” This is equivalent to “Every student in this class has not taken calculus,” which is just the universal quantification of the negation of the original propositional function, or, phrased in the language of quantifiers,

$$\forall x \neg Q(x).$$

This example illustrates the equivalence

$$\neg \exists x Q(x) \equiv \forall x \neg Q(x).$$

To show that $\neg \exists x Q(x)$ and $\forall x \neg Q(x)$ are logically equivalent no matter what $Q(x)$ is and what the domain is, first note that $\neg \exists x Q(x)$ is true if and only if $\exists x Q(x)$ is false. This is true if and only if no x exists in the domain for which $Q(x)$ is true. Next, note that no x exists in the domain for which $Q(x)$ is true if and only if $Q(x)$ is false for every x in the domain. Finally, note that $Q(x)$ is false for every x in the domain if and only if $\neg Q(x)$ is true for all x in the domain, which holds if and only if $\forall x \neg Q(x)$ is true. Putting these steps together, we see that $\neg \exists x Q(x)$ is true if and only if $\forall x \neg Q(x)$ is true. We conclude that $\neg \exists x Q(x)$ and $\forall x \neg Q(x)$ are logically equivalent.

The rules for negations for quantifiers are called **De Morgan’s laws for quantifiers**. These rules are summarized in Table 2.

Remark: When the domain of a predicate $P(x)$ consists of n elements, where n is a positive integer greater than one, the rules for negating quantified statements are exactly the same as De Morgan’s laws discussed in Section 1.3. This is why these rules are called De Morgan’s laws for quantifiers. When the domain has n elements $x_1, x_2, \dots, x_n$, it follows that $\neg \forall x P(x)$ is the same as $\neg(P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n))$, which is equivalent to $\neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg P(x_n)$ by De Morgan’s laws, and this is the same as $\exists x \neg P(x)$. Similarly, $\neg \exists x P(x)$ is the same as $\neg(P(x_1) \vee$

TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg\exists xP(x)$	$\forall x\neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall xP(x)$	$\exists x\neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

$P(x_2) \vee \cdots \vee P(x_n)$), which by De Morgan's laws is equivalent to $\neg P(x_1) \wedge \neg P(x_2) \wedge \cdots \wedge \neg P(x_n)$, and this is the same as $\forall x\neg P(x)$.

We illustrate the negation of quantified statements in Examples 20 and 21.

EXAMPLE 20 What are the negations of the statements “There is an honest politician” and “All Americans eat cheeseburgers”?

Solution: Let $H(x)$ denote “ x is honest.” Then the statement “There is an honest politician” is represented by $\exists xH(x)$, where the domain consists of all politicians. The negation of this statement is $\neg\exists xH(x)$, which is equivalent to $\forall x\neg H(x)$. This negation can be expressed as “Every politician is dishonest.” (Note: In English, the statement “All politicians are not honest” is ambiguous. In common usage, this statement often means “Not all politicians are honest.” Consequently, we do not use this statement to express this negation.)

Extra Examples ➤

Let $C(x)$ denote “ x eats cheeseburgers.” Then the statement “All Americans eat cheeseburgers” is represented by $\forall xC(x)$, where the domain consists of all Americans. The negation of this statement is $\neg\forall xC(x)$, which is equivalent to $\exists x\neg C(x)$. This negation can be expressed in several different ways, including “Some American does not eat cheeseburgers” and “There is an American who does not eat cheeseburgers.” ◀

EXAMPLE 21 What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Solution: The negation of $\forall x(x^2 > x)$ is the statement $\neg\forall x(x^2 > x)$, which is equivalent to $\exists x\neg(x^2 > x)$. This can be rewritten as $\exists x(x^2 \leq x)$. The negation of $\exists x(x^2 = 2)$ is the statement $\neg\exists x(x^2 = 2)$, which is equivalent to $\forall x\neg(x^2 = 2)$. This can be rewritten as $\forall x(x^2 \neq 2)$. The truth values of these statements depend on the domain. ◀

We use De Morgan's laws for quantifiers in Example 22.

EXAMPLE 22 Show that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution: By De Morgan's law for universal quantifiers, we know that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x\neg(P(x) \rightarrow Q(x))$ are logically equivalent. By the fifth logical equivalence in Table 7 in Section 1.3, we know that $\neg(P(x) \rightarrow Q(x))$ and $P(x) \wedge \neg Q(x)$ are logically equivalent for every x . Because we can substitute one logically equivalent expression for another in a logical equivalence, it follows that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent. ◀

1.4.10 Translating from English into Logical Expressions

Translating sentences in English (or other natural languages) into logical expressions is a crucial task in mathematics, logic programming, artificial intelligence, software engineering, and many other disciplines. We began studying this topic in Section 1.1, where we used propositions

to express sentences in logical expressions. In that discussion, we purposely avoided sentences whose translations required predicates and quantifiers. Translating from English to logical expressions becomes even more complex when quantifiers are needed. Furthermore, there can be many ways to translate a particular sentence. (As a consequence, there is no “cookbook” approach that can be followed step by step.) We will use some examples to illustrate how to translate sentences from English into logical expressions. The goal in this translation is to produce simple and useful logical expressions. In this section, we restrict ourselves to sentences that can be translated into logical expressions using a single quantifier; in the next section, we will look at more complicated sentences that require multiple quantifiers.

EXAMPLE 23 Express the statement “Every student in this class has studied calculus” using predicates and quantifiers.

Extra Examples 

Solution: First, we rewrite the statement so that we can clearly identify the appropriate quantifiers to use. Doing so, we obtain:

“For every student in this class, that student has studied calculus.”

Next, we introduce a variable x so that our statement becomes

“For every student x in this class, x has studied calculus.”

Continuing, we introduce $C(x)$, which is the statement “ x has studied calculus.” Consequently, if the domain for x consists of the students in the class, we can translate our statement as $\forall x C(x)$.

However, there are other correct approaches; different domains of discourse and other predicates can be used. The approach we select depends on the subsequent reasoning we want to carry out. For example, we may be interested in a wider group of people than only those in this class. If we change the domain to consist of all people, we will need to express our statement as

“For every person x , if person x is a student in this class, then x has studied calculus.”



If $S(x)$ represents the statement that person x is in this class, we see that our statement can be expressed as $\forall x(S(x) \rightarrow C(x))$. [*Caution!* Our statement *cannot* be expressed as $\forall x(S(x) \wedge C(x))$ because this statement says that all people are students in this class and have studied calculus!]

Finally, when we are interested in the background of people in subjects besides calculus, we may prefer to use the two-variable quantifier $Q(x, y)$ for the statement “Student x has studied subject y .” Then we would replace $C(x)$ by $Q(x, \text{calculus})$ in both approaches to obtain $\forall x Q(x, \text{calculus})$ or $\forall x(S(x) \rightarrow Q(x, \text{calculus}))$. 

In Example 23 we displayed different approaches for expressing the same statement using predicates and quantifiers. However, we should always adopt the simplest approach that is adequate for use in subsequent reasoning.

EXAMPLE 24 Express the statements “Some student in this class has visited Mexico” and “Every student in this class has visited either Canada or Mexico” using predicates and quantifiers.

Solution: The statement “Some student in this class has visited Mexico” means that

“There is a student in this class with the property that the student has visited Mexico.”

We can introduce a variable x , so that our statement becomes

“There is a student x in this class having the property that x has visited Mexico.”

We introduce $M(x)$, which is the statement “ x has visited Mexico.” If the domain for x consists of the students in this class, we can translate this first statement as $\exists xM(x)$.

However, if we are interested in people other than those in this class, we look at the statement a little differently. Our statement can be expressed as

“There is a person x having the properties that x is a student in this class and x has visited Mexico.”



In this case, the domain for the variable x consists of all people. We introduce $S(x)$ to represent “ x is a student in this class.” Our solution becomes $\exists x(S(x) \wedge M(x))$ because the statement is that there is a person x who is a student in this class and who has visited Mexico. [Caution! Our statement cannot be expressed as $\exists x(S(x) \rightarrow M(x))$, which is true when there is someone not in the class because, in that case, for such a person x , $S(x) \rightarrow M(x)$ becomes either $\mathbf{F} \rightarrow \mathbf{T}$ or $\mathbf{F} \rightarrow \mathbf{F}$, both of which are true.]

Similarly, the second statement can be expressed as

“For every x in this class, x has the property that x has visited Mexico or x has visited Canada.”

(Note that we are assuming the inclusive, rather than the exclusive, or here.) We let $C(x)$ be “ x has visited Canada.” Following our earlier reasoning, we see that if the domain for x consists of the students in this class, this second statement can be expressed as $\forall x(C(x) \vee M(x))$. However, if the domain for x consists of all people, our statement can be expressed as

“For every person x , if x is a student in this class, then x has visited Mexico or x has visited Canada.”

In this case, the statement can be expressed as $\forall x(S(x) \rightarrow (C(x) \vee M(x)))$.

Instead of using $M(x)$ and $C(x)$ to represent that x has visited Mexico and x has visited Canada, respectively, we could use a two-place predicate $V(x, y)$ to represent “ x has visited country y .” In this case, $V(x, \text{Mexico})$ and $V(x, \text{Canada})$ would have the same meaning as $M(x)$ and $C(x)$ and could replace them in our answers. If we are working with many statements that involve people visiting different countries, we might prefer to use this two-variable approach. Otherwise, for simplicity, we would stick with the one-variable predicates $M(x)$ and $C(x)$. 

1.4.11 Using Quantifiers in System Specifications

In Section 1.2 we used propositions to represent system specifications. However, many system specifications involve predicates and quantifications. This is illustrated in Example 25.

EXAMPLE 25

Extra
Examples 

Use predicates and quantifiers to express the system specifications “Every mail message larger than one megabyte will be compressed” and “If a user is active, at least one network link will be available.”

Solution: Let $S(m, y)$ be “Mail message m is larger than y megabytes,” where the variable x has the domain of all mail messages and the variable y is a positive real number, and let $C(m)$ denote “Mail message m will be compressed.” Then the specification “Every mail message larger than one megabyte will be compressed” can be represented as $\forall m(S(m, 1) \rightarrow C(m))$.

Let $A(u)$ represent “User u is active,” where the variable u has the domain of all users, let $S(n, x)$ denote “Network link n is in state x ,” where n has the domain of all network links and x has the domain of all possible states for a network link. Then the specification “If a user is active, at least one network link will be available” can be represented by $\exists uA(u) \rightarrow \exists nS(n, \text{available})$. 

Remember the rules of precedence for quantifiers and logical connectives!

1.4.12 Examples from Lewis Carroll

Lewis Carroll (really C. L. Dodgson writing under a pseudonym), the author of *Alice in Wonderland*, is also the author of several works on symbolic logic. His books contain many examples of reasoning using quantifiers. Examples 26 and 27 come from his book *Symbolic Logic*; other examples from that book are given in the exercises at the end of this section. These examples illustrate how quantifiers are used to express various types of statements.

EXAMPLE 26 Consider these statements. The first two are called *premises* and the third is called the *conclusion*. The entire set is called an *argument*.

“All lions are fierce.”
 “Some lions do not drink coffee.”
 “Some fierce creatures do not drink coffee.”

(In Section 1.6 we will discuss the issue of determining whether the conclusion is a valid consequence of the premises. In this example, it is.) Let $P(x)$, $Q(x)$, and $R(x)$ be the statements “ x is a lion,” “ x is fierce,” and “ x drinks coffee,” respectively. Assuming that the domain consists of all creatures, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.

Solution: We can express these statements as

$$\begin{aligned}\forall x(P(x) \rightarrow Q(x)). \\ \exists x(P(x) \wedge \neg R(x)). \\ \exists x(Q(x) \wedge \neg R(x)).\end{aligned}$$

Notice that the second statement cannot be written as $\exists x(P(x) \rightarrow \neg R(x))$. The reason is that $P(x) \rightarrow \neg R(x)$ is true whenever x is not a lion, so that $\exists x(P(x) \rightarrow \neg R(x))$ is true as long as there is at least one creature that is not a lion, even if every lion drinks coffee. Similarly, the third statement cannot be written as

$$\exists x(Q(x) \rightarrow \neg R(x)).$$

EXAMPLE 27 Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

“All hummingbirds are richly colored.”
 “No large birds live on honey.”
 “Birds that do not live on honey are dull in color.”
 “Hummingbirds are small.”

Links



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CHARLES LUTWIDGE DODGSON (1832–1898) We know Charles Dodgson as Lewis Carroll—the pseudonym he used in his literary works. Dodgson, the son of a clergyman, was the third of 11 children, all of whom stuttered. He was uncomfortable in the company of adults and is said to have spoken without stuttering only to young girls, many of whom he entertained, corresponded with, and photographed (sometimes in poses that today would be considered inappropriate). Although attracted to young girls, he was extremely puritanical and religious. His friendship with the three young daughters of Dean Liddell led to his writing *Alice in Wonderland*, which brought him money and fame.

Dodgson graduated from Oxford in 1854 and obtained his master of arts degree in 1857. He was appointed lecturer in mathematics at Christ Church College, Oxford, in 1855. He was ordained in the Church of England in 1861 but never practiced his ministry. His writings published under this real name include articles and books on geometry, determinants, and the mathematics of tournaments and elections. (He also used the pseudonym Lewis Carroll for his many works on recreational logic.)

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements “ x is a hummingbird,” “ x is large,” “ x lives on honey,” and “ x is richly colored,” respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$.

Solution: We can express the statements in the argument as

$$\begin{aligned}\forall x(P(x) \rightarrow S(x)). \\ \neg \exists x(Q(x) \wedge R(x)). \\ \forall x(\neg R(x) \rightarrow \neg S(x)). \\ \forall x(P(x) \rightarrow \neg Q(x)).\end{aligned}$$

(Note we have assumed that “small” is the same as “not large” and that “dull in color” is the same as “not richly colored.” To show that the fourth statement is a valid conclusion of the first three, we need to use rules of inference that will be discussed in Section 1.6.)

1.4.13 Logic Programming

An important type of programming language is designed to reason using the rules of predicate logic. Prolog (from *Programming in Logic*), developed in the 1970s by computer scientists working in the area of artificial intelligence, is an example of such a language. Prolog programs include a set of declarations consisting of two types of statements, **Prolog facts** and **Prolog rules**. Prolog facts define predicates by specifying the elements that satisfy these predicates. Prolog rules are used to define new predicates using those already defined by Prolog facts. Example 28 illustrates these notions.



EXAMPLE 28

Consider a Prolog program given facts telling it the instructor of each class and in which classes students are enrolled. The program uses these facts to answer queries concerning the professors who teach particular students. Such a program could use the predicates *instructor*(p, c) and *enrolled*(s, c) to represent that professor p is the instructor of course c and that student s is enrolled in course c , respectively. For example, the Prolog facts in such a program might include:

```
instructor(chan,math273)
instructor(patel,ee222)
instructor(grossman,cs301)
enrolled(kevin,math273)
enrolled(juana,ee222)
enrolled(juana,cs301)
enrolled(kiko,math273)
enrolled(kiko,cs301)
```

(Lowercase letters have been used for entries because Prolog considers names beginning with an uppercase letter to be variables.)

A new predicate *teaches*(p, s), representing that professor p teaches student s , can be defined using the Prolog rule

```
teaches(P,S) :- instructor(P,C), enrolled(S,C)
```

which means that *teaches*(p, s) is true if there exists a class c such that professor p is the instructor of class c and student s is enrolled in class c . (Note that a comma is used to represent a conjunction of predicates in Prolog. Similarly, a semicolon is used to represent a disjunction of predicates.)

Prolog answers queries using the facts and rules it is given. For example, using the facts and rules listed, the query

```
?enrolled(kevin,math273)
```

produces the response

```
yes
```

because the fact *enrolled*(kevin, math273) was provided as input. The query

```
?enrolled(X,math273)
```

produces the response

```
kevin
kiko
```

To produce this response, Prolog determines all possible values of X for which *enrolled*(X , math273) has been included as a Prolog fact. Similarly, to find all the professors who are instructors in classes being taken by Juana, we use the query

```
?teaches(X,juana)
```

This query returns

```
patel
grossman
```

Exercises

- Let $P(x)$ denote the statement " $x \leq 4$." What are these truth values?
 - $P(0)$
 - $P(4)$
 - $P(6)$
- Let $P(x)$ be the statement "The word x contains the letter a ." What are these truth values?
 - $P(\text{orange})$
 - $P(\text{lemon})$
 - $P(\text{true})$
 - $P(\text{false})$
- Let $Q(x, y)$ denote the statement " x is the capital of y ." What are these truth values?
 - $Q(\text{Denver, Colorado})$
 - $Q(\text{Detroit, Michigan})$
 - $Q(\text{Massachusetts, Boston})$
 - $Q(\text{New York, New York})$
- State the value of x after the statement **if** $P(x)$ **then** $x := 1$ is executed, where $P(x)$ is the statement " $x > 1$," if the value of x when this statement is reached is
 - $x = 0$.
 - $x = 1$.
 - $x = 2$.
- Let $P(x)$ be the statement " x spends more than five hours every weekday in class," where the domain for x consists of all students. Express each of these quantifications in English.
 - $\exists x P(x)$
 - $\forall x P(x)$
 - $\exists x \neg P(x)$
 - $\forall x \neg P(x)$
- Let $N(x)$ be the statement " x has visited North Dakota," where the domain consists of the students in your school. Express each of these quantifications in English.
 - $\exists x N(x)$
 - $\forall x N(x)$
 - $\neg \exists x N(x)$
 - $\exists x \neg N(x)$
 - $\neg \forall x N(x)$
 - $\forall x \neg N(x)$
- Translate these statements into English, where $C(x)$ is " x is a comedian" and $F(x)$ is " x is funny" and the domain consists of all people.
 - $\forall x (C(x) \rightarrow F(x))$
 - $\forall x (C(x) \wedge F(x))$
 - $\exists x (C(x) \rightarrow F(x))$
 - $\exists x (C(x) \wedge F(x))$
- Translate these statements into English, where $R(x)$ is " x is a rabbit" and $H(x)$ is " x hops" and the domain consists of all animals.
 - $\forall x (R(x) \rightarrow H(x))$
 - $\forall x (R(x) \wedge H(x))$
 - $\exists x (R(x) \rightarrow H(x))$
 - $\exists x (R(x) \wedge H(x))$
- Let $P(x)$ be the statement " x can speak Russian" and let $Q(x)$ be the statement " x knows the computer language C++." Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

- a) There is a student at your school who can speak Russian and who knows C++.
- b) There is a student at your school who can speak Russian but who doesn't know C++.
- c) Every student at your school either can speak Russian or knows C++.
- d) No student at your school can speak Russian or knows C++.
10. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.
- a) A student in your class has a cat, a dog, and a ferret.
- b) All students in your class have a cat, a dog, or a ferret.
- c) Some student in your class has a cat and a ferret, but not a dog.
- d) No student in your class has a cat, a dog, and a ferret.
- e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.
11. Let $P(x)$ be the statement " $x = x^2$." If the domain consists of the integers, what are these truth values?
- a) $P(0)$ b) $P(1)$ c) $P(2)$
d) $P(-1)$ e) $\exists xP(x)$ f) $\forall xP(x)$
12. Let $Q(x)$ be the statement " $x + 1 > 2x$." If the domain consists of all integers, what are these truth values?
- a) $Q(0)$ b) $Q(-1)$ c) $Q(1)$
d) $\exists xQ(x)$ e) $\forall xQ(x)$ f) $\exists x\neg Q(x)$
g) $\forall x\neg Q(x)$
13. Determine the truth value of each of these statements if the domain consists of all integers.
- a) $\forall n(n + 1 > n)$ b) $\exists n(2n = 3n)$
c) $\exists n(n = -n)$ d) $\forall n(3n \leq 4n)$
14. Determine the truth value of each of these statements if the domain consists of all real numbers.
- a) $\exists x(x^3 = -1)$ b) $\exists x(x^4 < x^2)$
c) $\forall x((-x)^2 = x^2)$ d) $\forall x(2x > x)$
15. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
- a) $\forall n(n^2 \geq 0)$ b) $\exists n(n^2 = 2)$
c) $\forall n(n^2 \geq n)$ d) $\exists n(n^2 < 0)$
16. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.
- a) $\exists x(x^2 = 2)$ b) $\exists x(x^2 = -1)$
c) $\forall x(x^2 + 2 \geq 1)$ d) $\forall x(x^2 \neq x)$
17. Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3, and 4. Write out each of these propositions using disjunctions, conjunctions, and negations.
- a) $\exists xP(x)$ b) $\forall xP(x)$ c) $\exists x\neg P(x)$
d) $\forall x\neg P(x)$ e) $\neg\exists xP(x)$ f) $\neg\forall xP(x)$
18. Suppose that the domain of the propositional function $P(x)$ consists of the integers $-2, -1, 0, 1$, and 2 . Write out each of these propositions using disjunctions, conjunctions, and negations.
- a) $\exists xP(x)$ b) $\forall xP(x)$ c) $\exists x\neg P(x)$
d) $\forall x\neg P(x)$ e) $\neg\exists xP(x)$ f) $\neg\forall xP(x)$
19. Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2, 3, 4, and 5. Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.
- a) $\exists xP(x)$ b) $\forall xP(x)$
c) $\neg\exists xP(x)$ d) $\neg\forall xP(x)$
e) $\forall x((x \neq 3) \rightarrow P(x)) \vee \exists x\neg P(x)$
20. Suppose that the domain of the propositional function $P(x)$ consists of $-5, -3, -1, 1, 3$, and 5 . Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.
- a) $\exists xP(x)$ b) $\forall xP(x)$
c) $\forall x((x \neq 1) \rightarrow P(x))$
d) $\exists x((x \geq 0) \wedge P(x))$
e) $\exists x(\neg P(x)) \wedge \forall x((x < 0) \rightarrow P(x))$
21. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.
- a) Everyone is studying discrete mathematics.
b) Everyone is older than 21 years.
c) Every two people have the same mother.
d) No two different people have the same grandmother.
22. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.
- a) Everyone speaks Hindi.
b) There is someone older than 21 years.
c) Every two people have the same first name.
d) Someone knows more than two other people.
23. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.
- a) Someone in your class can speak Hindi.
b) Everyone in your class is friendly.
c) There is a person in your class who was not born in California.
d) A student in your class has been in a movie.
e) No student in your class has taken a course in logic programming.
24. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.
- a) Everyone in your class has a cellular phone.
b) Somebody in your class has seen a foreign movie.
c) There is a person in your class who cannot swim.
d) All students in your class can solve quadratic equations.
e) Some student in your class does not want to be rich.
25. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.

- a) No one is perfect.
 b) Not everyone is perfect.
 c) All your friends are perfect.
 d) At least one of your friends is perfect.
 e) Everyone is your friend and is perfect.
 f) Not everybody is your friend or someone is not perfect.
26. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.
 a) Someone in your school has visited Uzbekistan.
 b) Everyone in your class has studied calculus and C++.
 c) No one in your school owns both a bicycle and a motorcycle.
 d) There is a person in your school who is not happy.
 e) Everyone in your school was born in the twentieth century.
27. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.
 a) A student in your school has lived in Vietnam.
 b) There is a student in your school who cannot speak Hindi.
 c) A student in your school knows Java, Prolog, and C++.
 d) Everyone in your class enjoys Thai food.
 e) Someone in your class does not play hockey.
28. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.
 a) Something is not in the correct place.
 b) All tools are in the correct place and are in excellent condition.
 c) Everything is in the correct place and in excellent condition.
 d) Nothing is in the correct place and is in excellent condition.
 e) One of your tools is not in the correct place, but it is in excellent condition.
29. Express each of these statements using logical operators, predicates, and quantifiers.
 a) Some propositions are tautologies.
 b) The negation of a contradiction is a tautology.
 c) The disjunction of two contingencies can be a tautology.
 d) The conjunction of two tautologies is a tautology.
30. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.
- a) $\exists x P(x, 3)$
 b) $\forall y P(1, y)$
 c) $\exists y \neg P(2, y)$
 d) $\forall x \neg P(x, 2)$
31. Suppose that the domain of $Q(x, y, z)$ consists of triples x, y, z , where $x = 0, 1$, or 2 , $y = 0$ or 1 , and $z = 0$ or 1 . Write out these propositions using disjunctions and conjunctions.
 a) $\forall y Q(0, y, 0)$
 b) $\exists x Q(x, 1, 1)$
 c) $\exists z \neg Q(0, 0, z)$
 d) $\exists x \neg Q(x, 0, 1)$
32. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
 a) All dogs have fleas.
 b) There is a horse that can add.
 c) Every koala can climb.
 d) No monkey can speak French.
 e) There exists a pig that can swim and catch fish.
33. Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
 a) Some old dogs can learn new tricks.
 b) No rabbit knows calculus.
 c) Every bird can fly.
 d) There is no dog that can talk.
 e) There is no one in this class who knows French and Russian.
34. Express the negation of these propositions using quantifiers, and then express the negation in English.
 a) Some drivers do not obey the speed limit.
 b) All Swedish movies are serious.
 c) No one can keep a secret.
 d) There is someone in this class who does not have a good attitude.
35. Express the negation of each of these statements in terms of quantifiers without using the negation symbol.
 a) $\forall x (x > 1)$
 b) $\forall x (x \leq 2)$
 c) $\exists x (x \geq 4)$
 d) $\exists x (x < 0)$
 e) $\forall x ((x < -1) \vee (x > 2))$
 f) $\exists x ((x < 4) \vee (x > 7))$
36. Express the negation of each of these statements in terms of quantifiers without using the negation symbol.
 a) $\forall x (-2 < x < 3)$
 b) $\forall x (0 \leq x < 5)$
 c) $\exists x (-4 \leq x \leq 1)$
 d) $\exists x (-5 < x < -1)$
37. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
 a) $\forall x (x^2 \geq x)$
 b) $\forall x (x > 0 \vee x < 0)$
 c) $\forall x (x = 1)$

38. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all real numbers.

- a) $\forall x(x^2 \neq x)$ b) $\forall x(x^2 \neq 2)$
c) $\forall x(|x| > 0)$

39. Express each of these statements using predicates and quantifiers.

- a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year.
b) A man qualifies for the marathon if his best previous time is less than 3 hours and a woman qualifies for the marathon if her best previous time is less than 3.5 hours.
c) A student must take at least 60 course hours, or at least 45 course hours and write a master's thesis, and receive a grade no lower than a B in all required courses, to receive a master's degree.
d) There is a student who has taken more than 21 credit hours in a semester and received all A's.

Exercises 40–44 deal with the translation between system specification and logical expressions involving quantifiers.

40. Translate these system specifications into English, where the predicate $S(x, y)$ is “ x is in state y ” and where the domain for x and y consists of all systems and all possible states, respectively.

- a) $\exists xS(x, \text{open})$
b) $\forall x(S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$
c) $\exists xS(x, \text{open}) \vee \exists xS(x, \text{diagnostic})$
d) $\exists x\neg S(x, \text{available})$
e) $\forall x\neg S(x, \text{working})$

41. Translate these specifications into English, where $F(p)$ is “Printer p is out of service,” $B(p)$ is “Printer p is busy,” $L(j)$ is “Print job j is lost,” and $Q(j)$ is “Print job j is queued.”

- a) $\exists p(F(p) \wedge B(p)) \rightarrow \exists jL(j)$
b) $\forall pB(p) \rightarrow \exists jQ(j)$
c) $\exists j(Q(j) \wedge L(j)) \rightarrow \exists pF(p)$
d) $(\forall pB(p) \wedge \forall jQ(j)) \rightarrow \exists jL(j)$

42. Express each of these system specifications using predicates, quantifiers, and logical connectives.

- a) When there is less than 30 megabytes free on the hard disk, a warning message is sent to all users.
b) No directories in the file system can be opened and no files can be closed when system errors have been detected.
c) The file system cannot be backed up if there is a user currently logged on.
d) Video on demand can be delivered when there are at least 8 megabytes of memory available and the connection speed is at least 56 kilobits per second.

43. Express each of these system specifications using predicates, quantifiers, and logical connectives.

- a) At least one mail message, among the nonempty set of messages, can be saved if there is a disk with more than 10 kilobytes of free space.

- b) Whenever there is an active alert, all queued messages are transmitted.
c) The diagnostic monitor tracks the status of all systems except the main console.
d) Each participant on the conference call whom the host of the call did not put on a special list was billed.

44. Express each of these system specifications using predicates, quantifiers, and logical connectives.

- a) Every user has access to an electronic mailbox.
b) The system mailbox can be accessed by everyone in the group if the file system is locked.
c) The firewall is in a diagnostic state only if the proxy server is in a diagnostic state.
d) At least one router is functioning normally if the throughput is between 100 kbps and 500 kbps and the proxy server is not in diagnostic mode.

45. Determine whether $\forall x(P(x) \rightarrow Q(x))$ and $\forall xP(x) \rightarrow \forall xQ(x)$ are logically equivalent. Justify your answer.

46. Determine whether $\forall x(P(x) \leftrightarrow Q(x))$ and $\forall x P(x) \leftrightarrow \forall xQ(x)$ are logically equivalent. Justify your answer.

47. Show that $\exists x(P(x) \vee Q(x))$ and $\exists xP(x) \vee \exists xQ(x)$ are logically equivalent.

Exercises 48–51 establish rules for **null quantification** that we can use when a quantified variable does not appear in part of a statement.

48. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $(\forall xP(x)) \vee A \equiv \forall x(P(x) \vee A)$
b) $(\exists xP(x)) \vee A \equiv \exists x(P(x) \vee A)$

49. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $(\forall xP(x)) \wedge A \equiv \forall x(P(x) \wedge A)$
b) $(\exists xP(x)) \wedge A \equiv \exists x(P(x) \wedge A)$

50. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $\forall x(A \rightarrow P(x)) \equiv A \rightarrow \forall xP(x)$
b) $\exists x(A \rightarrow P(x)) \equiv A \rightarrow \exists xP(x)$

51. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

- a) $\forall x(P(x) \rightarrow A) \equiv \exists xP(x) \rightarrow A$
b) $\exists x(P(x) \rightarrow A) \equiv \forall xP(x) \rightarrow A$

52. Show that $\forall xP(x) \vee \forall xQ(x)$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

53. Show that $\exists xP(x) \wedge \exists xQ(x)$ and $\exists x(P(x) \wedge Q(x))$ are not logically equivalent.

54. As mentioned in the text, the notation $\exists!xP(x)$ denotes

“There exists a unique x such that $P(x)$ is true.”

If the domain consists of all integers, what are the truth values of these statements?

- a) $\exists!x(x > 1)$ b) $\exists!x(x^2 = 1)$
c) $\exists!x(x + 3 = 2x)$ d) $\exists!x(x = x + 1)$

55. What are the truth values of these statements?

- a) $\exists!xP(x) \rightarrow \exists xP(x)$
- b) $\forall xP(x) \rightarrow \exists!xP(x)$
- c) $\exists!x\neg P(x) \rightarrow \neg\forall xP(x)$

56. Write out $\exists!xP(x)$, where the domain consists of the integers 1, 2, and 3, in terms of negations, conjunctions, and disjunctions.

57. Given the Prolog facts in Example 28, what would Prolog return given these queries?

- a) `?instructor(chan, math273)`
- b) `?instructor(patel, cs301)`
- c) `?enrolled(X, cs301)`
- d) `?enrolled(kiko, Y)`
- e) `?teaches(grossman, Y)`

58. Given the Prolog facts in Example 28, what would Prolog return when given these queries?

- a) `?enrolled(kevin, ee222)`
- b) `?enrolled(kiko, math273)`
- c) `?instructor(grossman, X)`
- d) `?instructor(X, cs301)`
- e) `?teaches(X, kevin)`

59. Suppose that Prolog facts are used to define the predicates *mother*(*M*, *Y*) and *father*(*F*, *X*), which represent that *M* is the mother of *Y* and *F* is the father of *X*, respectively. Give a Prolog rule to define the predicate *sibling*(*X*, *Y*), which represents that *X* and *Y* are siblings (that is, have the same mother and the same father).

60. Suppose that Prolog facts are used to define the predicates *mother*(*M*, *Y*) and *father*(*F*, *X*), which represent that *M* is the mother of *Y* and *F* is the father of *X*, respectively. Give a Prolog rule to define the predicate *grandfather*(*X*, *Y*), which represents that *X* is the grandfather of *Y*. [Hint: You can write a disjunction in Prolog either by using a semicolon to separate predicates or by putting these predicates on separate lines.]

Exercises 61–64 are based on questions found in the book *Symbolic Logic* by Lewis Carroll.

61. Let *P*(*x*), *Q*(*x*), and *R*(*x*) be the statements “*x* is a professor,” “*x* is ignorant,” and “*x* is vain,” respectively. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), and *R*(*x*), where the domain consists of all people.

a) No professors are ignorant.

b) All ignorant people are vain.

c) No professors are vain.

d) Does (c) follow from (a) and (b)?

62. Let *P*(*x*), *Q*(*x*), and *R*(*x*) be the statements “*x* is a clear explanation,” “*x* is satisfactory,” and “*x* is an excuse,” respectively. Suppose that the domain for *x* consists of all English text. Express each of these statements using quantifiers, logical connectives, and *P*(*x*), *Q*(*x*), and *R*(*x*).

a) All clear explanations are satisfactory.

b) Some excuses are unsatisfactory.

c) Some excuses are not clear explanations.

*d) Does (c) follow from (a) and (b)?

63. Let *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*) be the statements “*x* is a baby,” “*x* is logical,” “*x* is able to manage a crocodile,” and “*x* is despised,” respectively. Suppose that the domain consists of all people. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*).

a) Babies are illogical.

b) Nobody is despised who can manage a crocodile.

c) Illogical persons are despised.

d) Babies cannot manage crocodiles.

*e) Does (d) follow from (a), (b), and (c)? If not, is there a correct conclusion?

64. Let *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*) be the statements “*x* is a duck,” “*x* is one of my poultry,” “*x* is an officer,” and “*x* is willing to waltz,” respectively. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*).

a) No ducks are willing to waltz.

b) No officers ever decline to waltz.

c) All my poultry are ducks.

d) My poultry are not officers.

*e) Does (d) follow from (a), (b), and (c)? If not, is there a correct conclusion?

1.5 Nested Quantifiers

1.5.1 Introduction

In Section 1.4 we defined the existential and universal quantifiers and showed how they can be used to represent mathematical statements. We also explained how they can be used to translate English sentences into logical expressions. However, in Section 1.4 we avoided **nested quantifiers**, where one quantifier is within the scope of another, such as

$$\forall x \exists y (x + y = 0).$$

Note that everything within the scope of a quantifier can be thought of as a propositional function. For example,

$$\forall x \exists y (x + y = 0)$$

is the same thing as $\forall x Q(x)$, where $Q(x)$ is $\exists y P(x, y)$, where $P(x, y)$ is $x + y = 0$.

Nested quantifiers commonly occur in mathematics and computer science. Although nested quantifiers can sometimes be difficult to understand, the rules we have already studied in Section 1.4 can help us use them. In this section we will gain experience working with nested quantifiers. We will see how to use nested quantifiers to express mathematical statements such as “The sum of two positive integers is always positive.” We will show how nested quantifiers can be used to translate English sentences such as “Everyone has exactly one best friend” into logical statements. Moreover, we will gain experience working with the negations of statements involving nested quantifiers.

1.5.2 Understanding Statements Involving Nested Quantifiers

To understand statements involving nested quantifiers, we need to unravel what the quantifiers and predicates that appear mean. This is illustrated in Examples 1 and 2.

EXAMPLE 1 Assume that the domain for the variables x and y consists of all real numbers. The statement

$$\forall x \forall y (x + y = y + x)$$

**Extra
Examples** 

says that $x + y = y + x$ for all real numbers x and y . This is the commutative law for addition of real numbers. Likewise, the statement

$$\forall x \exists y (x + y = 0)$$

says that for every real number x there is a real number y such that $x + y = 0$. This states that every real number has an additive inverse. Similarly, the statement

$$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$$

is the associative law for addition of real numbers. 

EXAMPLE 2 Translate into English the statement

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0)),$$

where the domain for both variables consists of all real numbers.

Solution: This statement says that for every real number x and for every real number y , if $x > 0$ and $y < 0$, then $xy < 0$. That is, this statement says that for real numbers x and y , if x is positive and y is negative, then xy is negative. This can be stated more succinctly as “The product of a positive real number and a negative real number is always a negative real number.” 

THINKING OF QUANTIFICATION AS LOOPS In working with quantifications of more than one variable, it is sometimes helpful to think in terms of nested loops. (If there are infinitely many elements in the domain of some variable, we cannot actually loop through all values. Nevertheless, this way of thinking is helpful in understanding nested quantifiers.) For example, to see whether $\forall x \forall y P(x, y)$ is true, we loop through the values for x , and for each x we loop through the values for y . If we find that for all values of x that $P(x, y)$ is true for all values of y ,

we have determined that $\forall x \forall y P(x, y)$ is true. If we ever hit a value x for which we hit a value y for which $P(x, y)$ is false, we have shown that $\forall x \forall y P(x, y)$ is false.

Similarly, to determine whether $\forall x \exists y P(x, y)$ is true, we loop through the values for x . For each x we loop through the values for y until we find a y for which $P(x, y)$ is true. If for every x we hit such a y , then $\forall x \exists y P(x, y)$ is true; if for some x we never hit such a y , then $\forall x \exists y P(x, y)$ is false.

To see whether $\exists x \forall y P(x, y)$ is true, we loop through the values for x until we find an x for which $P(x, y)$ is always true when we loop through all values for y . Once we find such an x , we know that $\exists x \forall y P(x, y)$ is true. If we never hit such an x , then we know that $\exists x \forall y P(x, y)$ is false.

Finally, to see whether $\exists x \exists y P(x, y)$ is true, we loop through the values for x , where for each x we loop through the values for y until we hit an x for which we hit a y for which $P(x, y)$ is true. The statement $\exists x \exists y P(x, y)$ is false only if we never hit an x for which we hit a y such that $P(x, y)$ is true.

1.5.3 The Order of Quantifiers

Many mathematical statements involve multiple quantifications of propositional functions involving more than one variable. It is important to note that the order of the quantifiers is important, unless all the quantifiers are universal quantifiers or all are existential quantifiers.

These remarks are illustrated by Examples 3–5.

EXAMPLE 3 Let $P(x, y)$ be the statement “ $x + y = y + x$.” What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$, where the domain for all variables consists of all real numbers?

Extra Examples ➤

Solution: The quantification

$$\forall x \forall y P(x, y)$$

denotes the proposition

“For all real numbers x , for all real numbers y , $x + y = y + x$.”

Because $P(x, y)$ is true for all real numbers x and y (it is the commutative law for addition, which is an axiom for the real numbers—see Appendix 1), the proposition $\forall x \forall y P(x, y)$ is true. Note that the statement $\forall y \forall x P(x, y)$ says “For all real numbers y , for all real numbers x , $x + y = y + x$.” This has the same meaning as the statement “For all real numbers x , for all real numbers y , $x + y = y + x$.” That is, $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ have the same meaning, and both are true. This illustrates the principle that the order of nested universal quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement. ◀

EXAMPLE 4 Let $Q(x, y)$ denote “ $x + y = 0$.” What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

Solution: The quantification

$$\exists y \forall x Q(x, y)$$

denotes the proposition

“There is a real number y such that for every real number x , $Q(x, y)$.”

No matter what value of y is chosen, there is only one value of x for which $x + y = 0$. Because there is no real number y such that $x + y = 0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

The quantification

$$\forall x \exists y Q(x, y)$$

denotes the proposition

“For every real number x there is a real number y such that $Q(x, y)$.”

Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$. Hence, the statement $\forall x \exists y Q(x, y)$ is true. 

Be careful with the order of existential and universal quantifiers!

Example 4 illustrates that the order in which quantifiers appear makes a difference. The statements $\exists y \forall x P(x, y)$ and $\forall x \exists y P(x, y)$ are not logically equivalent. The statement $\exists y \forall x P(x, y)$ is true if and only if there is a y that makes $P(x, y)$ true for every x . So, for this statement to be true, there must be a particular value of y for which $P(x, y)$ is true regardless of the choice of x . On the other hand, $\forall x \exists y P(x, y)$ is true if and only if for every value of x there is a value of y for which $P(x, y)$ is true. So, for this statement to be true, no matter which x you choose, there must be a value of y (possibly depending on the x you choose) for which $P(x, y)$ is true. In other words, in the second case, y can depend on x , whereas in the first case, y is a constant independent of x .

From these observations, it follows that if $\exists y \forall x P(x, y)$ is true, then $\forall x \exists y P(x, y)$ must also be true. However, if $\forall x \exists y P(x, y)$ is true, it is not necessary for $\exists y \forall x P(x, y)$ to be true. (See Supplementary Exercises 30 and 31.)

Table 1 summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of more than two variables are also common, as Example 5 illustrates.

EXAMPLE 5 Let $Q(x, y, z)$ be the statement “ $x + y = z$.” What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Solution: Suppose that x and y are assigned values. Then, there exists a real number z such that $x + y = z$. Consequently, the quantification

$$\forall x \forall y \exists z Q(x, y, z),$$

which is the statement

“For all real numbers x and for all real numbers y there is a real number z such that $x + y = z$,”

TABLE 1 Quantifications of Two Variables.

Statement	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

is true. The order of the quantification here is important, because the quantification

$$\exists z \forall x \forall y Q(x, y, z),$$

which is the statement

“There is a real number z such that for all real numbers x and for all real numbers y it is true that $x + y = z$,”

is false, because there is no value of z that satisfies the equation $x + y = z$ for all values of x and y . 

1.5.4 Translating Mathematical Statements into Statements Involving Nested Quantifiers

Mathematical statements expressed in English can be translated into logical expressions, as Examples 6–8 show.

EXAMPLE 6 Translate the statement “The sum of two positive integers is always positive” into a logical expression.

Extra Examples 

Solution: To translate this statement into a logical expression, we first rewrite it so that the implied quantifiers and a domain are shown: “For every two integers, if these integers are both positive, then the sum of these integers is positive.” Next, we introduce the variables x and y to obtain “For all positive integers x and y , $x + y$ is positive.” Consequently, we can express this statement as

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x + y > 0)),$$

where the domain for both variables consists of all integers. Note that we could also translate this using the positive integers as the domain. Then the statement “The sum of two positive integers is always positive” becomes “For every two positive integers, the sum of these integers is positive.” We can express this as

$$\forall x \forall y (x + y > 0),$$

where the domain for both variables consists of all positive integers. 

EXAMPLE 7 Translate the statement “Every real number except zero has a multiplicative inverse.” (A **multiplicative inverse** of a real number x is a real number y such that $xy = 1$.)

Solution: We first rewrite this as “For every real number x except zero, x has a multiplicative inverse.” We can rewrite this as “For every real number x , if $x \neq 0$, then there exists a real number y such that $xy = 1$.” This can be rewritten as

$$\forall x ((x \neq 0) \rightarrow \exists y (xy = 1)).$$


One example that you may be familiar with is the concept of limit, which is important in calculus.

EXAMPLE 8 (*Requires calculus*) Use quantifiers to express the definition of the limit of a real-valued function $f(x)$ of a real variable x at a point a in its domain.

Solution: Recall that the definition of the statement

$$\lim_{x \rightarrow a} f(x) = L$$

is: For every real number $\epsilon > 0$ there exists a real number $\delta > 0$ such that $|f(x) - L| < \epsilon$ whenever $0 < |x - a| < \delta$. This definition of a limit can be phrased in terms of quantifiers by

$$\forall \epsilon \exists \delta \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon),$$

where the domain for the variables δ and ϵ consists of all positive real numbers and for x consists of all real numbers.

This definition can also be expressed as

$$\forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

when the domain for the variables ϵ and δ consists of all real numbers, rather than just the positive real numbers. [Here, restricted quantifiers have been used. Recall that $\forall x > 0 P(x)$ means that for all x with $x > 0$, $P(x)$ is true.] ◀

1.5.5 Translating from Nested Quantifiers into English

Expressions with nested quantifiers expressing statements in English can be quite complicated. The first step in translating such an expression is to write out what the quantifiers and predicates in the expression mean. The next step is to express this meaning in a simpler sentence. This process is illustrated in Examples 9 and 10.

EXAMPLE 9 Translate the statement

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

into English, where $C(x)$ is “ x has a computer,” $F(x, y)$ is “ x and y are friends,” and the domain for both x and y consists of all students in your school.

Solution: The statement says that for every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends. In other words, every student in your school has a computer or has a friend who has a computer. ◀

EXAMPLE 10 Translate the statement

$$\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

into English, where $F(a, b)$ means a and b are friends and the domain for x , y , and z consists of all students in your school.

Solution: We first examine the expression $(F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z)$. This expression says that if students x and y are friends, and students x and z are friends, and furthermore, if y and z are not the same student, then y and z are not friends. It follows that the original statement, which is triply quantified, says that there is a student x such that for all students y and all students z other than y , if x and y are friends and x and z are friends, then y and z are not friends. In other words, there is a student none of whose friends are also friends with each other. ◀

1.5.6 Translating English Sentences into Logical Expressions

In Section 1.4 we showed how quantifiers can be used to translate sentences into logical expressions. However, we avoided sentences whose translation into logical expressions required the use of nested quantifiers. We now address the translation of such sentences.

EXAMPLE 11 Express the statement “If a person is female and is a parent, then this person is someone’s mother” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The statement “If a person is female and is a parent, then this person is someone’s mother” can be expressed as “For every person x , if person x is female and person x is a parent, then there exists a person y such that person x is the mother of person y .” We introduce the propositional functions $F(x)$ to represent “ x is female,” $P(x)$ to represent “ x is a parent,” and $M(x, y)$ to represent “ x is the mother of y .” The original statement can be represented as

$$\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y)).$$

Using the null quantification rule in part (b) of Exercise 49 in Section 1.4, we can move $\exists y$ to the left so that it appears just after $\forall x$, because y does not appear in $F(x) \wedge P(x)$. We obtain the logically equivalent expression

$$\forall x \exists y ((F(x) \wedge P(x)) \rightarrow M(x, y)).$$

EXAMPLE 12 Express the statement “Everyone has exactly one best friend” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The statement “Everyone has exactly one best friend” can be expressed as “For every person x , person x has exactly one best friend.” Introducing the universal quantifier, we see that this statement is the same as “ $\forall x$ (person x has exactly one best friend),” where the domain consists of all people.

To say that x has exactly one best friend means that there is a person y who is the best friend of x , and furthermore, that for every person z , if person z is not person y , then z is not the best friend of x . When we introduce the predicate $B(x, y)$ to be the statement “ y is the best friend of x ,” the statement that x has exactly one best friend can be represented as

$$\exists y(B(x, y) \wedge \forall z((z \neq y) \rightarrow \neg B(x, z))).$$

Consequently, our original statement can be expressed as

$$\forall x \exists y (B(x, y) \wedge \forall z((z \neq y) \rightarrow \neg B(x, z))).$$

[Note that we can write this statement as $\forall x \exists! y B(x, y)$, where $\exists!$ is the “uniqueness quantifier” defined in Section 1.4.]

EXAMPLE 13 Use quantifiers to express the statement “There is a woman who has taken a flight on every airline in the world.”

Solution: Let $P(w, f)$ be “ w has taken f ” and $Q(f, a)$ be “ f is a flight on a .” We can express the statement as

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a)),$$

where the domains of discourse for w, f , and a consist of all the women in the world, all airplane flights, and all airlines, respectively.

The statement could also be expressed as

$$\exists w \forall a \exists f R(w, f, a),$$

where $R(w, f, a)$ is “ w has taken f on a .” Although this is more compact, it somewhat obscures the relationships among the variables. Consequently, the first solution is usually preferable. 

1.5.7 Negating Nested Quantifiers

Assessment

Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier. This is illustrated in Examples 14–16.

EXAMPLE 14

Extra Examples

Express the negation of the statement $\forall x \exists y (xy = 1)$ so that no negation precedes a quantifier.

Solution: By successively applying De Morgan’s laws for quantifiers in Table 2 of Section 1.4, we can move the negation in $\neg \forall x \exists y (xy = 1)$ inside all the quantifiers. We find that $\neg \forall x \exists y (xy = 1)$ is equivalent to $\exists x \neg \exists y (xy = 1)$, which is equivalent to $\exists x \forall y \neg (xy = 1)$. Because $\neg (xy = 1)$ can be expressed more simply as $xy \neq 1$, we conclude that our negated statement can be expressed as $\exists x \forall y (xy \neq 1)$. 

EXAMPLE 15

Use quantifiers to express the statement that “There does not exist a woman who has taken a flight on every airline in the world.”

Solution: This statement is the negation of the statement “There is a woman who has taken a flight on every airline in the world” from Example 13. By Example 13, our statement can be expressed as $\neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$, where $P(w, f)$ is “ w has taken f ” and $Q(f, a)$ is “ f is a flight on a .” By successively applying De Morgan’s laws for quantifiers in Table 2 of Section 1.4 to move the negation inside successive quantifiers and by applying De Morgan’s law for negating a conjunction in the last step, we find that our statement is equivalent to each of this sequence of statements:

$$\begin{aligned} \neg \exists w \neg \forall a \neg \exists f (P(w, f) \wedge Q(f, a)) &\equiv \forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a)). \end{aligned}$$

This last statement states “For every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline.” 

EXAMPLE 16

(Requires calculus) Use quantifiers and predicates to express the fact that $\lim_{x \rightarrow a} f(x)$ does not exist where $f(x)$ is a real-valued function of a real variable x and a belongs to the domain of f

Solution: To say that $\lim_{x \rightarrow a} f(x)$ does not exist means that for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$. By using Example 8, the statement $\lim_{x \rightarrow a} f(x) \neq L$ can be expressed as

$$\neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon).$$

Successively applying the rules for negating quantified expressions, we construct this sequence of equivalent statements:

$$\begin{aligned}
 & \neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \neg \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \neg \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \exists x \neg (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).
 \end{aligned}$$

In the last step we used the equivalence $\neg(p \rightarrow q) \equiv p \wedge \neg q$, which follows from the fifth equivalence in Table 7 of Section 1.3.

Because the statement “ $\lim_{x \rightarrow a} f(x)$ does not exist” means for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$, this can be expressed as

$$\forall L \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).$$

This last statement says that for every real number L there is a real number $\epsilon > 0$ such that for every real number $\delta > 0$, there exists a real number x such that $0 < |x - a| < \delta$ and $|f(x) - L| \geq \epsilon$. ◀

Exercises

- Translate these statements into English, where the domain for each variable consists of all real numbers.
 - $\forall x \exists y (x < y)$
 - $\forall x \forall y (((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0))$
 - $\forall x \forall y \exists z (xy = z)$
- Translate these statements into English, where the domain for each variable consists of all real numbers.
 - $\exists x \forall y (xy = y)$
 - $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$
 - $\forall x \forall y \exists z (x = y + z)$
- Let $Q(x, y)$ be the statement “ x has sent an e-mail message to y ,” where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

a) $\exists x \exists y Q(x, y)$	b) $\exists x \forall y Q(x, y)$
c) $\forall x \exists y Q(x, y)$	d) $\exists y \forall x Q(x, y)$
e) $\forall y \exists x Q(x, y)$	f) $\forall x \forall y Q(x, y)$
- Let $P(x, y)$ be the statement “Student x has taken class y ,” where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.

a) $\exists x \exists y P(x, y)$	b) $\exists x \forall y P(x, y)$
c) $\forall x \exists y P(x, y)$	d) $\exists y \forall x P(x, y)$
e) $\forall y \exists x P(x, y)$	f) $\forall x \forall y P(x, y)$
- Let $W(x, y)$ mean that student x has visited website y , where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.
 - $W(\text{Sarah Smith}, \text{www.att.com})$
 - $\exists x W(x, \text{www.imdb.org})$
 - $\exists y W(\text{José Orez}, y)$
 - $\exists y (W(\text{Ashok Puri}, y) \wedge W(\text{Cindy Yoon}, y))$
 - $\exists y \forall z (y \neq (\text{David Belcher}) \wedge (W(\text{David Belcher}, z) \rightarrow W(y, z)))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (W(x, z) \leftrightarrow W(y, z)))$
- Let $C(x, y)$ mean that student x is enrolled in class y , where the domain for x consists of all students in your school and the domain for y consists of all classes being given at your school. Express each of these statements by a simple English sentence.
 - $C(\text{Randy Goldberg}, \text{CS 252})$
 - $\exists x C(x, \text{Math 695})$
 - $\exists y C(\text{Carol Sitea}, y)$
 - $\exists x (C(x, \text{Math 222}) \wedge C(x, \text{CS 252}))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (C(x, z) \rightarrow C(y, z)))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (C(x, z) \leftrightarrow C(y, z)))$
- Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.
 - $\neg T(\text{Abdallah Hussein}, \text{Japanese})$

- b) $\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$
 c) $\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$
 d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$
 e) $\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$
 f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$
8. Let $Q(x, y)$ be the statement “Student x has been a contestant on quiz show y .” Express each of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.
- There is a student at your school who has been a contestant on a television quiz show.
 - No student at your school has ever been a contestant on a television quiz show.
 - There is a student at your school who has been a contestant on *Jeopardy!* and on *Wheel of Fortune*.
 - Every television quiz show has had a student from your school as a contestant.
 - At least two students from your school have been contestants on *Jeopardy!*.
9. Let $L(x, y)$ be the statement “ x loves y ,” where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.
- Everybody loves Jerry.
 - Everybody loves somebody.
 - There is somebody whom everybody loves.
 - Nobody loves everybody.
 - There is somebody whom Lydia does not love.
 - There is somebody whom no one loves.
 - There is exactly one person whom everybody loves.
 - There are exactly two people whom Lynn loves.
 - Everyone loves himself or herself.
 - There is someone who loves no one besides himself or herself.
10. Let $F(x, y)$ be the statement “ x can fool y ,” where the domain consists of all people in the world. Use quantifiers to express each of these statements.
- Everybody can fool Fred.
 - Evelyn can fool everybody.
 - Everybody can fool somebody.
 - There is no one who can fool everybody.
 - Everyone can be fooled by somebody.
 - No one can fool both Fred and Jerry.
 - Nancy can fool exactly two people.
 - There is exactly one person whom everybody can fool.
 - No one can fool himself or herself.
 - There is someone who can fool exactly one person besides himself or herself.
11. Let $S(x)$ be the predicate “ x is a student,” $F(x)$ the predicate “ x is a faculty member,” and $A(x, y)$ the predicate “ x has asked y a question,” where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.
- Lois has asked Professor Michaels a question.
 - Every student has asked Professor Gross a question.
 - Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.
 - Some student has not asked any faculty member a question.
 - There is a faculty member who has never been asked a question by a student.
 - Some student has asked every faculty member a question.
 - There is a faculty member who has asked every other faculty member a question.
 - Some student has never been asked a question by a faculty member.
12. Let $I(x)$ be the statement “ x has an Internet connection” and $C(x, y)$ be the statement “ x and y have chatted over the Internet,” where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of these statements.
- Jerry does not have an Internet connection.
 - Rachel has not chatted over the Internet with Chelsea.
 - Jan and Sharon have never chatted over the Internet.
 - No one in the class has chatted with Bob.
 - Sanjay has chatted with everyone except Joseph.
 - Someone in your class does not have an Internet connection.
 - Not everyone in your class has an Internet connection.
 - Exactly one student in your class has an Internet connection.
 - Everyone except one student in your class has an Internet connection.
 - Everyone in your class with an Internet connection has chatted over the Internet with at least one other student in your class.
 - Someone in your class has an Internet connection but has not chatted with anyone else in your class.
 - There are two students in your class who have not chatted with each other over the Internet.
 - There is a student in your class who has chatted with everyone in your class over the Internet.
 - There are at least two students in your class who have not chatted with the same person in your class.
 - There are two students in the class who between them have chatted with everyone else in the class.
13. Let $M(x, y)$ be “ x has sent y an e-mail message” and $T(x, y)$ be “ x has telephoned y ,” where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all e-mail messages that were sent are received, which is not the way things often work.)
- Chou has never sent an e-mail message to Koko.
 - Arlene has never sent an e-mail message to or telephoned Sarah.
 - José has never received an e-mail message from Deborah.
 - Every student in your class has sent an e-mail message to Ken.
 - No one in your class has telephoned Nina.

- f) Everyone in your class has either telephoned Avi or sent him an e-mail message.
- g) There is a student in your class who has sent everyone else in your class an e-mail message.
- h) There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class.
- i) There are two different students in your class who have sent each other e-mail messages.
- j) There is a student who has sent himself or herself an e-mail message.
- k) There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.
- l) Every student in the class has either received an e-mail message or received a telephone call from another student in the class.
- m) There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student.
- n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class.
14. Use quantifiers and predicates with more than one variable to express these statements.
- a) There is a student in this class who can speak Hindi.
- b) Every student in this class plays some sport.
- c) Some student in this class has visited Alaska but has not visited Hawaii.
- d) All students in this class have learned at least one programming language.
- e) There is a student in this class who has taken every course offered by one of the departments in this school.
- f) Some student in this class grew up in the same town as exactly one other student in this class.
- g) Every student in this class has chatted with at least one other student in at least one chat group.
15. Use quantifiers and predicates with more than one variable to express these statements.
- a) Every computer science student needs a course in discrete mathematics.
- b) There is a student in this class who owns a personal computer.
- c) Every student in this class has taken at least one computer science course.
- d) There is a student in this class who has taken at least one course in computer science.
- e) Every student in this class has been in every building on campus.
- f) There is a student in this class who has been in every room of at least one building on campus.
- g) Every student in this class has been in at least one room of every building on campus.
16. A discrete mathematics class contains 1 mathematics major who is a freshman, 12 mathematics majors who are sophomores, 15 computer science majors who are sophomores, 2 mathematics majors who are juniors, 2 computer science majors who are juniors, and 1 computer science major who is a senior. Express each of these statements in terms of quantifiers and then determine its truth value.
- a) There is a student in the class who is a junior.
- b) Every student in the class is a computer science major.
- c) There is a student in the class who is neither a mathematics major nor a junior.
- d) Every student in the class is either a sophomore or a computer science major.
- e) There is a major such that there is a student in the class in every year of study with that major.
17. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.
- a) Every user has access to exactly one mailbox.
- b) There is a process that continues to run during all error conditions only if the kernel is working correctly.
- c) All users on the campus network can access all websites whose url has a .edu extension.
- *d) There are exactly two systems that monitor every remote server.
18. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.
- a) At least one console must be accessible during every fault condition.
- b) The e-mail address of every user can be retrieved whenever the archive contains at least one message sent by every user on the system.
- c) For every security breach there is at least one mechanism that can detect that breach if and only if there is a process that has not been compromised.
- d) There are at least two paths connecting every two distinct endpoints on the network.
- e) No one knows the password of every user on the system except for the system administrator, who knows all passwords.
19. Express each of these statements using mathematical and logical operators, predicates, and quantifiers, where the domain consists of all integers.
- a) The sum of two negative integers is negative.
- b) The difference of two positive integers is not necessarily positive.
- c) The sum of the squares of two integers is greater than or equal to the square of their sum.
- d) The absolute value of the product of two integers is the product of their absolute values.
20. Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.
- a) The product of two negative integers is positive.
- b) The average of two positive integers is positive.

- c) The difference of two negative integers is not necessarily negative.
- d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.
21. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that every positive integer is the sum of the squares of four integers.
22. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that there is a positive integer that is not the sum of three squares.
23. Express each of these mathematical statements using predicates, quantifiers, logical connectives, and mathematical operators.
- a) The product of two negative real numbers is positive.
- b) The difference of a real number and itself is zero.
- c) Every positive real number has exactly two square roots.
- d) A negative real number does not have a square root that is a real number.
24. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.
- a) $\exists x \forall y (x + y = y)$
- b) $\forall x \forall y ((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0)$
- c) $\exists x \exists y ((x \leq 0) \wedge (y \leq 0) \wedge (x - y > 0))$
- d) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$
25. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.
- a) $\exists x \forall y (xy = y)$
- b) $\forall x \forall y ((x < 0) \wedge (y < 0)) \rightarrow (xy > 0)$
- c) $\exists x \exists y ((x^2 > y) \wedge (x < y))$
- d) $\forall x \forall y \exists z (x + y = z)$
26. Let $Q(x, y)$ be the statement “ $x + y = x - y$.” If the domain for both variables consists of all integers, what are the truth values?
- a) $Q(1, 1)$
- b) $Q(2, 0)$
- c) $\forall y Q(1, y)$
- d) $\exists x Q(x, 2)$
- e) $\exists x \exists y Q(x, y)$
- f) $\forall x \exists y Q(x, y)$
- g) $\exists y \forall x Q(x, y)$
- h) $\forall y \exists x Q(x, y)$
- i) $\forall x \forall y Q(x, y)$
27. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
- a) $\forall n \exists m (n^2 < m)$
- b) $\exists n \forall m (n < m^2)$
- c) $\forall n \exists m (n + m = 0)$
- d) $\exists n \forall m (nm = m)$
- e) $\exists n \exists m (n^2 + m^2 = 5)$
- f) $\exists n \exists m (n^2 + m^2 = 6)$
- g) $\exists n \exists m (n + m = 4 \wedge n - m = 1)$
- h) $\exists n \exists m (n + m = 4 \wedge n - m = 2)$
- i) $\forall n \forall m \exists p (p = (m + n)/2)$
28. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.
- a) $\forall x \exists y (x^2 = y)$
- b) $\forall x \exists y (x = y^2)$
- c) $\exists x \forall y (xy = 0)$
- d) $\exists x \exists y (x + y \neq y + x)$
- e) $\forall x (x \neq 0 \rightarrow \exists y (xy = 1))$
- f) $\exists x \forall y (y \neq 0 \rightarrow xy = 1)$
- g) $\forall x \exists y (x + y = 1)$
- h) $\exists x \exists y (x + 2y = 2 \wedge 2x + 4y = 5)$
- i) $\forall x \exists y (x + y = 2 \wedge 2x - y = 1)$
- j) $\forall x \forall y \exists z (z = (x + y)/2)$
29. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.
- a) $\forall x \forall y P(x, y)$
- b) $\exists x \exists y P(x, y)$
- c) $\exists x \forall y P(x, y)$
- d) $\forall y \exists x P(x, y)$
30. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).
- a) $\neg \exists y \exists x P(x, y)$
- b) $\neg \forall x \exists y P(x, y)$
- c) $\neg \exists y (Q(y) \wedge \forall x \neg R(x, y))$
- d) $\neg \exists y (\exists x R(x, y) \vee \forall x S(x, y))$
- e) $\neg \exists y (\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$
31. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\forall x \exists y \forall z T(x, y, z)$
- b) $\forall x \exists y P(x, y) \vee \forall x \exists y Q(x, y)$
- c) $\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$
- d) $\forall x \exists y (P(x, y) \rightarrow Q(x, y))$
32. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\exists z \forall y \forall x T(x, y, z)$
- b) $\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y)$
- c) $\exists x \exists y (Q(x, y) \leftrightarrow Q(y, x))$
- d) $\forall y \exists x \exists z (T(x, y, z) \vee Q(x, y))$
33. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).
- a) $\neg \forall x \forall y P(x, y)$
- b) $\neg \forall y \exists x P(x, y)$
- c) $\neg \forall y \forall x (P(x, y) \vee Q(x, y))$
- d) $\neg (\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y))$
- e) $\neg \forall x (\exists y \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z))$
34. Find a common domain for the variables x, y , and z for which the statement $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z = x) \vee (z = y)))$ is true and another domain for which it is false.
35. Find a common domain for the variables x, y, z , and w for which the statement $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is true and another common domain for these variables for which it is false.
36. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
- a) No one has lost more than one thousand dollars playing the lottery.
- b) There is a student in this class who has chatted with exactly one other student.

- c) No student in this class has sent e-mail to exactly two other students in this class.
 - d) Some student has solved every exercise in this book.
 - e) No student has solved at least one exercise in every section of this book.
37. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
- a) Every student in this class has taken exactly two mathematics classes at this school.
 - b) Someone has visited every country in the world except Libya.
 - c) No one has climbed every mountain in the Himalayas.
 - d) Every movie actor has either been in a movie with Kevin Bacon or has been in a movie with someone who has been in a movie with Kevin Bacon.
38. Express the negations of these propositions using quantifiers, and in English.
- a) Every student in this class likes mathematics.
 - b) There is a student in this class who has never seen a computer.
 - c) There is a student in this class who has taken every mathematics course offered at this school.
 - d) There is a student in this class who has been in at least one room of every building on campus.
39. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x \forall y (x^2 = y^2 \rightarrow x = y)$
 - b) $\forall x \exists y (y^2 = x)$
 - c) $\forall x \forall y (xy \geq x)$
40. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x \exists y (x = 1/y)$
 - b) $\forall x \exists y (y^2 - x < 100)$
 - c) $\forall x \forall y (x^2 \neq y^3)$
41. Use quantifiers to express the associative law for multiplication of real numbers.
42. Use quantifiers to express the distributive laws of multiplication over addition for real numbers.
43. Use quantifiers and logical connectives to express the fact that every linear polynomial (that is, polynomial of degree 1) with real coefficients and where the coefficient of x is nonzero, has exactly one real root.
44. Use quantifiers and logical connectives to express the fact that a quadratic polynomial with real number coefficients has at most two real roots.
45. Determine the truth value of the statement $\forall x \exists y (xy = 1)$ if the domain for the variables consists of

- a) the nonzero real numbers.
- b) the nonzero integers.
- c) the positive real numbers.

46. Determine the truth value of the statement $\exists x \forall y (x \leq y^2)$ if the domain for the variables consists of

- a) the positive real numbers.
- b) the integers.
- c) the nonzero real numbers.

47. Show that the two statements $\neg \exists x \forall y P(x, y)$ and $\forall x \exists y \neg P(x, y)$, where both quantifiers over the first variable in $P(x, y)$ have the same domain, and both quantifiers over the second variable in $P(x, y)$ have the same domain, are logically equivalent.

*48. Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x \forall y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain, are logically equivalent. (The new variable y is used to combine the quantifications correctly.)

*49. a) Show that $\forall x P(x) \wedge \exists x Q(x)$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same nonempty domain.

b) Show that $\forall x P(x) \vee \exists x Q(x)$ is equivalent to $\forall x \exists y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain.

A statement is in **prenex normal form (PNF)** if and only if it is of the form

$$Q_1 x_1 Q_2 x_2 \cdots Q_k x_k P(x_1, x_2, \dots, x_k),$$

where each Q_i , $i = 1, 2, \dots, k$, is either the existential quantifier or the universal quantifier, and $P(x_1, \dots, x_k)$ is a predicate involving no quantifiers. For example, $\exists x \forall y (P(x, y) \wedge Q(y))$ is in prenex normal form, whereas $\exists x P(x) \vee \forall x Q(x)$ is not (because the quantifiers do not all occur first).

Every statement formed from propositional variables, predicates, **T**, and **F** using logical connectives and quantifiers is equivalent to a statement in prenex normal form. Exercise 51 asks for a proof of this fact.

*50. Put these statements in prenex normal form. [Hint: Use logical equivalence from Tables 6 and 7 in Section 1.3, Table 2 in Section 1.4, Example 19 in Section 1.4, Exercises 47 and 48 in Section 1.4, and Exercises 48 and 49.]

- a) $\exists x P(x) \vee \exists x Q(x) \vee A$, where A is a proposition not involving any quantifiers
- b) $\neg(\forall x P(x) \vee \forall x Q(x))$
- c) $\exists x P(x) \rightarrow \exists x Q(x)$

**51. Show how to transform an arbitrary statement to a statement in prenex normal form that is equivalent to the given statement. (Note: A formal solution of this exercise requires use of structural induction, covered in Section 5.3.)

*52. Express the quantification $\exists! x P(x)$, introduced in Section 1.4, using universal quantifications, existential quantifications, and logical operators.

1.6 Rules of Inference

1.6.1 Introduction

Later in this chapter we will study proofs. Proofs in mathematics are valid arguments that establish the truth of mathematical statements. By an **argument**, we mean a sequence of statements that end with a conclusion. By **valid**, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or **premises**, of the argument. That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false. To deduce new statements from statements we already have, we use rules of inference which are templates for constructing valid arguments. Rules of inference are our basic tools for establishing the truth of statements.

Before we study mathematical proofs, we will look at arguments that involve only compound propositions. We will define what it means for an argument involving compound propositions to be valid. Then we will introduce a collection of rules of inference in propositional logic. These rules of inference are among the most important ingredients in producing valid arguments. After we illustrate how rules of inference are used to produce valid arguments, we will describe some common forms of incorrect reasoning, called **fallacies**, which lead to invalid arguments.

After studying rules of inference in propositional logic, we will introduce rules of inference for quantified statements. We will describe how these rules of inference can be used to produce valid arguments. These rules of inference for statements involving existential and universal quantifiers play an important role in proofs in computer science and mathematics, although they are often used without being explicitly mentioned.

Finally, we will show how rules of inference for propositions and for quantified statements can be combined. These combinations of rule of inference are often used together in complicated arguments.

1.6.2 Valid Arguments in Propositional Logic

Consider the following argument involving propositions (which, by definition, is a sequence of propositions):

“If you have a current password, then you can log onto the network.”

“You have a current password.”

Therefore,

“You can log onto the network.”

We would like to determine whether this is a valid argument. That is, we would like to determine whether the conclusion “You can log onto the network” must be true when the premises “If you have a current password, then you can log onto the network” and “You have a current password” are both true.

Before we discuss the validity of this particular argument, we will look at its form. Use p to represent “You have a current password” and q to represent “You can log onto the network.” Then, the argument has the form

$$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

where $\therefore$ is the symbol that denotes “therefore.”

We know that when p and q are propositional variables, the statement $((p \rightarrow q) \wedge p) \rightarrow q$ is a tautology (see Exercise 12(c) in Section 1.3). In particular, when both $p \rightarrow q$ and p are true, we know that q must also be true. We say this form of argument is **valid** because whenever all its premises (all statements in the argument other than the final one, the conclusion) are true, the conclusion must also be true. Now suppose that both “If you have a current password, then you can log onto the network” and “You have a current password” are true statements. When we replace p by “You have a current password” and q by “You can log onto the network,” it necessarily follows that the conclusion “You can log onto the network” is true. This argument is **valid** because its form is valid. Note that whenever we replace p and q by propositions where $p \rightarrow q$ and p are both true, then q must also be true.

What happens when we replace p and q in this argument form by propositions where not both p and $p \rightarrow q$ are true? For example, suppose that p represents “You have access to the network” and q represents “You can change your grade” and that p is true, but $p \rightarrow q$ is false. The argument we obtain by substituting these values of p and q into the argument form is

“If you have access to the network, then you can change your grade.”
 “You have access to the network.”

$\therefore$ “You can change your grade.”

The argument we obtained is a valid argument, but because one of the premises, namely the first premise, is false, we cannot conclude that the conclusion is true. (Most likely, this conclusion is false.)

In our discussion, to analyze an argument, we replaced propositions by propositional variables. This changed an argument to an **argument form**. We saw that the validity of an argument follows from the validity of the form of the argument. We summarize the terminology used to discuss the validity of arguments with our definition of the key notions.

Definition 1

An *argument* in propositional logic is a sequence of propositions. All but the final proposition in the argument are called *premises* and the final proposition is called the *conclusion*. An argument is *valid* if the truth of all its premises implies that the conclusion is true.

An *argument form* in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is *valid* if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Remark: From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \dots, p_n$ and conclusion q is valid exactly when $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.

The key to showing that an argument in propositional logic is valid is to show that its argument form is valid. Consequently, we would like techniques to show that argument forms are valid. We will now develop methods for accomplishing this task.

1.6.3 Rules of Inference for Propositional Logic

We can always use a truth table to show that an argument form is valid. We do this by showing that whenever the premises are true, the conclusion must also be true. However, this can be a tedious approach. For example, when an argument form involves 10 different propositional variables, to use a truth table to show this argument form is valid requires $2^{10} = 1024$ different rows. Fortunately, we do not have to resort to truth tables. Instead, we can first establish the

validity of some relatively simple argument forms, called **rules of inference**. These rules of inference can be used as building blocks to construct more complicated valid argument forms. We will now introduce the most important rules of inference in propositional logic.

The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ is the basis of the rule of inference called **modus ponens**, or the **law of detachment**. (Modus ponens is Latin for *mode that affirms*.) This tautology leads to the following valid argument form, which we have already seen in our initial discussion about arguments (where, as before, the symbol $\therefore$ denotes “therefore”):

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Using this notation, the hypotheses are written in a column, followed by a horizontal bar, followed by a line that begins with the therefore symbol and ends with the conclusion. In particular, modus ponens tells us that if a conditional statement and the hypothesis of this conditional statement are both true, then the conclusion must also be true. Example 1 illustrates the use of modus ponens.

EXAMPLE 1 Suppose that the conditional statement “If it snows today, then we will go skiing” and its hypothesis, “It is snowing today,” are true. Then, by modus ponens, it follows that the conclusion of the conditional statement, “We will go skiing,” is true. ◀

As we mentioned earlier, a valid argument can lead to an incorrect conclusion if one or more of its premises is false. We illustrate this again in Example 2.

EXAMPLE 2 Determine whether the argument given here is valid and determine whether its conclusion must be true because of the validity of the argument.

$$\begin{array}{l} \text{“If } \sqrt{2} > \frac{3}{2}, \text{ then } (\sqrt{2})^2 > \left(\frac{3}{2}\right)^2. \text{ We know that } \sqrt{2} > \frac{3}{2}. \text{ Consequently,} \\ (\sqrt{2})^2 = 2 > \left(\frac{3}{2}\right)^2 = \frac{9}{4}.” \end{array}$$

Solution: Let p be the proposition “ $\sqrt{2} > \frac{3}{2}$ ” and q the proposition “ $2 > (\frac{3}{2})^2$.” The premises of the argument are $p \rightarrow q$ and p , and q is its conclusion. This argument is valid because it is constructed by using modus ponens, a valid argument form. However, one of its premises, $\sqrt{2} > \frac{3}{2}$, is false. Consequently, we cannot conclude that the conclusion is true. Furthermore, note that the conclusion of this argument is false, because $2 < \frac{9}{4}$. ◀

There are many useful rules of inference for propositional logic. Perhaps the most widely used of these are listed in Table 1. Exercises 13–16, 25, 33, and 34 in Section 1.3 ask for the verifications that these rules of inference are valid argument forms. We now give examples of arguments that use these rules of inference. In each argument, we first use propositional variables to express the propositions in the argument. We then show that the resulting argument form is a rule of inference from Table 1.

EXAMPLE 3 State which rule of inference is the basis of the following argument: “It is below freezing now. Therefore, it is below freezing or raining now.”

Solution: Let p be the proposition “It is below freezing now,” and let q be the proposition “It is raining now.” Then this argument is of the form

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$$

TABLE 1 Rules of Inference.		
Rule of Inference	Tautology	Name
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

This is an argument that uses the addition rule. 

EXAMPLE 4 State which rule of inference is the basis of the following argument: “It is below freezing and raining now. Therefore, it is below freezing now.”

Solution: Let p be the proposition “It is below freezing now,” and let q be the proposition “It is raining now.” This argument is of the form

$$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$$

This argument uses the simplification rule. 

EXAMPLE 5 State which rule of inference is used in the argument:

If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Solution: Let p be the proposition “It is raining today,” let q be the proposition “We will not have a barbecue today,” and let r be the proposition “We will have a barbecue tomorrow.” Then this argument is of the form

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Hence, this argument is a hypothetical syllogism. 

1.6.4 Using Rules of Inference to Build Arguments

When there are many premises, several rules of inference are often needed to show that an argument is valid. This is illustrated by Examples 6 and 7, where the steps of arguments are displayed on separate lines, with the reason for each step explicitly stated. These examples also show how arguments in English can be analyzed using rules of inference.

EXAMPLE 6

Extra Examples 

Show that the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”

Solution: Let p be the proposition “It is sunny this afternoon,” q the proposition “It is colder than yesterday,” r the proposition “We will go swimming,” s the proposition “We will take a canoe trip,” and t the proposition “We will be home by sunset.” Then the premises become $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$. The conclusion is simply t . We need to give a valid argument with premises $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$ and conclusion t .

We construct an argument to show that our premises lead to the desired conclusion as follows.

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Note that we could have used a truth table to show that whenever each of the four hypotheses is true, the conclusion is also true. However, because we are working with five propositional variables, p , q , r , s , and t , such a truth table would have 32 rows. 

EXAMPLE 7

Show that the premises “If you send me an e-mail message, then I will finish writing the program,” “If you do not send me an e-mail message, then I will go to sleep early,” and “If I go to sleep early, then I will wake up feeling refreshed” lead to the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed.”

Solution: Let p be the proposition “You send me an e-mail message,” q the proposition “I will finish writing the program,” r the proposition “I will go to sleep early,” and s the proposition “I

will wake up feeling refreshed.” Then the premises are $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$. The desired conclusion is $\neg q \rightarrow s$. We need to give a valid argument with premises $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$ and conclusion $\neg q \rightarrow s$.

This argument form shows that the premises lead to the desired conclusion.

Step	Reason
1. $p \rightarrow q$	Premise
2. $\neg q \rightarrow \neg p$	Contrapositive of (1)
3. $\neg p \rightarrow r$	Premise
4. $\neg q \rightarrow r$	Hypothetical syllogism using (2) and (3)
5. $r \rightarrow s$	Premise
6. $\neg q \rightarrow s$	Hypothetical syllogism using (4) and (5)

1.6.5 Resolution

Computer programs have been developed to automate the task of reasoning and proving theorems. Many of these programs make use of a rule of inference known as **resolution**. This rule of inference is based on the tautology

Links

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r).$$

(Exercise 34 in Section 1.3 asks for the verification that this is a tautology.) The final disjunction in the resolution rule, $q \vee r$, is called the **resolvent**. When we let $q = r$ in this tautology, we obtain $(p \vee q) \wedge (\neg p \vee q) \rightarrow q$. Furthermore, when we let $r = \mathbf{F}$, we obtain $(p \vee q) \wedge (\neg p) \rightarrow q$ (because $q \vee \mathbf{F} \equiv q$), which is the tautology on which the rule of disjunctive syllogism is based.

EXAMPLE 8

Use resolution to show that the hypotheses “Jasmine is skiing or it is not snowing” and “It is snowing or Bart is playing hockey” imply that “Jasmine is skiing or Bart is playing hockey.”

Extra Examples

Solution: Let p be the proposition “It is snowing,” q the proposition “Jasmine is skiing,” and r the proposition “Bart is playing hockey.” We can represent the hypotheses as $\neg p \vee q$ and $p \vee r$, respectively. Using resolution, the proposition $q \vee r$, “Jasmine is skiing or Bart is playing hockey,” follows.

Resolution plays an important role in programming languages based on the rules of logic, such as Prolog (where resolution rules for quantified statements are applied). Furthermore, it can be used to build automatic theorem proving systems. To construct proofs in propositional logic using resolution as the only rule of inference, the hypotheses and the conclusion must be expressed as **clauses**, where a clause is a disjunction of variables or negations of these variables. We can replace a statement in propositional logic that is not a clause by one or more equivalent statements that are clauses. For example, suppose we have a statement of the form $p \vee (q \wedge r)$. Because $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$, we can replace the single statement $p \vee (q \wedge r)$ by two statements $p \vee q$ and $p \vee r$, each of which is a clause. We can replace a statement of the form $\neg(p \vee q)$ by the two statements $\neg p$ and $\neg q$ because De Morgan’s law tells us that $\neg(p \vee q) \equiv \neg p \wedge \neg q$. We can also replace a conditional statement $p \rightarrow q$ with the equivalent disjunction $\neg p \vee q$.

EXAMPLE 9

Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

Solution: We can rewrite the premises $(p \wedge q) \vee r$ as two clauses, $p \vee r$ and $q \vee r$. We can also replace $r \rightarrow s$ by the equivalent clause $\neg r \vee s$. Using the two clauses $p \vee r$ and $\neg r \vee s$, we can use resolution to conclude $p \vee s$.

1.6.6 Fallacies

Several common fallacies arise in incorrect arguments. These fallacies resemble rules of inference, but are based on contingencies rather than tautologies. These are discussed here to show the distinction between correct and incorrect reasoning.

Links ▶ The proposition $((p \rightarrow q) \wedge q) \rightarrow p$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology. In other words, they treat the argument with premises $p \rightarrow q$ and q and conclusion p as a valid argument form, which it is not. This type of incorrect reasoning is called the **fallacy of affirming the conclusion**.

EXAMPLE 10 Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

Solution: Let p be the proposition “You did every problem in this book.” Let q be the proposition “You learned discrete mathematics.” Then this argument is of the form: if $p \rightarrow q$ and q , then p . This is an example of an incorrect argument using the fallacy of affirming the conclusion. Indeed, it is possible for you to learn discrete mathematics in some way other than by doing every problem in this book. (You may learn discrete mathematics by reading, listening to lectures, doing some, but not all, the problems in this book, and so on.) ◀

The proposition $((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this incorrectly as a rule of inference. This type of incorrect reasoning is called the **fallacy of denying the hypothesis**.

EXAMPLE 11 Let p and q be as in Example 10. If the conditional statement $p \rightarrow q$ is true, and $\neg p$ is true, is it correct to conclude that $\neg q$ is true? In other words, is it correct to assume that you did not learn discrete mathematics if you did not do every problem in the book, assuming that if you do every problem in this book, then you will learn discrete mathematics?

Solution: It is possible that you learned discrete mathematics even if you did not do every problem in this book. This incorrect argument is of the form $p \rightarrow q$ and $\neg p$ imply $\neg q$, which is an example of the fallacy of denying the hypothesis. ◀

1.6.7 Rules of Inference for Quantified Statements

We have discussed rules of inference for propositions. We will now describe some important rules of inference for statements involving quantifiers. These rules of inference are used extensively in mathematical arguments, often without being explicitly mentioned.

Universal instantiation is the rule of inference used to conclude that $P(c)$ is true, where c is a particular member of the domain, given the premise $\forall xP(x)$. Universal instantiation is used when we conclude from the statement “All women are wise” that “Lisa is wise,” where Lisa is a member of the domain of all women.

Universal generalization is the rule of inference that states that $\forall xP(x)$ is true, given the premise that $P(c)$ is true for all elements c in the domain. Universal generalization is used when we show that $\forall xP(x)$ is true by taking an arbitrary element c from the domain and showing that $P(c)$ is true. The element c that we select must be an arbitrary, and not a specific, element of the domain. That is, when we assert from $\forall xP(x)$ the existence of an element c in the domain,

TABLE 2 Rules of Inference for Quantified Statements.

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall xP(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$	Universal generalization
$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$	Existential generalization

we have no control over c and cannot make any other assumptions about c other than it comes from the domain. Universal generalization is used implicitly in many proofs in mathematics and is seldom mentioned explicitly. However, the error of adding unwarranted assumptions about the arbitrary element c when universal generalization is used is all too common in incorrect reasoning.

Existential instantiation is the rule that allows us to conclude that there is an element c in the domain for which $P(c)$ is true if we know that $\exists xP(x)$ is true. We cannot select an arbitrary value of c here, but rather it must be a c for which $P(c)$ is true. Usually we have no knowledge of what c is, only that it exists. Because it exists, we may give it a name (c) and continue our argument.

Existential generalization is the rule of inference that is used to conclude that $\exists xP(x)$ is true when a particular element c with $P(c)$ true is known. That is, if we know one element c in the domain for which $P(c)$ is true, then we know that $\exists xP(x)$ is true.

We summarize these rules of inference in Table 2. We will illustrate how some of these rules of inference for quantified statements are used in Examples 12 and 13.

EXAMPLE 12 Show that the premises “Everyone in this discrete mathematics class has taken a course in computer science” and “Marla is a student in this class” imply the conclusion “Marla has taken a course in computer science.”

Extra Examples ➤

Solution: Let $D(x)$ denote “ x is in this discrete mathematics class,” and let $C(x)$ denote “ x has taken a course in computer science.” Then the premises are $\forall x(D(x) \rightarrow C(x))$ and $D(\text{Marla})$. The conclusion is $C(\text{Marla})$.

The following steps can be used to establish the conclusion from the premises.

Step	Reason
1. $\forall x(D(x) \rightarrow C(x))$	Premise
2. $D(\text{Marla}) \rightarrow C(\text{Marla})$	Universal instantiation from (1)
3. $D(\text{Marla})$	Premise
4. $C(\text{Marla})$	Modus ponens from (2) and (3)

EXAMPLE 13 Show that the premises “A student in this class has not read the book,” and “Everyone in this class passed the first exam” imply the conclusion “Someone who passed the first exam has not read the book.”

Solution: Let $C(x)$ be “ x is in this class,” $B(x)$ be “ x has read the book,” and $P(x)$ be “ x passed the first exam.” The premises are $\exists x(C(x) \wedge \neg B(x))$ and $\forall x(C(x) \rightarrow P(x))$. The conclusion is $\exists x(P(x) \wedge \neg B(x))$. These steps can be used to establish the conclusion from the premises.

Step	Reason
1. $\exists x(C(x) \wedge \neg B(x))$	Premise
2. $C(a) \wedge \neg B(a)$	Existential instantiation from (1)
3. $C(a)$	Simplification from (2)
4. $\forall x(C(x) \rightarrow P(x))$	Premise
5. $C(a) \rightarrow P(a)$	Universal instantiation from (4)
6. $P(a)$	Modus ponens from (3) and (5)
7. $\neg B(a)$	Simplification from (2)
8. $P(a) \wedge \neg B(a)$	Conjunction from (6) and (7)
9. $\exists x(P(x) \wedge \neg B(x))$	Existential generalization from (8)

1.6.8 Combining Rules of Inference for Propositions and Quantified Statements

We have developed rules of inference both for propositions and for quantified statements. Note that in our arguments in Examples 12 and 13 we used both universal instantiation, a rule of inference for quantified statements, and modus ponens, a rule of inference for propositional logic. We will often need to use this combination of rules of inference. Because universal instantiation and modus ponens are used so often together, this combination of rules is sometimes called **universal modus ponens**. This rule tells us that if $\forall x(P(x) \rightarrow Q(x))$ is true, and if $P(a)$ is true for a particular element a in the domain of the universal quantifier, then $Q(a)$ must also be true. To see this, note that by universal instantiation, $P(a) \rightarrow Q(a)$ is true. Then, by modus ponens, $Q(a)$ must also be true. We can describe universal modus ponens as follows:

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ P(a), \text{ where } a \text{ is a particular element in the domain} \\ \hline \therefore Q(a) \end{array}$$

Universal modus ponens is commonly used in mathematical arguments. This is illustrated in Example 14.

EXAMPLE 14 Assume that “For all positive integers n , if n is greater than 4, then n^2 is less than 2^n ” is true. Use universal modus ponens to show that $100^2 < 2^{100}$.

Solution: Let $P(n)$ denote “ $n > 4$ ” and $Q(n)$ denote “ $n^2 < 2^n$.” The statement “For all positive integers n , if n is greater than 4, then n^2 is less than 2^n ” can be represented by $\forall n(P(n) \rightarrow Q(n))$, where the domain consists of all positive integers. We are assuming that $\forall n(P(n) \rightarrow Q(n))$ is true. Note that $P(100)$ is true because $100 > 4$. It follows by universal modus ponens that $Q(100)$ is true, namely, that $100^2 < 2^{100}$.

Another useful combination of a rule of inference from propositional logic and a rule of inference for quantified statements is **universal modus tollens**. Universal modus tollens combines universal instantiation and modus tollens and can be expressed in the following way:

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ \neg Q(a), \text{ where } a \text{ is a particular element in the domain} \\ \hline \therefore \neg P(a) \end{array}$$

The verification of universal modus tollens is left as Exercise 25. Exercises 26–29 develop additional combinations of rules of inference in propositional logic and quantified statements.

Exercises

- Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If Socrates is human, then Socrates is mortal.
Socrates is human.

∴ Socrates is mortal.

- Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If George does not have eight legs, then he is not a spider.
George is a spider.

∴ George has eight legs.

- What rule of inference is used in each of these arguments?
 - Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
 - Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
 - If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
 - If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
 - If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.
- What rule of inference is used in each of these arguments?
 - Kangaroos live in Australia and are marsupials. Therefore, kangaroos are marsupials.
 - It is either hotter than 100 degrees today or the pollution is dangerous. It is less than 100 degrees outside today. Therefore, the pollution is dangerous.
 - Linda is an excellent swimmer. If Linda is an excellent swimmer, then she can work as a lifeguard. Therefore, Linda can work as a lifeguard.
 - Steve will work at a computer company this summer. Therefore, this summer Steve will work at a computer company or he will be a beach bum.
 - If I work all night on this homework, then I can answer all the exercises. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material.

- Use rules of inference to show that the hypotheses “Randy works hard,” “If Randy works hard, then he is a dull boy,” and “If Randy is a dull boy, then he will not get the job” imply the conclusion “Randy will not get the job.”
- Use rules of inference to show that the hypotheses “If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on,” “If the sailing race is held, then the trophy will be awarded,” and “The trophy was not awarded” imply the conclusion “It rained.”
- What rules of inference are used in this famous argument? “All men are mortal. Socrates is a man. Therefore, Socrates is mortal.”
- What rules of inference are used in this argument? “No man is an island. Manhattan is an island. Therefore, Manhattan is not a man.”
- For each of these collections of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.
 - “If I take the day off, it either rains or snows.” “I took Tuesday off or I took Thursday off.” “It was sunny on Tuesday.” “It did not snow on Thursday.”
 - “If I eat spicy foods, then I have strange dreams.” “I have strange dreams if there is thunder while I sleep.” “I did not have strange dreams.”
 - “I am either clever or lucky.” “I am not lucky.” “If I am lucky, then I will win the lottery.”
 - “Every computer science major has a personal computer.” “Ralph does not have a personal computer.” “Ann has a personal computer.”
 - “What is good for corporations is good for the United States.” “What is good for the United States is good for you.” “What is good for corporations is for you to buy lots of stuff.”
 - “All rodents gnaw their food.” “Mice are rodents.” “Rabbits do not gnaw their food.” “Bats are not rodents.”
- For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.
 - “If I play hockey, then I am sore the next day.” “I use the whirlpool if I am sore.” “I did not use the whirlpool.”
 - “If I work, it is either sunny or partly sunny.” “I worked last Monday or I worked last Friday.” “It was not sunny on Tuesday.” “It was not partly sunny on Friday.”
 - “All insects have six legs.” “Dragonflies are insects.” “Spiders do not have six legs.” “Spiders eat dragonflies.”

- d) “Every student has an Internet account.” “Homer does not have an Internet account.” “Maggie has an Internet account.”
- e) “All foods that are healthy to eat do not taste good.” “Tofu is healthy to eat.” “You only eat what tastes good.” “You do not eat tofu.” “Cheeseburgers are not healthy to eat.”
- f) “I am either dreaming or hallucinating.” “I am not dreaming.” “If I am hallucinating, I see elephants running down the road.”
11. Show that the argument form with premises $p_1, p_2, \dots, p_n$ and conclusion $q \rightarrow r$ is valid if the argument form with premises $p_1, p_2, \dots, p_n, q$, and conclusion r is valid.
12. Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow p$, and $\neg s$ and conclusion $q \rightarrow r$ is valid by first using Exercise 11 and then using rules of inference from Table 1.
13. For each of these arguments, explain which rules of inference are used for each step.
- “Doug, a student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job.”
 - “Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution.”
 - “Each of the 93 students in this class owns a personal computer. Everyone who owns a personal computer can use a word processing program. Therefore, Zeke, a student in this class, can use a word processing program.”
 - “Everyone in New Jersey lives within 50 miles of the ocean. Someone in New Jersey has never seen the ocean. Therefore, someone who lives within 50 miles of the ocean has never seen the ocean.”
14. For each of these arguments, explain which rules of inference are used for each step.
- “Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class has gotten a speeding ticket.”
 - “Each of five roommates, Melissa, Aaron, Ralph, Veneesha, and Keeshawn, has taken a course in discrete mathematics. Every student who has taken a course in discrete mathematics can take a course in algorithms. Therefore, all five roommates can take a course in algorithms next year.”
 - “All movies produced by John Sayles are wonderful. John Sayles produced a movie about coal miners. Therefore, there is a wonderful movie about coal miners.”
 - “There is someone in this class who has been to France. Everyone who goes to France visits the Louvre. Therefore, someone in this class has visited the Louvre.”
15. For each of these arguments determine whether the argument is correct or incorrect and explain why.
- All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.
 - Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.
 - All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.
 - Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.
16. For each of these arguments determine whether the argument is correct or incorrect and explain why.
- Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university.
 - A convertible car is fun to drive. Isaac’s car is not a convertible. Therefore, Isaac’s car is not fun to drive.
 - Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie.
 - All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore, Hamilton sets at least a dozen traps.
17. What is wrong with this argument? Let $H(x)$ be “ x is happy.” Given the premise $\exists x H(x)$, we conclude that $H(\text{Lola})$. Therefore, Lola is happy.
18. What is wrong with this argument? Let $S(x, y)$ be “ x is shorter than y .” Given the premise $\exists s S(s, \text{Max})$, it follows that $S(\text{Max}, \text{Max})$. Then by existential generalization it follows that $\exists x S(x, x)$, so that someone is shorter than himself.
19. Determine whether each of these arguments is valid. If an argument is correct, what rule of inference is being used? If it is not, what logical error occurs?
- If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$. Then $n > 1$.
 - If n is a real number with $n > 3$, then $n^2 > 9$. Suppose that $n^2 \leq 9$. Then $n \leq 3$.
 - If n is a real number with $n > 2$, then $n^2 > 4$. Suppose that $n \leq 2$. Then $n^2 \leq 4$.
20. Determine whether these are valid arguments.
- If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.
 - If $x^2 \neq 0$, where x is a real number, then $x \neq 0$. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$.
21. Which rules of inference are used to establish the conclusion of Lewis Carroll’s argument described in Example 26 of Section 1.4?
22. Which rules of inference are used to establish the conclusion of Lewis Carroll’s argument described in Example 27 of Section 1.4?

23. Identify the error or errors in this argument that supposedly shows that if $\exists xP(x) \wedge \exists xQ(x)$ is true then $\exists x(P(x) \wedge Q(x))$ is true.
1. $\exists xP(x) \vee \exists xQ(x)$ Premise
 2. $\exists xP(x)$ Simplification from (1)
 3. $P(c)$ Existential instantiation from (2)
 4. $\exists xQ(x)$ Simplification from (1)
 5. $Q(c)$ Existential instantiation from (4)
 6. $P(c) \wedge Q(c)$ Conjunction from (3) and (5)
 7. $\exists x(P(x) \wedge Q(x))$ Existential generalization
24. Identify the error or errors in this argument that supposedly shows that if $\forall x(P(x) \vee Q(x))$ is true then $\forall xP(x) \vee \forall xQ(x)$ is true.
1. $\forall x(P(x) \vee Q(x))$ Premise
 2. $P(c) \vee Q(c)$ Universal instantiation from (1)
 3. $P(c)$ Simplification from (2)
 4. $\forall xP(x)$ Universal generalization from (3)
 5. $Q(c)$ Simplification from (2)
 6. $\forall xQ(x)$ Universal generalization from (5)
 7. $\forall x(P(x) \vee \forall xQ(x))$ Conjunction from (4) and (6)
25. Justify the rule of universal modus tollens by showing that the premises $\forall x(P(x) \rightarrow Q(x))$ and $\neg Q(a)$ for a particular element a in the domain, imply $\neg P(a)$.
26. Justify the rule of **universal transitivity**, which states that if $\forall x(P(x) \rightarrow Q(x))$ and $\forall x(Q(x) \rightarrow R(x))$ are true, then $\forall x(P(x) \rightarrow R(x))$ is true, where the domains of all quantifiers are the same.
27. Use rules of inference to show that if $\forall x(P(x) \rightarrow (Q(x) \wedge S(x)))$ and $\forall x(P(x) \wedge R(x))$ are true, then $\forall x(R(x) \wedge S(x))$ is true.
28. Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$ and $\forall x((\neg P(x) \wedge Q(x)) \rightarrow R(x))$ are true, then $\forall x(\neg R(x) \rightarrow P(x))$ is also true, where the domains of all quantifiers are the same.
29. Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$, $\forall x(\neg Q(x) \vee S(x))$, $\forall x(R(x) \rightarrow \neg S(x))$, and $\exists x\neg P(x)$ are true, then $\exists x\neg R(x)$ is true.
30. Use resolution to show the hypotheses “Allen is a bad boy or Hillary is a good girl” and “Allen is a good boy or David is happy” imply the conclusion “Hillary is a good girl or David is happy.”
31. Use resolution to show that the hypotheses “It is not raining or Yvette has her umbrella,” “Yvette does not have her umbrella or she does not get wet,” and “It is raining or Yvette does not get wet” imply that “Yvette does not get wet.”
32. Show that the equivalence $p \wedge \neg p \equiv \mathbf{F}$ can be derived using resolution together with the fact that a conditional statement with a false hypothesis is true. [Hint: Let $q = r = \mathbf{F}$ in resolution.]
33. Use resolution to show that the compound proposition $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is not satisfiable.
- *34. The Logic Problem, taken from *WFF'N PROOF, The Game of Logic*, has these two assumptions:
1. “Logic is difficult or not many students like logic.”
 2. “If mathematics is easy, then logic is not difficult.”
- By translating these assumptions into statements involving propositional variables and logical connectives, determine whether each of the following are valid conclusions of these assumptions:
- a) That mathematics is not easy, if many students like logic.
 - b) That not many students like logic, if mathematics is not easy.
 - c) That mathematics is not easy or logic is difficult.
 - d) That logic is not difficult or mathematics is not easy.
 - e) That if not many students like logic, then either mathematics is not easy or logic is not difficult.
- *35. Determine whether this argument, taken from Kalish and Montague [KaMo64], is valid.
- If Superman were able and willing to prevent evil, he would do so. If Superman were unable to prevent evil, he would be impotent; if he were unwilling to prevent evil, he would be malevolent. Superman does not prevent evil. If Superman exists, he is neither impotent nor malevolent. Therefore, Superman does not exist.

1.7 Introduction to Proofs

1.7.1 Introduction

In this section we introduce the notion of a proof and describe methods for constructing proofs. A proof is a valid argument that establishes the truth of a mathematical statement. A proof can use the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven theorems. Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.

In our discussion we move from formal proofs of theorems toward more informal proofs. The arguments we introduced in Section 1.6 to show that statements involving propositions and quantified statements are true were formal proofs, where all steps were supplied, and the rules for each step in the argument were given. However, formal proofs of useful theorems can

be extremely long and hard to follow. In practice, the proofs of theorems designed for human consumption are almost always **informal proofs**, where more than one rule of inference may be used in each step, where steps may be skipped, where the axioms being assumed and the rules of inference used are not explicitly stated. Informal proofs can often explain to humans why theorems are true, while computers are perfectly happy producing formal proofs using automated reasoning systems.

The methods of proof discussed in this chapter are important not only because they are used to prove mathematical theorems, but also for their many applications to computer science. These applications include verifying that computer programs are correct, establishing that operating systems are secure, making inferences in artificial intelligence, showing that system specifications are consistent, and so on. Consequently, understanding the techniques used in proofs is essential both in mathematics and in computer science.

1.7.2 Some Terminology

Links 

Formally, a **theorem** is a statement that can be shown to be true. In mathematical writing, the term theorem is usually reserved for a statement that is considered at least somewhat important. Less important theorems sometimes are called **propositions**. (Theorems can also be referred to as **facts** or **results**.) A theorem may be the universal quantification of a conditional statement with one or more premises and a conclusion. However, it may be some other type of logical statement, as the examples later in this chapter will show. We demonstrate that a theorem is true with a **proof**. A proof is a valid argument that establishes the truth of a theorem. The statements used in a proof can include **axioms** (or **postulates**), which are statements we assume to be true (for example, the axioms for the real numbers, given in Appendix 1, and the axioms of plane geometry), the premises, if any, of the theorem, and previously proven theorems. Axioms may be stated using primitive terms that do not require definition, but all other terms used in theorems and their proofs must be defined. Rules of inference, together with definitions of terms, are used to draw conclusions from other assertions, tying together the steps of a proof. In practice, the final step of a proof is usually just the conclusion of the theorem. However, for clarity, we will often recap the statement of the theorem as the final step of a proof.

A less important theorem that is helpful in the proof of other results is called a **lemma** (plural *lemmas* or *lemmata*). Complicated proofs are usually easier to understand when they are proved using a series of lemmas, where each lemma is proved individually. A **corollary** is a theorem that can be established directly from a theorem that has been proved. A **conjecture** is a statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert. When a proof of a conjecture is found, the conjecture becomes a theorem. Many times conjectures are shown to be false, so they are not theorems.

1.7.3 Understanding How Theorems Are Stated

Extra Examples 

Before we introduce methods for proving theorems, we need to understand how many mathematical theorems are stated. Many theorems assert that a property holds for all elements in a domain, such as the integers or the real numbers. Although the precise statement of such theorems needs to include a universal quantifier, the standard convention in mathematics is to omit it. For example, the statement

“If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$ ”

really means

“For all positive real numbers x and y , if $x > y$, then $x^2 > y^2$.”

Furthermore, when theorems of this type are proved, the first step of the proof usually involves selecting a general element of the domain. Subsequent steps show that this element has the property in question. Finally, universal generalization implies that the theorem holds for all members of the domain.

1.7.4 Methods of Proving Theorems

Assessment

Proving mathematical theorems can be difficult. To construct proofs we need all available ammunition, including a powerful battery of different proof methods. These methods provide the overall approach and strategy of proofs. Understanding these methods is a key component of learning how to read and construct mathematical proofs. Once we have chosen a proof method, we use axioms, definitions of terms, previously proved results, and rules of inference to complete the proof. Note that in this book we will always assume the axioms for real numbers found in Appendix 1. We will also assume the usual axioms whenever we prove a result about geometry. When you construct your own proofs, be careful not to use anything but these axioms, definitions, and previously proved results as facts!

To prove a theorem of the form $\forall x(P(x) \rightarrow Q(x))$, our goal is to show that $P(c) \rightarrow Q(c)$ is true, where c is an arbitrary element of the domain, and then apply universal generalization. In this proof, we need to show that a conditional statement is true. Because of this, we now focus on methods that show that conditional statements are true. Recall that $p \rightarrow q$ is true unless p is true but q is false. Note that to prove the statement $p \rightarrow q$, we need only show that q is true if p is true. The following discussion will give the most common techniques for proving conditional statements. Later we will discuss methods for proving other types of statements. In this section, and in Section 1.8, we will develop a large arsenal of proof techniques that can be used to prove a wide variety of theorems.

When you read proofs, you will often find the words “obviously” or “clearly.” These words indicate that steps have been omitted that the author expects the reader to be able to fill in. Unfortunately, this assumption is often not warranted and readers are not at all sure how to fill in the gaps. We will assiduously try to avoid using these words and try not to omit too many steps. However, if we included all steps in proofs, our proofs would often be excruciatingly long.

1.7.5 Direct Proofs

A **direct proof** of a conditional statement $p \rightarrow q$ is constructed when the first step is the assumption that p is true; subsequent steps are constructed using rules of inference, with the final step showing that q must also be true. A direct proof shows that a conditional statement $p \rightarrow q$ is true by showing that if p is true, then q must also be true, so that the combination p true and q false never occurs. In a direct proof, we assume that p is true and use axioms, definitions, and previously proven theorems, together with rules of inference, to show that q must also be true. You will find that direct proofs of many results are quite straightforward. Starting with the hypothesis and leading to the conclusion, the way forward is essentially dictated by the premises available at that step. However, direct proofs sometimes require particular insights and can be quite tricky. The first direct proofs we present here are quite straightforward; later in the text you will see some that require some insight.

We will provide examples of several different direct proofs. Before we give the first example, we need to define some terminology.

Definition 1

The integer n is *even* if there exists an integer k such that $n = 2k$, and n is *odd* if there exists an integer k such that $n = 2k + 1$. (Note that every integer is either even or odd, and no integer is both even and odd.) Two integers have the *same parity* when both are even or both are odd; they have *opposite parity* when one is even and the other is odd.

EXAMPLE 1 Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Extra
Examples

Solution: Note that this theorem states $\forall n P(n) \rightarrow Q(n)$, where $P(n)$ is “ n is an odd integer” and $Q(n)$ is “ n^2 is odd.” As we have said, we will follow the usual convention in mathematical proofs by showing that $P(n)$ implies $Q(n)$, and not explicitly using universal instantiation. To begin a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that n is odd. By the definition of an odd integer, it follows that $n = 2k + 1$, where k is some integer. We want to show that n^2 is also odd. We can square both sides of the equation $n = 2k + 1$ to obtain a new equation that expresses n^2 . When we do this, we find that $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. By the definition of an odd integer, we can conclude that n^2 is an odd integer (it is one more than twice an integer). Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer. ◀

EXAMPLE 2 Give a direct proof that if m and n are both perfect squares, then mn is also a perfect square. (An integer a is a **perfect square** if there is an integer b such that $a = b^2$.)

Solution: To produce a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that m and n are both perfect squares. By the definition of a perfect square, it follows that there are integers s and t such that $m = s^2$ and $n = t^2$. The goal of the proof is to show that mn must also be a perfect square when m and n are; looking ahead we see how we can show this by substituting s^2 for m and t^2 for n into mn . This tells us that $mn = s^2 t^2$. Hence, $mn = s^2 t^2 = (ss)(tt) = (st)(st) = (st)^2$, using commutativity and associativity of multiplication. By the definition of perfect square, it follows that mn is also a perfect square, because it is the square of st , which is an integer. We have proved that if m and n are both perfect squares, then mn is also a perfect square. ◀

1.7.6 Proof by Contraposition

Direct proofs lead from the premises of a theorem to the conclusion. They begin with the premises, continue with a sequence of deductions, and end with the conclusion. However, we will see that attempts at direct proofs often reach dead ends. We need other methods of proving theorems of the form $\forall x (P(x) \rightarrow Q(x))$. Proofs of theorems of this type that are not direct proofs, that is, that do not start with the premises and end with the conclusion, are called **indirect proofs**.

An extremely useful type of indirect proof is known as **proof by contraposition**. Proofs by contraposition make use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive, $\neg q \rightarrow \neg p$. This means that the conditional statement $p \rightarrow q$ can be proved by showing that its contrapositive, $\neg q \rightarrow \neg p$, is true. In a proof by contraposition of $p \rightarrow q$, we take $\neg q$ as a premise, and using axioms, definitions, and previously proven theorems, together with rules of inference, we show that $\neg p$ must follow. We will illustrate proof by contraposition with two examples. These examples show that proof by contraposition can succeed when we cannot easily find a direct proof.

EXAMPLE 3 Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Extra
Examples

Solution: We first attempt a direct proof. To construct a direct proof, we first assume that $3n + 2$ is an odd integer. From the definition of an odd integer, we know that $3n + 2 = 2k + 1$ for some integer k . Can we use this fact to show that n is odd? We see that $3n + 1 = 2k$, but there does not seem to be any direct way to conclude that n is odd. Because our attempt at a direct proof failed, we next try a proof by contraposition.

The first step in a proof by contraposition is to assume that the conclusion of the conditional statement “If $3n + 2$ is odd, then n is odd” is false; namely, assume that n is even. Then, by the definition of an even integer, $n = 2k$ for some integer k . Substituting $2k$ for n , we find that $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. This tells us that $3n + 2$ is even (because it is a multiple of 2), and therefore not odd. This is the negation of the premise of the theorem. Because the negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true. Our proof by contraposition succeeded; we have proved the theorem “If $3n + 2$ is odd, then n is odd.” ◀

EXAMPLE 4 Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Solution: Because there is no obvious way of showing that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ directly from the equation $n = ab$, where a and b are positive integers, we attempt a proof by contraposition.

The first step in a proof by contraposition is to assume that the conclusion of the conditional statement “If $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ ” is false. That is, we assume that the statement $(a \leq \sqrt{n}) \vee (b \leq \sqrt{n})$ is false. Using the meaning of disjunction together with De Morgan’s law, we see that this implies that both $a \leq \sqrt{n}$ and $b \leq \sqrt{n}$ are false. This implies that $a > \sqrt{n}$ and $b > \sqrt{n}$. We can multiply these inequalities together (using the fact that if $0 < s < t$ and $0 < u < v$, then $su < tv$) to obtain $ab > \sqrt{n} \cdot \sqrt{n} = n$. This shows that $ab \neq n$, which contradicts the statement $n = ab$.

Because the negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true. Our proof by contraposition succeeded; we have proved that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. ◀

VACUOUS AND TRIVIAL PROOFS We can quickly prove that a conditional statement $p \rightarrow q$ is true when we know that p is false, because $p \rightarrow q$ must be true when p is false. Consequently, if we can show that p is false, then we have a proof, called a **vacuous proof**, of the conditional statement $p \rightarrow q$. Vacuous proofs are often used to establish special cases of theorems that state that a conditional statement is true for all positive integers [i.e., a theorem of the kind $\forall n P(n)$, where $P(n)$ is a propositional function]. Proof techniques for theorems of this kind will be discussed in Section 5.1.

EXAMPLE 5 Show that the proposition $P(0)$ is true, where $P(n)$ is “If $n > 1$, then $n^2 > n$ ” and the domain consists of all integers.

Solution: Note that $P(0)$ is “If $0 > 1$, then $0^2 > 0$.” We can show $P(0)$ using a vacuous proof. Indeed, the hypothesis $0 > 1$ is false. This tells us that $P(0)$ is automatically true. ◀

Remark: The fact that the conclusion of this conditional statement, $0^2 > 0$, is false is irrelevant to the truth value of the conditional statement, because a conditional statement with a false hypothesis is guaranteed to be true.

EXAMPLE 6 Prove that if n is an integer with $10 \leq n \leq 15$ which is a perfect square, then n is also a perfect cube.

Solution: Note that there are no perfect squares n with $10 \leq n \leq 15$, because $3^2 = 9$ and $4^2 = 16$. Hence, the statement that n is an integer with $10 \leq n \leq 15$ which is a perfect square is false for all integers n . Consequently, the statement to be proved is true for all integers n . ◀

We can also quickly prove a conditional statement $p \rightarrow q$ if we know that the conclusion q is true. By showing that q is true, it follows that $p \rightarrow q$ must also be true. A proof of $p \rightarrow q$ that uses the fact that q is true is called a **trivial proof**. Trivial proofs are often important when special cases of theorems are proved (see the discussion of proof by cases in Section 1.8) and in mathematical induction, which is a proof technique discussed in Section 5.1.

EXAMPLE 7 Let $P(n)$ be “If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$,” where the domain consists of all nonnegative integers. Show that $P(0)$ is true.

Solution: The proposition $P(0)$ is “If $a \geq b$, then $a^0 \geq b^0$.” Because $a^0 = b^0 = 1$, the conclusion of the conditional statement “If $a \geq b$, then $a^0 \geq b^0$ ” is true. Hence, this conditional statement, which is $P(0)$, is true. This is an example of a trivial proof. Note that the hypothesis, which is the statement “ $a \geq b$,” was not needed in this proof. 

A LITTLE PROOF STRATEGY We have described two important approaches for proving theorems of the form $\forall x(P(x) \rightarrow Q(x))$: direct proof and proof by contraposition. We have also given examples that show how each is used. However, when you are presented with a theorem of the form $\forall x(P(x) \rightarrow Q(x))$, which method should you use to attempt to prove it? We will provide a few rules of thumb here; in Section 1.8 we will discuss proof strategy at greater length.

When you want to prove a statement of the form $\forall x(P(x) \rightarrow Q(x))$, first evaluate whether a direct proof looks promising. Begin by expanding the definitions in the hypotheses. Start to reason using these hypotheses, together with axioms and available theorems. If a direct proof does not seem to go anywhere, for instance when there is no clear way to use hypotheses as in Examples 3 and 4 to reach the conclusion, try the same thing with a proof by contraposition.

(Hypotheses such as x is irrational or $x \neq 0$ that are difficult to reason from are a clue that an indirect proof might be your best bet.)

Recall that in a proof by contraposition you assume that the conclusion of the conditional statement is false and use a direct proof to show this implies that the hypothesis must be false. Often, you will find that a proof by contraposition is easily constructed from the negation of the conclusion. We illustrate this strategy in Examples 7 and 8. In each example, note how straightforward a proof by contraposition is, while there is no clear way to provide a direct proof.

Before we present our next example, we need a definition.

Definition 2

The real number r is *rational* if there exist integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called *irrational*.

EXAMPLE 8 Prove that the sum of two rational numbers is rational. (Note that if we include the implicit quantifiers here, the theorem we want to prove is “For every real number r and every real number s , if r and s are rational numbers, then $r + s$ is rational.”)

Extra Examples 

Solution: We first attempt a direct proof. To begin, suppose that r and s are rational numbers. From the definition of a rational number, it follows that there are integers p and q , with $q \neq 0$, such that $r = p/q$, and integers t and u , with $u \neq 0$, such that $s = t/u$. Can we use this information to show that $r + s$ is rational? That is, can we find integers v and w such that $r + s = v/w$ and $w \neq 0$?

With the goal of finding these integers v and w , we add $r = p/q$ and $s = t/u$, using qu as the common denominator. We find that

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu}.$$

Because $q \neq 0$ and $u \neq 0$, it follows that $qu \neq 0$. Consequently, we have expressed $r + s$ as the ratio of two integers, $v = pu + qt$ and $w = qu$, where $w \neq 0$. This means that $r + s$ is rational. We have proved that the sum of two rational numbers is rational; our attempt to find a direct proof succeeded. 

EXAMPLE 9 Prove that if n is an integer and n^2 is odd, then n is odd.

Solution: We first attempt a direct proof. Suppose that n is an integer and n^2 is odd. From the definition of an odd integer, there exists an integer k such that $n^2 = 2k + 1$. Can we use this information to show that n is odd? There seems to be no obvious approach to show that n is odd because solving for n produces the equation $n = \pm\sqrt{2k + 1}$, which is not terribly useful.

Because this attempt to use a direct proof did not bear fruit, we next attempt a proof by contraposition. We take as our hypothesis the statement that n is not odd. Because every integer is odd or even, this means that n is even. This implies that there exists an integer k such that $n = 2k$. To prove the theorem, we need to show that this hypothesis implies the conclusion that n^2 is not odd, that is, that n^2 is even. Can we use the equation $n = 2k$ to achieve this? By squaring both sides of this equation, we obtain $n^2 = 4k^2 = 2(2k^2)$, which implies that n^2 is also even because $n^2 = 2t$, where $t = 2k^2$. We have proved that if n is an integer and n^2 is odd, then n is odd. Our attempt to find a proof by contraposition succeeded. 

1.7.7 Proofs by Contradiction

Suppose we want to prove that a statement p is true. Furthermore, suppose that we can find a contradiction q such that $\neg p \rightarrow q$ is true. Because q is false, but $\neg p \rightarrow q$ is true, we can conclude that $\neg p$ is false, which means that p is true. How can we find a contradiction q that might help us prove that p is true in this way?

Because the statement $r \wedge \neg r$ is a contradiction whenever r is a proposition, we can prove that p is true if we can show that $\neg p \rightarrow (r \wedge \neg r)$ is true for some proposition r . Proofs of this type are called **proofs by contradiction**. Because a proof by contradiction does not prove a result directly, it is another type of indirect proof. We provide three examples of proof by contradiction. The first is an example of an application of the pigeonhole principle, a combinatorial technique that we will cover in depth in Section 6.2.

EXAMPLE 10 Show that at least four of any 22 days must fall on the same day of the week.

Extra Examples 

Solution: Let p be the proposition “At least four of 22 chosen days fall on the same day of the week.” Suppose that $\neg p$ is true. This means that at most three of the 22 days fall on the same day of the week. Because there are seven days of the week, this implies that at most 21 days could have been chosen, as for each of the days of the week, at most three of the chosen days could fall on that day. This contradicts the premise that we have 22 days under consideration. That is, if r is the statement that 22 days are chosen, then we have shown that $\neg p \rightarrow (r \wedge \neg r)$. Consequently, we know that p is true. We have proved that at least four of 22 chosen days fall on the same day of the week. 

EXAMPLE 11 Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Solution: Let p be the proposition “ $\sqrt{2}$ is irrational.” To start a proof by contradiction, we suppose that $\neg p$ is true. Note that $\neg p$ is the statement “It is not the case that $\sqrt{2}$ is irrational,” which says that $\sqrt{2}$ is rational. We will show that assuming that $\neg p$ is true leads to a contradiction.

If $\sqrt{2}$ is rational, there exist integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors (so that the fraction a/b is in lowest terms). (Here, we are using the fact that every rational number can be written in lowest terms.) Because $\sqrt{2} = a/b$, when both sides of this equation are squared, it follows that

$$2 = \frac{a^2}{b^2}.$$

Hence,

$$2b^2 = a^2.$$

By the definition of an even integer it follows that a^2 is even. We next use the fact that if a^2 is even, a must also be even, which follows by Exercise 18. Furthermore, because a is even, by the definition of an even integer, $a = 2c$ for some integer c . Thus,

$$2b^2 = 4c^2.$$

Dividing both sides of this equation by 2 gives

$$b^2 = 2c^2.$$

By the definition of even, this means that b^2 is even. Again using the fact that if the square of an integer is even, then the integer itself must be even, we conclude that b must be even as well.

We have now shown that the assumption of $\neg p$ leads to the equation $\sqrt{2} = a/b$, where a and b have no common factors, but both a and b are even, that is, 2 divides both a and b . Note that the statement that $\sqrt{2} = a/b$, where a and b have no common factors, means, in particular, that 2 does not divide both a and b . Because our assumption of $\neg p$ leads to the contradiction that 2 divides both a and b and 2 does not divide both a and b , $\neg p$ must be false. That is, the statement p , “ $\sqrt{2}$ is irrational,” is true. We have proved that $\sqrt{2}$ is irrational. 

Proof by contradiction can be used to prove conditional statements. In such proofs, we first assume the negation of the conclusion. We then use the premises of the theorem and the negation of the conclusion to arrive at a contradiction. (The reason that such proofs are valid rests on the logical equivalence of $p \rightarrow q$ and $(p \wedge \neg q) \rightarrow \mathbf{F}$. To see that these statements are equivalent, simply note that each is false in exactly one case, namely, when p is true and q is false.)

Note that we can rewrite a proof by contraposition of a conditional statement as a proof by contradiction. In a proof of $p \rightarrow q$ by contraposition, we assume that $\neg q$ is true. We then show that $\neg p$ must also be true. To rewrite a proof by contraposition of $p \rightarrow q$ as a proof by contradiction, we suppose that both p and $\neg q$ are true. Then, we use the steps from the proof of $\neg q \rightarrow \neg p$ to show that $\neg p$ is true. This leads to the contradiction $p \wedge \neg p$, completing the proof. Example 11 illustrates how a proof by contraposition of a conditional statement can be rewritten as a proof by contradiction.

EXAMPLE 12 Give a proof by contradiction of the theorem “If $3n + 2$ is odd, then n is odd.”

Solution: Let p be “ $3n + 2$ is odd” and q be “ n is odd.” To construct a proof by contradiction, assume that both p and $\neg q$ are true. That is, assume that $3n + 2$ is odd and that n is not odd. Because n is not odd, we know that it is even. Because n is even, there is an integer k such that $n = 2k$. This implies that $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. Because $3n + 2$ is $2t$, where $t = 3k + 1$, $3n + 2$ is even. Note that the statement “ $3n + 2$ is even” is equivalent to the statement $\neg p$, because an integer is even if and only if it is not odd. Because both p and $\neg p$ are

true, we have a contradiction. This completes the proof by contradiction, proving that if $3n + 2$ is odd, then n is odd. 

Note that we can also prove by contradiction that $p \rightarrow q$ is true by assuming that p and $\neg q$ are true, and showing that q must be also be true. This implies that $\neg q$ and q are both true, a contradiction. This observation tells us that we can turn a direct proof into a proof by contradiction.

PROOFS OF EQUIVALENCE To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true. The validity of this approach is based on the tautology

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p).$$

EXAMPLE 13 Prove the theorem “If n is an integer, then n is odd if and only if n^2 is odd.”

Extra Examples 

Solution: This theorem has the form “ p if and only if q ,” where p is “ n is odd” and q is “ n^2 is odd.” (As usual, we do not explicitly deal with the universal quantification.) To prove this theorem, we need to show that $p \rightarrow q$ and $q \rightarrow p$ are true.

We have already shown (in Example 1) that $p \rightarrow q$ is true and (in Example 8) that $q \rightarrow p$ is true.

Because we have shown that both $p \rightarrow q$ and $q \rightarrow p$ are true, we have shown that the theorem is true. 

Sometimes a theorem states that several propositions are equivalent. Such a theorem states that propositions $p_1, p_2, p_3, \dots, p_n$ are equivalent. This can be written as

$$p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n,$$

which states that all n propositions have the same truth values, and consequently, that for all i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$, p_i and p_j are equivalent. One way to prove these are mutually equivalent is to use the tautology

$$p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n \leftrightarrow (p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \dots \wedge (p_n \rightarrow p_1).$$

This shows that if the n conditional statements $p_1 \rightarrow p_2, p_2 \rightarrow p_3, \dots, p_n \rightarrow p_1$ can be shown to be true, then the propositions $p_1, p_2, \dots, p_n$ are all equivalent.

This is much more efficient than proving that $p_i \rightarrow p_j$ for all $i \neq j$ with $1 \leq i \leq n$ and $1 \leq j \leq n$. (Note that there are $n^2 - n$ such conditional statements.)

When we prove that a group of statements are equivalent, we can establish any chain of conditional statements we choose as long as it is possible to work through the chain to go from any one of these statements to any other statement. For example, we can show that p_1, p_2 , and p_3 are equivalent by showing that $p_1 \rightarrow p_3, p_3 \rightarrow p_2$, and $p_2 \rightarrow p_1$.

EXAMPLE 14 Show that these statements about the integer n are equivalent:

- p_1 : n is even.
- p_2 : $n - 1$ is odd.
- p_3 : n^2 is even.

Solution: We will show that these three statements are equivalent by showing that the conditional statements $p_1 \rightarrow p_2, p_2 \rightarrow p_3$, and $p_3 \rightarrow p_1$ are true.

We use a direct proof to show that $p_1 \rightarrow p_2$. Suppose that n is even. Then $n = 2k$ for some integer k . Consequently, $n - 1 = 2k - 1 = 2(k - 1) + 1$. This means that $n - 1$ is odd because it is of the form $2m + 1$, where m is the integer $k - 1$.

We also use a direct proof to show that $p_2 \rightarrow p_3$. Now suppose $n - 1$ is odd. Then $n - 1 = 2k + 1$ for some integer k . Hence, $n = 2k + 2$ so that $n^2 = (2k + 2)^2 = 4k^2 + 8k + 4 = 2(2k^2 + 4k + 2)$. This means that n^2 is twice the integer $2k^2 + 4k + 2$, and hence is even.

To prove $p_3 \rightarrow p_1$, we use a proof by contraposition. That is, we prove that if n is not even, then n^2 is not even. This is the same as proving that if n is odd, then n^2 is odd, which we have already done in Example 1. This completes the proof. 

COUNTEREXAMPLES In Section 1.4 we stated that to show that a statement of the form $\forall x P(x)$ is false, we need only find a **counterexample**, that is, an example x for which $P(x)$ is false. When presented with a statement of the form $\forall x P(x)$, which we believe to be false or which has resisted all proof attempts, we look for a counterexample. We illustrate the use of counterexamples in Example 15.

EXAMPLE 15 Show that the statement “Every positive integer is the sum of the squares of two integers” is false.

Extra Examples 

Solution: To show that this statement is false, we look for a counterexample, which is a particular integer that is not the sum of the squares of two integers. It does not take long to find a counterexample, because 3 cannot be written as the sum of the squares of two integers. To show this is the case, note that the only perfect squares not exceeding 3 are $0^2 = 0$ and $1^2 = 1$. Furthermore, there is no way to get 3 as the sum of two terms each of which is 0 or 1. Consequently, we have shown that “Every positive integer is the sum of the squares of two integers” is false. 

1.7.8 Mistakes in Proofs

There are many common errors made in constructing mathematical proofs. We will briefly describe some of these here. Among the most common errors are mistakes in arithmetic and basic algebra. Even professional mathematicians make such errors, especially when working with complicated formulae. Whenever you use such computations you should check them as carefully as possible. (You should also review any troublesome aspects of basic algebra, especially before you study Section 5.1.)

Links 

Each step of a mathematical proof needs to be correct and the conclusion needs to follow logically from the steps that precede it. Many mistakes result from the introduction of steps that do not logically follow from those that precede it. This is illustrated in Examples 16–18.

EXAMPLE 16 What is wrong with this famous supposed “proof” that $1 = 2$?

“Proof”: We use these steps, where a and b are two equal positive integers.

Step	Reason
1. $a = b$	Given
2. $a^2 = ab$	Multiply both sides of (1) by a
3. $a^2 - b^2 = ab - b^2$	Subtract b^2 from both sides of (2)
4. $(a - b)(a + b) = b(a - b)$	Factor both sides of (3)
5. $a + b = b$	Divide both sides of (4) by $a - b$
6. $2b = b$	Replace a by b in (5) because $a = b$ and simplify
7. $2 = 1$	Divide both sides of (6) by b

Solution: Every step is valid except for step 5, where we divided both sides by $a - b$. The error is that $a - b$ equals zero; division of both sides of an equation by the same quantity is valid as long as this quantity is not zero. ◀

EXAMPLE 17 What is wrong with this “proof”?

“Theorem”: If n^2 is positive, then n is positive.

“Proof”: Suppose that n^2 is positive. Because the conditional statement “If n is positive, then n^2 is positive” is true, we can conclude that n is positive.

Solution: Let $P(n)$ be “ n is positive” and $Q(n)$ be “ n^2 is positive.” Then our hypothesis is $Q(n)$. The statement “If n is positive, then n^2 is positive” is the statement $\forall n(P(n) \rightarrow Q(n))$. From the hypothesis $Q(n)$ and the statement $\forall n(P(n) \rightarrow Q(n))$ we cannot conclude $P(n)$, because we are not using a valid rule of inference. Instead, this is an example of the fallacy of affirming the conclusion. A counterexample is supplied by $n = -1$ for which $n^2 = 1$ is positive, but n is negative. ◀

EXAMPLE 18 What is wrong with this “proof”?

“Theorem”: If n is not positive, then n^2 is not positive. (This is the contrapositive of the “theorem” in Example 17.)

“Proof”: Suppose that n is not positive. Because the conditional statement “If n is positive, then n^2 is positive” is true, we can conclude that n^2 is not positive.

Solution: Let $P(n)$ and $Q(n)$ be as in the solution of Example 17. Then our hypothesis is $\neg P(n)$ and the statement “If n is positive, then n^2 is positive” is the statement $\forall n(P(n) \rightarrow Q(n))$. From the hypothesis $\neg P(n)$ and the statement $\forall n(P(n) \rightarrow Q(n))$ we cannot conclude $\neg Q(n)$, because we are not using a valid rule of inference. Instead, this is an example of the fallacy of denying the hypothesis. A counterexample is supplied by $n = -1$, as in Example 17. ◀

Finally, we briefly discuss a particularly nasty type of error. Many incorrect arguments are based on a fallacy called **begging the question**. This fallacy occurs when one or more steps of a proof are based on the truth of the statement being proved. In other words, this fallacy arises when a statement is proved using itself, or a statement equivalent to it. That is why this fallacy is also called **circular reasoning**.

EXAMPLE 19 Is the following argument correct? It supposedly shows that n is an even integer whenever n^2 is an even integer.

Suppose that n^2 is even. Then $n^2 = 2k$ for some integer k . Let $n = 2l$ for some integer l . This shows that n is even.

Solution: This argument is incorrect. The statement “let $n = 2l$ for some integer l ” occurs in the proof. No argument has been given to show that n can be written as $2l$ for some integer l . This is circular reasoning because this statement is equivalent to the statement being proved, namely, “ n is even.” The result itself is correct; only the method of proof is wrong. ◀

Making mistakes in proofs is part of the learning process. When you make a mistake that someone else finds, you should carefully analyze where you went wrong and make sure that you do not make the same mistake again. Even professional mathematicians make mistakes in proofs. More than a few incorrect proofs of important results have fooled people for many years before subtle errors in them were found.

1.7.9 Just a Beginning

We have now developed a basic arsenal of proof methods. In the next section we will introduce other important proof methods. We will also introduce several important proof techniques in Chapter 5, including mathematical induction, which can be used to prove results that hold for all positive integers. In Chapter 6 we will introduce the notion of combinatorial proofs.

In this section we introduced several methods for proving theorems of the form $\forall x(P(x) \rightarrow Q(x))$, including direct proofs and proofs by contraposition. There are many theorems of this type whose proofs are easy to construct by directly working through the hypotheses and definitions of the terms of the theorem. However, it is often difficult to prove a theorem without resorting to a clever use of a proof by contraposition or a proof by contradiction, or some other proof technique. In Section 1.8 we will address proof strategy. We will describe various approaches that can be used to find proofs when straightforward approaches do not work. Constructing proofs is an art that can be learned only through experience, including writing proofs, having your proofs critiqued, and reading and analyzing other proofs.

Exercises

- Use a direct proof to show that the sum of two odd integers is even.
- Use a direct proof to show that the sum of two even integers is even.
- Show that the square of an even number is an even number using a direct proof.
- Show that the additive inverse, or negative, of an even number is an even number using a direct proof.
- Prove that if $m + n$ and $n + p$ are even integers, where m , n , and p are integers, then $m + p$ is even. What kind of proof did you use?
- Use a direct proof to show that the product of two odd numbers is odd.
- Use a direct proof to show that every odd integer is the difference of two squares. [*Hint:* Find the difference of the squares of $k + 1$ and k where k is a positive integer.]
- Prove that if n is a perfect square, then $n + 2$ is not a perfect square.
- Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.
- Use a direct proof to show that the product of two rational numbers is rational.
- Prove or disprove that the product of two irrational numbers is irrational.
- Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.
- Prove that if x is irrational, then $1/x$ is irrational.
- Prove that if x is rational and $x \neq 0$, then $1/x$ is rational.
- Prove that if x is an irrational number and $x > 0$, then $\sqrt{x}$ is also irrational.
- Prove that if x , y , and z are integers and $x + y + z$ is odd, then at least one of x , y , and z is odd.
- Use a proof by contraposition to show that if $x + y \geq 2$, where x and y are real numbers, then $x \geq 1$ or $y \geq 1$.
-  Prove that if m and n are integers and mn is even, then m is even or n is even.
- Show that if n is an integer and $n^3 + 5$ is odd, then n is even using
 - a proof by contraposition.
 - a proof by contradiction.
- Prove that if n is an integer and $3n + 2$ is even, then n is even using
 - a proof by contraposition.
 - a proof by contradiction.
- Prove the proposition $P(0)$, where $P(n)$ is the proposition “If n is a positive integer greater than 1, then $n^2 > n$.” What kind of proof did you use?
- Prove the proposition $P(1)$, where $P(n)$ is the proposition “If n is a positive integer, then $n^2 \geq n$.” What kind of proof did you use?
- Let $P(n)$ be the proposition “If a and b are positive real numbers, then $(a + b)^n \geq a^n + b^n$.” Prove that $P(1)$ is true. What kind of proof did you use?
- Show that if you pick three socks from a drawer containing just blue socks and black socks, you must get either a pair of blue socks or a pair of black socks.

25. Show that at least ten of any 64 days chosen must fall on the same day of the week.
26. Show that at least three of any 25 days chosen must fall in the same month of the year.
27. Use a proof by contradiction to show that there is no rational number r for which $r^3 + r + 1 = 0$. [Hint: Assume that $r = a/b$ is a root, where a and b are integers and a/b is in lowest terms. Obtain an equation involving integers by multiplying by b^3 . Then look at whether a and b are each odd or even.]
28. Prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even.
29. Prove that if n is a positive integer, then n is odd if and only if $5n + 6$ is odd.
30. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.
31. Prove or disprove that if m and n are integers such that $mn = 1$, then either $m = 1$ and $n = 1$, or else $m = -1$ and $n = -1$.
32. Show that these three statements are equivalent, where a and b are real numbers: (i) a is less than b , (ii) the average of a and b is greater than a , and (iii) the average of a and b is less than b .
33. Show that these statements about the integer x are equivalent: (i) $3x + 2$ is even, (ii) $x + 5$ is odd, (iii) x^2 is even.
34. Show that these statements about the real number x are equivalent: (i) x is rational, (ii) $x/2$ is rational, (iii) $3x - 1$ is rational.
35. Show that these statements about the real number x are equivalent: (i) x is irrational, (ii) $3x + 2$ is irrational, (iii) $x/2$ is irrational.
36. Is this reasoning for finding the solutions of the equation $\sqrt{2x^2 - 1} = x$ correct? (1) $\sqrt{2x^2 - 1} = x$ is given; (2) $2x^2 - 1 = x^2$, obtained by squaring both sides of (1); (3) $x^2 - 1 = 0$, obtained by subtracting x^2 from both sides of (2); (4) $(x - 1)(x + 1) = 0$, obtained by factoring the left-hand side of $x^2 - 1$; (5) $x = 1$ or $x = -1$, which follows because $ab = 0$ implies that $a = 0$ or $b = 0$.
37. Are these steps for finding the solutions of $\sqrt{x + 3} = 3 - x$ correct? (1) $\sqrt{x + 3} = 3 - x$ is given; (2) $x + 3 = x^2 - 6x + 9$, obtained by squaring both sides of (1); (3) $0 = x^2 - 7x + 6$, obtained by subtracting $x + 3$ from both sides of (2); (4) $0 = (x - 1)(x - 6)$, obtained by factoring the right-hand side of (3); (5) $x = 1$ or $x = 6$, which follows from (4) because $ab = 0$ implies that $a = 0$ or $b = 0$.
38. Show that the propositions p_1, p_2, p_3 , and p_4 can be shown to be equivalent by showing that $p_1 \leftrightarrow p_4, p_2 \leftrightarrow p_3$, and $p_1 \leftrightarrow p_3$.
39. Show that the propositions p_1, p_2, p_3, p_4 , and p_5 can be shown to be equivalent by proving that the conditional statements $p_1 \rightarrow p_4, p_3 \rightarrow p_1, p_4 \rightarrow p_2, p_2 \rightarrow p_5$, and $p_5 \rightarrow p_3$ are true.
40. Find a counterexample to the statement that every positive integer can be written as the sum of the squares of three integers.
41. Prove that at least one of the real numbers $a_1, a_2, \dots, a_n$ is greater than or equal to the average of these numbers. What kind of proof did you use?
42. Use Exercise 41 to show that if the first 10 positive integers are placed around a circle, in any order, there exist three integers in consecutive locations around the circle that have a sum greater than or equal to 17.
43. Prove that if n is an integer, these four statements are equivalent: (i) n is even, (ii) $n + 1$ is odd, (iii) $3n + 1$ is odd, (iv) $3n$ is even.
44. Prove that these four statements about the integer n are equivalent: (i) n^2 is odd, (ii) $1 - n$ is even, (iii) n^3 is odd, (iv) $n^2 + 1$ is even.

1.8 Proof Methods and Strategy

1.8.1 Introduction

Assessment

In Section 1.7 we introduced many methods of proof and illustrated how each method can be used. In this section we continue this effort. We will introduce several other commonly used proof methods, including the method of proving a theorem by considering different cases separately. We will also discuss proofs where we prove the existence of objects with desired properties.

In Section 1.7 we briefly discussed the strategy behind constructing proofs. This strategy includes selecting a proof method and then successfully constructing an argument step by step, based on this method. In this section, after we have developed a versatile arsenal of proof methods, we will study some aspects of the art and science of proofs. We will provide advice on how to find a proof of a theorem. We will describe some tricks of the trade, including how proofs can be found by working backward and by adapting existing proofs.

When mathematicians work, they formulate conjectures and attempt to prove or disprove them. We will briefly describe this process here by proving results about tiling checkerboards with dominoes and other types of pieces. Looking at tilings of this kind, we will be able to quickly formulate conjectures and prove theorems without first developing a theory.

We will conclude the section by discussing the role of open questions. In particular, we will discuss some interesting problems either that have been solved after remaining open for hundreds of years or that still remain open.

1.8.2 Exhaustive Proof and Proof by Cases

Sometimes we cannot prove a theorem using a single argument that holds for all possible cases. We now introduce a method that can be used to prove a theorem by considering different cases separately. This method is based on a rule of inference that we will now introduce. To prove a conditional statement of the form

$$(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q$$

the tautology

$$[(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \cdots \wedge (p_n \rightarrow q)]$$

can be used as a rule of inference. This shows that the original conditional statement with a hypothesis made up of a disjunction of the propositions $p_1, p_2, \dots, p_n$ can be proved by proving each of the n conditional statements $p_i \rightarrow q$, $i = 1, 2, \dots, n$, individually. Such an argument is called a **proof by cases**. Sometimes to prove that a conditional statement $p \rightarrow q$ is true, it is convenient to use a disjunction $p_1 \vee p_2 \vee \cdots \vee p_n$ instead of p as the hypothesis of the conditional statement, where p and $p_1 \vee p_2 \vee \cdots \vee p_n$ are equivalent.

EXHAUSTIVE PROOF Some theorems can be proved by examining a relatively small number of examples. Such proofs are called **exhaustive proofs**, or **proofs by exhaustion** because these proofs proceed by exhausting all possibilities. An exhaustive proof is a special type of proof by cases where each case involves checking a single example. We now provide some illustrations of exhaustive proofs.

EXAMPLE 1 Prove that $(n + 1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

Extra Examples 

Solution: We use a proof by exhaustion. We only need verify the inequality $(n + 1)^3 \geq 3^n$ when $n = 1, 2, 3$, and 4 . For $n = 1$, we have $(n + 1)^3 = 2^3 = 8$ and $3^n = 3^1 = 3$; for $n = 2$, we have $(n + 1)^3 = 3^3 = 27$ and $3^n = 3^2 = 9$; for $n = 3$, we have $(n + 1)^3 = 4^3 = 64$ and $3^n = 3^3 = 27$; and for $n = 4$, we have $(n + 1)^3 = 5^3 = 125$ and $3^n = 3^4 = 81$. In each of these four cases, we see that $(n + 1)^3 \geq 3^n$. We have used the method of exhaustion to prove that $(n + 1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$. 

EXAMPLE 2 Prove that the only consecutive positive integers not exceeding 100 that are perfect powers are 8 and 9. (An integer n is a **perfect power** if it equals m^a , where m is an integer and a is an integer greater than 1.)

Solution: We use a proof by exhaustion. In particular, we can prove this fact by examining positive integers n not exceeding 100, first checking whether n is a perfect power, and if it is, checking whether $n + 1$ is also a perfect power. A quicker way to do this is simply to look at all perfect powers not exceeding 100 and checking whether the next largest integer is also a perfect power. The squares of positive integers not exceeding 100 are 1, 4, 9, 16, 25, 36, 49, 64, 81, and

100. The cubes of positive integers not exceeding 100 are 1, 8, 27, and 64. The fourth powers of positive integers not exceeding 100 are 1, 16, and 81. The fifth powers of positive integers not exceeding 100 are 1 and 32. The sixth powers of positive integers not exceeding 100 are 1 and 64. There are no powers of positive integers higher than the sixth power not exceeding 100, other than 1. Looking at this list of perfect powers not exceeding 100, we see that $n = 8$ is the only perfect power n for which $n + 1$ is also a perfect power. That is, $2^3 = 8$ and $3^2 = 9$ are the only two consecutive perfect powers not exceeding 100. ◀

Proofs by exhaustion can tire out people and computers when the number of cases challenges the available processing power!

People can carry out exhaustive proofs when it is necessary to check only a relatively small number of instances of a statement. Computers do not complain when they are asked to check a much larger number of instances of a statement, but they still have limitations. Note that not even a computer can check all instances when it is impossible to list all instances to check.

PROOF BY CASES A proof by cases must cover all possible cases that arise in a theorem. We illustrate proof by cases with a couple of examples. In each example, you should check that all possible cases are covered.

EXAMPLE 3 Prove that if n is an integer, then $n^2 \geq n$.

Extra Examples ▶

Solution: We can prove that $n^2 \geq n$ for every integer by considering three cases, when $n = 0$, when $n \geq 1$, and when $n \leq -1$. We split the proof into three cases because it is straightforward to prove the result by considering zero, positive integers, and negative integers separately.

Case (i): When $n = 0$, because $0^2 = 0$, we see that $0^2 \geq 0$. It follows that $n^2 \geq n$ is true in this case.

Case (ii): When $n \geq 1$, when we multiply both sides of the inequality $n \geq 1$ by the positive integer n , we obtain $n \cdot n \geq n \cdot 1$. This implies that $n^2 \geq n$ for $n \geq 1$.

Case (iii): In this case $n \leq -1$. However, $n^2 \geq 0$. It follows that $n^2 \geq n$.

Because the inequality $n^2 \geq n$ holds in all three cases, we can conclude that if n is an integer, then $n^2 \geq n$. ◀

EXAMPLE 4 Use a proof by cases to show that $|xy| = |x||y|$, where x and y are real numbers. (Recall that $|a|$, the absolute value of a , equals a when $a \geq 0$ and equals $-a$ when $a \leq 0$.)

Solution: In our proof of this theorem, we remove absolute values using the fact that $|a| = a$ when $a \geq 0$ and $|a| = -a$ when $a \leq 0$. Because both $|x|$ and $|y|$ occur in our formula, we will need four cases: (i) x and y both nonnegative, (ii) x nonnegative and y negative, (iii) x negative and y nonnegative, and (iv) x negative and y negative. We denote by p_1 , p_2 , p_3 , and p_4 , the proposition stating the assumption for each of these four cases, respectively.

(Note that we can remove the absolute value signs by making the appropriate choice of signs within each case.)

Case (i): We see that $p_1 \rightarrow q$ because $xy \geq 0$ when $x \geq 0$ and $y \geq 0$, so that $|xy| = xy = |x||y|$.

Case (ii): To see that $p_2 \rightarrow q$, note that if $x \geq 0$ and $y < 0$, then $xy \leq 0$, so that $|xy| = -xy = x(-y) = |x||y|$. (Here, because $y < 0$, we have $|y| = -y$.)

Case (iii): To see that $p_3 \rightarrow q$, we follow the same reasoning as the previous case with the roles of x and y reversed.

Case (iv): To see that $p_4 \rightarrow q$, note that when $x < 0$ and $y < 0$, it follows that $xy > 0$. Hence, $|xy| = xy = (-x)(-y) = |x||y|$.

Because $|xy| = |x||y|$ holds in each of the four cases and these cases exhaust all possibilities, we can conclude that $|xy| = |x||y|$, whenever x and y are real numbers. 

LEVERAGING PROOF BY CASES The examples we have presented illustrating proof by cases provide some insight into when to use this method of proof. In particular, when it is not possible to consider all cases of a proof at the same time, a proof by cases should be considered. When should you use such a proof? Generally, look for a proof by cases when there is no obvious way to begin a proof, but when extra information in each case helps move the proof forward. Example 5 illustrates how the method of proof by cases can be used effectively.

EXAMPLE 5 Formulate a conjecture about the final decimal digit of the square of an integer and prove your result.

Solution: The smallest perfect squares are 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, and so on. We notice that the digits that occur as the final digit of a square are 0, 1, 4, 5, 6, and 9, with 2, 3, 7, and 8 never appearing as the final digit of a square. We conjecture this theorem: The final decimal digit of a perfect square is 0, 1, 4, 5, 6, or 9. How can we prove this theorem?

We first note that we can express an integer n as $10a + b$, where a and b are positive integers and b is 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9. Here a is the integer obtained by subtracting the final decimal digit of n from n and dividing by 10. Next, note that $(10a + b)^2 = 100a^2 + 20ab + b^2 = 10(10a^2 + 2b) + b^2$, so that the final decimal digit of n^2 is the same as the final decimal digit of b^2 . Furthermore, note that the final decimal digit of b^2 is the same as the final decimal digit of $(10 - b)^2 = 100 - 20b + b^2$. Consequently, we can reduce our proof to the consideration of six cases.

Case (i): The final digit of n is 1 or 9. Then the final decimal digit of n^2 is the final decimal digit of $1^2 = 1$ or $9^2 = 81$, namely, 1.

Case (ii): The final digit of n is 2 or 8. Then the final decimal digit of n^2 is the final decimal digit of $2^2 = 4$ or $8^2 = 64$, namely, 4.

Case (iii): The final digit of n is 3 or 7. Then the final decimal digit of n^2 is the final decimal digit of $3^2 = 9$ or $7^2 = 49$, namely, 9.

Case (iv): The final digit of n is 4 or 6. Then the final decimal digit of n^2 is the final decimal digit of $4^2 = 16$ or $6^2 = 36$, namely, 6.

Case (v): The final decimal digit of n is 5. Then the final decimal digit of n^2 is the final decimal digit of $5^2 = 25$, namely, 5.

Case (vi): The final decimal digit of n is 0. Then the final decimal digit of n^2 is the final decimal digit of $0^2 = 0$, namely, 0.

Because we have considered all six cases, we can conclude that the final decimal digit of n^2 , where n is an integer is either 0, 1, 2, 4, 5, 6, or 9. 

Sometimes we can eliminate all but a few examples in a proof by cases, as Example 6 illustrates.

EXAMPLE 6 Show that there are no solutions in integers x and y of $x^2 + 3y^2 = 8$.

Solution: We can quickly reduce a proof to checking just a few simple cases because $x^2 > 8$ when $|x| \geq 3$ and $3y^2 > 8$ when $|y| \geq 2$. This leaves the cases when x equals $-2, -1, 0, 1$, or 2 and y equals $-1, 0$, or 1 . We can finish using an exhaustive proof. To dispense with the remaining cases, we note that possible values for x^2 are 0, 1, and 4, and possible values for $3y^2$ are 0 and 3, and the largest sum of possible values for x^2 and $3y^2$ is 7. Consequently, it is impossible for $x^2 + 3y^2 = 8$ to hold when x and y are integers. 

WITHOUT LOSS OF GENERALITY In the proof in Example 4, we dismissed case (iii), where $x < 0$ and $y \geq 0$, because it is the same as case (ii), where $x \geq 0$ and $y < 0$, with the roles of x and y reversed. To shorten the proof, we could have proved cases (ii) and (iii) together by assuming, **without loss of generality**, that $x \geq 0$ and $y < 0$. Implicit in this statement is that we can complete the case with $x < 0$ and $y \geq 0$ using the same argument as we used for the case with $x \geq 0$ and $y < 0$, but with the obvious changes.

In general, when the phrase “without loss of generality” is used in a proof (often abbreviated as WLOG), we assert that by proving one case of a theorem, no additional argument is required to prove other specified cases. That is, other cases follow by making straightforward changes to the argument, or by filling in some straightforward initial step. Proofs by cases can often be made much more efficient when the notion of without loss of generality is employed. Incorrect use of this principle, however, can lead to unfortunate errors. Sometimes assumptions are made that lead to a loss in generality. Such assumptions can be made that do not take into account that one case may be substantially different from others. This can lead to an incomplete, and possibly unsalvageable, proof. In fact, many incorrect proofs of famous theorems turned out to rely on arguments that used the idea of “without loss of generality” to establish cases that could not be quickly proved from simpler cases.

We now illustrate a proof where without loss of generality is used effectively together with other proof techniques.



In a proof by cases be sure not to omit any cases and check that you have proved all cases correctly!

EXAMPLE 7 Show that if x and y are integers and both xy and $x + y$ are even, then both x and y are even.

Solution: We will use proof by contraposition, the notion of without loss of generality, and proof by cases. First, suppose that x and y are not both even. That is, assume that x is odd or that y is odd (or both). Without loss of generality, we assume that x is odd, so that $x = 2m + 1$ for some integer k .

To complete the proof, we need to show that xy is odd or $x + y$ is odd. Consider two cases: (i) y is even, and (ii) y is odd. In (i), $y = 2n$ for some integer n , so that $x + y = (2m + 1) + 2n = 2(m + n) + 1$ is odd. In (ii), $y = 2n + 1$ for some integer n , so that $xy = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$ is odd. This completes the proof by contraposition. (Note that our use of without loss of generality within the proof is justified because the proof when y is odd can be obtained by simply interchanging the roles of x and y in the proof we have given.)

COMMON ERRORS WITH EXHAUSTIVE PROOF AND PROOF BY CASES A common error of reasoning is to draw incorrect conclusions from examples. No matter how many separate examples are considered, a theorem is not proved by considering examples unless every possible case is covered. The problem of proving a theorem is analogous to showing that a computer program always produces the output desired. No matter how many input values are tested, unless all input values are tested, we cannot conclude that the program always produces the correct output.

EXAMPLE 8 Is it true that every positive integer is the sum of 18 fourth powers of integers?

Solution: To determine whether a positive integer n can be written as the sum of 18 fourth powers of integers, we might begin by examining whether n is the sum of 18 fourth powers of integers for the smallest positive integers. Because the fourth powers of integers are 0, 1, 16, 81, ..., if we can select 18 terms from these numbers that add up to n , then n is the sum of 18 fourth powers. We can show that all positive integers up to 78 can be written as the sum of 18 fourth powers. (The details are left to the reader.) However, if we decided this was enough checking, we would come to the wrong conclusion. It is not true that every positive integer is

the sum of 18 fourth powers because 79 is not the sum of 18 fourth powers (as the reader can verify). 

Another common error involves making unwarranted assumptions that lead to incorrect proofs by cases where not all cases are considered. This is illustrated in Example 9.

EXAMPLE 9 What is wrong with this “proof”?

“Theorem”: If x is a real number, then x^2 is a positive real number.

“Proof”: Let p_1 be “ x is positive,” let p_2 be “ x is negative,” and let q be “ x^2 is positive.” To show that $p_1 \rightarrow q$ is true, note that when x is positive, x^2 is positive because it is the product of two positive numbers, x and x . To show that $p_2 \rightarrow q$, note that when x is negative, x^2 is positive because it is the product of two negative numbers, x and x . This completes the proof.

Solution: The problem with this “proof” is that we missed the case of $x = 0$. When $x = 0$, $x^2 = 0$ is not positive, so the supposed theorem is false. If p is “ x is a real number,” then we can prove results where p is the hypothesis with three cases, p_1 , p_2 , and p_3 , where p_1 is “ x is positive,” p_2 is “ x is negative,” and p_3 is “ $x = 0$ ” because of the equivalence $p \leftrightarrow p_1 \vee p_2 \vee p_3$. 

1.8.3 Existence Proofs

Many theorems are assertions that objects of a particular type exist. A theorem of this type is a proposition of the form $\exists xP(x)$, where P is a predicate. A proof of a proposition of the form $\exists xP(x)$ is called an **existence proof**. There are several ways to prove a theorem of this type. Sometimes an existence proof of $\exists xP(x)$ can be given by finding an element a , called a **witness**, such that $P(a)$ is true. This type of existence proof is called **constructive**. It is also possible to give an existence proof that is **nonconstructive**; that is, we do not find an element a such that $P(a)$ is true, but rather prove that $\exists xP(x)$ is true in some other way. One common method of giving a nonconstructive existence proof is to use proof by contradiction and show that the negation of the existential quantification implies a contradiction. The concept of a constructive existence proof is illustrated by Example 10 and the concept of a nonconstructive existence proof is illustrated by Example 11.

EXAMPLE 10 **A Constructive Existence Proof** Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways.

Extra Examples 

Solution: After considerable computation (such as a computer search) we find that

$$1729 = 10^3 + 9^3 = 12^3 + 1^3.$$

Because we have displayed a positive integer that can be written as the sum of cubes in two different ways, we are done.

There is an interesting story pertaining to this example. The English mathematician G. H. Hardy, when visiting the ailing Indian prodigy Ramanujan in the hospital, remarked that 1729, the number of the cab he took, was rather dull. Ramanujan replied “No, it is a very interesting number; it is the smallest number expressible as the sum of cubes in two different ways.” 

EXAMPLE 11 **A Nonconstructive Existence Proof** Show that there exist irrational numbers x and y such that x^y is rational.

Solution: By Example 11 in Section 1.7 we know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$. If it is rational, we have two irrational numbers x and y with x^y rational, namely, $x = \sqrt{2}$ and $y = \sqrt{2}$. On the other hand if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2} \cdot \sqrt{2})} = \sqrt{2}^2 = 2$.

This proof is an example of a nonconstructive existence proof because we have not found irrational numbers x and y such that x^y is rational. Rather, we have shown that either the pair $x = \sqrt{2}, y = \sqrt{2}$ or the pair $x = \sqrt{2}^{\sqrt{2}}, y = \sqrt{2}$ have the desired property, but we do not know which of these two pairs works! ▶

Remark: Exercise 11 in Section 4.3 provides a constructive existence proof that there are irrational numbers x and y such that x^y is rational.

Nonconstructive existence proofs often are quite subtle, as Example 12 illustrates.

EXAMPLE 12

Chomp is a game played by two players. In this game, cookies are laid out on a rectangular grid. The cookie in the top left position is poisoned, as shown in Figure 1(a). The two players take turns making moves; at each move, a player is required to eat a remaining cookie, together with all cookies to the right and/or below it (see Figure 1(b), for example). The loser is the player who has no choice but to eat the poisoned cookie. We ask whether one of the two players has a winning strategy. That is, can one of the players always make moves that are guaranteed to lead to a win?

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GODFREY HAROLD HARDY (1877–1947) Hardy, born in Cranleigh, Surrey, England, was the older of two children of Isaac Hardy and Sophia Hall Hardy. His father was the geography and drawing master at the Cranleigh School and also gave singing lessons and played soccer. His mother gave piano lessons and helped run a boardinghouse for young students. Hardy's parents were devoted to their children's education. Hardy demonstrated his numerical ability at the early age of two when he began writing down numbers into the millions. He had a private mathematics tutor rather than attending regular classes at the Cranleigh School. He moved to Winchester College, a private high school, when he was 13 and was awarded a scholarship. He excelled in his studies and demonstrated a strong interest in mathematics. He entered Trinity College, Cambridge, in 1896 on a scholarship and won several prizes during his time there, graduating in 1899.

Hardy held the position of lecturer in mathematics at Trinity College at Cambridge University from 1906 to 1919, when he was appointed to the Sullivan chair of geometry at Oxford. He had become unhappy with Cambridge over the dismissal of the noted philosopher and mathematician Bertrand Russell from Trinity for antiwar activities and did not like a heavy load of administrative duties. In 1931 he returned to Cambridge as the Sadleirian professor of pure mathematics, where he remained until his retirement in 1942. He was a pure mathematician and held an elitist view of mathematics, hoping that his research could never be applied. Ironically, he is perhaps best known as one of the developers of the Hardy–Weinberg law, which predicts patterns of inheritance. His work in this area appeared as a letter to the journal *Science* in which he used simple algebraic ideas to demonstrate errors in an article on genetics. Hardy worked primarily in number theory and function theory, exploring such topics as the Riemann zeta function, Fourier series, and the distribution of primes. He made many important contributions to many important problems, such as Waring's problem about representing positive integers as sums of k th powers and the problem of representing odd integers as sums of three primes. Hardy is also remembered for his collaborations with John E. Littlewood, a colleague at Cambridge, with whom he wrote more than 100 papers, and the celebrated Indian mathematical prodigy Srinivasa Ramanujan. His collaboration with Littlewood led to the joke that there were only three important English mathematicians at that time, Hardy, Littlewood, and Hardy–Littlewood. However, some people believed that Hardy had invented a fictitious person, Littlewood, because Littlewood was seldom seen away from Cambridge. Hardy had the wisdom of recognizing Ramanujan's genius from unconventional but extremely creative writings Ramanujan sent him, while other mathematicians failed to see the genius. Hardy brought Ramanujan to Cambridge and collaborated on important joint papers, establishing new results on the number of partitions of an integer. Hardy was interested in mathematics education, and his book *A Course of Pure Mathematics* had a profound effect on undergraduate instruction in mathematics in the first half of the twentieth century. Hardy also wrote *A Mathematician's Apology*, in which he gives his answer to the question of whether it is worthwhile to devote one's life to the study of mathematics. It presents Hardy's view of what mathematics is and what a mathematician does.

Hardy had a strong interest in sports. He was an avid cricket fan and followed scores closely. One peculiar trait he had was that he did not like his picture taken (only five snapshots are known) and disliked mirrors, covering them with towels immediately upon entering a hotel room.

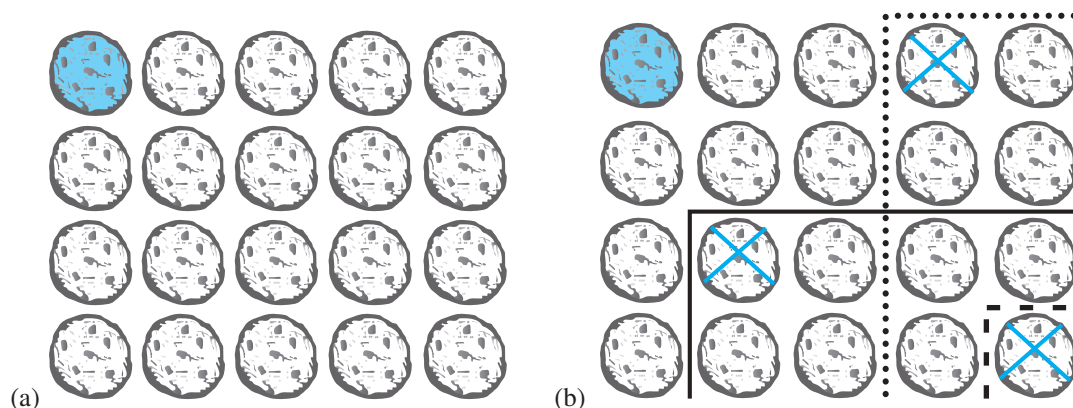


FIGURE 1 (a) Chomp (top left cookie poisoned). (b) Three possible moves.

Solution: We will give a nonconstructive existence proof of a winning strategy for the first player. That is, we will show that the first player always has a winning strategy without explicitly describing the moves this player must follow.

First, note that the game ends and cannot finish in a draw because with each move at least one cookie is eaten, so after no more than $m \times n$ moves the game ends, where the initial grid is $m \times n$. Now, suppose that the first player begins the game by eating just the cookie in the bottom right corner. There are two possibilities, this is the first move of a winning strategy for the first player, or the second player can make a move that is the first move of a winning strategy for the second player. In this second case, instead of eating just the cookie in the bottom right corner, the first player could have made the same move that the second player made as the first move of a winning strategy (and then continued to follow that winning strategy). This would guarantee a win for the first player.

Note that we showed that a winning strategy exists, but we did not specify an actual winning strategy. Consequently, the proof is a nonconstructive existence proof. In fact, no one has been able to describe a winning strategy for Chomp that applies for all rectangular grids by describing the moves that the first player should follow. However, winning strategies can be described for certain special cases, such as when the grid is square and when the grid only has two rows of cookies (see Exercises 15 and 16 in Section 5.2). ◀

1.8.4 Uniqueness Proofs

Some theorems assert the existence of a unique element with a particular property. In other words, these theorems assert that there is exactly one element with this property. To prove a statement of this type we need to show that an element with this property exists and that no other element has this property. The two parts of a **uniqueness proof** are:

Existence: We show that an element x with the desired property exists.

Uniqueness: We show that if x and y both have the desired property, then $x = y$.

Remark: Showing that there is a unique element x such that $P(x)$ is the same as proving the statement $\exists x(P(x) \wedge \forall y(y \neq x \rightarrow \neg P(y)))$.

We illustrate the elements of a uniqueness proof in Example 13.

EXAMPLE 13 Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Solution: First, note that the real number $r = -b/a$ is a solution of $ar + b = 0$ because $a(-b/a) + b = -b + b = 0$. Consequently, a real number r exists for which $ar + b = 0$. This is the existence part of the proof.

Extra Examples ➤

Second, suppose that s is a real number such that $as + b = 0$. Then $ar + b = as + b$, where $r = -b/a$. Subtracting b from both sides, we find that $ar = as$. Dividing both sides of this last equation by a , which is nonzero, we see that $r = s$. This establishes the uniqueness part of the proof. ◀

1.8.5 Proof Strategies

Finding proofs can be a challenging business. When you are confronted with a statement to prove, you should first replace terms by their definitions and then carefully analyze what the hypotheses and the conclusion mean. After doing so, you can attempt to prove the result using one of the available methods of proof. We have already provided some proof strategies in Section 1.7 for theorems of the form $\forall x(P(x) \rightarrow Q(x))$, when we introduced direct proof, proof by contraposition, and proof by contradiction. If the statement is a conditional statement, try a direct proof first as long as the hypotheses provide a good starting point; if this fails, try a proof

Links ➤



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SRINIVASA RAMANUJAN (1887–1920) The renowned mathematical prodigy Ramanujan was born and raised in southern India near the city of Madras (now called Chennai). His father was a clerk in a cloth shop. His mother contributed to the family income by singing at a local temple. Ramanujan studied at the local English language school, displaying his talent and interest for mathematics. At the age of 13 he mastered a textbook used by college students. When he was 15, a university student lent him a copy of *Synopsis of Pure Mathematics*. Ramanujan decided to work out the over 6000 results in this book, stated without proof or explanation, writing on sheets later collected to form notebooks. He graduated from high school in 1904, winning a scholarship to the University of Madras. Enrolling in a fine arts curriculum, he neglected his subjects other than mathematics and lost his scholarship. He failed to pass examinations at the university four times from 1904 to 1907, doing well only in mathematics. During this time he filled his notebooks with original writings, sometimes rediscovering already published work and at other times making new discoveries.

Without a university degree, it was difficult for Ramanujan to find a decent job. To survive, he had to depend on the goodwill of his friends. He tutored students in mathematics, but his unconventional ways of thinking and failure to stick to the syllabus caused problems. He was married in 1909 in an arranged marriage to a young woman nine years his junior. Needing to support himself and his wife, he moved to Madras and sought a job. He showed his notebooks of mathematical writings to his potential employers, but the books bewildered them. However, a professor at the Presidency College recognized his genius and supported him, and in 1912 he found work as an accounts clerk, earning a small salary.

Ramanujan continued his mathematical work during this time and published his first paper in 1910 in an Indian journal. He realized that his work was beyond that of the Indian mathematicians of his day and decided to write to leading English mathematicians. The first mathematicians he wrote to turned down his request for help. But in January 1913 he wrote to G. H. Hardy, who was inclined to turn Ramanujan down, but the mathematical statements in the letter, although stated without proof, puzzled Hardy. He decided to examine them closely with the help of his colleague and collaborator J. E. Littlewood. They decided, after careful study, that Ramanujan was probably a genius, because his statements “could only be written down by a mathematician of the highest class; they must be true, because if they were not true, no one would have the imagination to invent them.”

Hardy arranged a scholarship for Ramanujan, bringing him to England in 1914. Hardy personally tutored him in mathematical analysis, and they collaborated for five years, proving significant theorems about the number of partitions of integers. During this time, Ramanujan made important contributions to number theory and also worked on continued fractions, infinite series, and elliptic functions. Ramanujan had amazing insight involving certain types of functions and series, but his purported theorems on prime numbers were often wrong, illustrating his vague idea of what constitutes a correct proof. He was one of the youngest members ever appointed a Fellow of the Royal Society. Unfortunately, in 1917 Ramanujan became extremely ill. At the time, it was thought that he had trouble with the English climate and had contracted tuberculosis. It is now thought that he suffered from a vitamin deficiency, brought on by Ramanujan’s strict vegetarianism and shortages in wartime England. He returned to India in 1919, continuing to do mathematics even when confined to his bed. He was religious and thought his mathematical talent came from his family deity, Namagiri. He considered mathematics and religion to be linked. He said that “an equation for me has no meaning unless it expresses a thought of God.” His short life came to an end in April 1920, when he was 32 years old. Ramanujan left several notebooks of unpublished results. The writings in these notebooks illustrate Ramanujan’s insights but are quite sketchy. Many mathematicians have devoted many years of work to explaining and justifying the results in these notebooks. An excellent movie, *The Man Who Knew Infinity*, about the life of Ramanujan was released in 2015.

by contraposition. If neither of these approaches works, you might try a proof by contradiction. However, we did not provide any further guidance how to create such proofs. We now present some strategies that you can use to develop new proofs.

FORWARD AND BACKWARD REASONING Whichever method you choose, you need a starting point for your proof. To begin a direct proof of a conditional statement, you start with the premises. Using these premises, together with axioms and known theorems, you can construct a proof using a sequence of steps that leads to the conclusion. This type of reasoning, called *forward reasoning*, is the most common type of reasoning used to prove relatively simple results. Similarly, with indirect reasoning you can start with the negation of the conclusion and, using a sequence of steps, obtain the negation of the premises.

Unfortunately, forward reasoning is often difficult to use to prove more complicated results, because the reasoning needed to reach the desired conclusion may be far from obvious. In such cases it may be helpful to use *backward reasoning*. To reason backward to prove a statement q , we find a statement p that we can prove with the property that $p \rightarrow q$. (Note that it is not helpful to find a statement r that you can prove such that $q \rightarrow r$, because it is the fallacy of begging the question to conclude from $q \rightarrow r$ and r that q is true.) Backward reasoning is illustrated in Examples 14 and 15.

EXAMPLE 14

Extra
Examples

Given two positive real numbers x and y , their **arithmetic mean** is $(x + y)/2$ and their **geometric mean** is $\sqrt{xy}$. When we compare the arithmetic and geometric means of pairs of distinct positive real numbers, we find that the arithmetic mean is always greater than the geometric mean. [For example, when $x = 4$ and $y = 6$, we have $5 = (4 + 6)/2 > \sqrt{4 \cdot 6} = \sqrt{24}$.] Can we prove that this inequality is always true?

Solution: To prove that $(x + y)/2 > \sqrt{xy}$ when x and y are distinct positive real numbers, we can work backward. We construct a sequence of equivalent inequalities. The equivalent inequalities are

$$\begin{aligned}(x + y)/2 &> \sqrt{xy}, \\(x + y)^2/4 &> xy, \\(x + y)^2 &> 4xy, \\x^2 + 2xy + y^2 &> 4xy, \\x^2 - 2xy + y^2 &> 0, \\(x - y)^2 &> 0.\end{aligned}$$

Because $(x - y)^2 > 0$ when $x \neq y$, it follows that the final inequality is true. Because all these inequalities are equivalent, it follows that $(x + y)/2 > \sqrt{xy}$ when $x \neq y$. Once we have carried out this backward reasoning, we can build a proof based on reversing the steps. This produces construct a proof using forward reasoning. (Note that the steps of our backward reasoning will not be part of the final proof. These steps serve as our guide for putting this proof together.)

Proof: Suppose that x and y are distinct positive real numbers. Then $(x - y)^2 > 0$ because the square of a nonzero real number is positive (see Appendix 1). Because $(x - y)^2 = x^2 - 2xy + y^2$, this implies that $x^2 - 2xy + y^2 > 0$. Adding $4xy$ to both sides, we obtain $x^2 + 2xy + y^2 > 4xy$. Because $x^2 + 2xy + y^2 = (x + y)^2$, this means that $(x + y)^2 \geq 4xy$. Dividing both sides of this equation by 4, we see that $(x + y)^2/4 > xy$. Finally, taking square roots of both sides (which preserves the inequality because both sides are positive) yields $(x + y)/2 > \sqrt{xy}$. We conclude that if x and y are distinct positive real numbers, then their arithmetic mean $(x + y)/2$ is greater than their geometric mean $\sqrt{xy}$. 

EXAMPLE 15 Suppose that two people play a game taking turns removing one, two, or three stones at a time from a pile that begins with 15 stones. The person who removes the last stone wins the game. Show that the first player can win the game no matter what the second player does.

Solution: To prove that the first player can always win the game, we work backward. At the last step, the first player can win if this player is left with a pile containing one, two, or three stones. The second player will be forced to leave one, two, or three stones if this player has to remove stones from a pile containing four stones. Consequently, one way for the first person to win is to leave four stones for the second player on the next-to-last move. The first person can leave four stones when there are five, six, or seven stones left at the beginning of this player's move, which happens when the second player has to remove stones from a pile with eight stones. Consequently, to force the second player to leave five, six, or seven stones, the first player should leave eight stones for the second player at the second-to-last move for the first player. This means that there are nine, ten, or eleven stones when the first player makes this move. Similarly, the first player should leave twelve stones when this player makes the first move. We can reverse this argument to show that the first player can always make moves so that this player wins the game no matter what the second player does. These moves successively leave twelve, eight, and four stones for the second player. 

ADAPTING EXISTING PROOFS An excellent way to look for possible approaches that can be used to prove a statement is to take advantage of existing proofs of similar results. Often an existing proof can be adapted to prove other facts. Even when this is not the case, some of the ideas used in existing proofs may be helpful. Because existing proofs provide clues for new proofs, you should read and understand the proofs you encounter in your studies. This process is illustrated in Example 16.

EXAMPLE 16 In Example 11 of Section 1.7 we proved that $\sqrt{2}$ is irrational. We now conjecture that $\sqrt{3}$ is irrational. Can we adapt the proof in Example 11 in Section 1.7 to show that $\sqrt{3}$ is irrational? 

Extra
Examples

Solution: To adapt the proof in Example 11 in Section 1.7, we begin by mimicking the steps in that proof, but with $\sqrt{2}$ replaced with $\sqrt{3}$. First, we suppose that $\sqrt{3} = c/d$ where the fraction c/d is in lowest terms. Squaring both sides tells us that $3 = c^2/d^2$, so that $3d^2 = c^2$. Can we use this equation to show that 3 must be a factor of both c and d , similar to how we used the equation $2b^2 = a^2$ in Example 11 in Section 1.7 to show that 2 must be a factor of both a and b ? (Recall that an integer s is a factor of the integer t if t/s is an integer. An integer n is even if and only if 2 is a factor of n .) It turns out that we can, but we need some ammunition from number theory, which we will develop in Chapter 4. We sketch out the remainder of the proof, but leave the justification of these steps until Chapter 4. Because 3 is a factor of c^2 , it must also be a factor of c . Furthermore, because 3 is a factor of c , 9 is a factor of c^2 , which means that 9 is a factor of $3d^2$. This implies that 3 is a factor of d^2 , which means that 3 is a factor of that d . This makes 3 a factor of both c and d , which contradicts the assumption that c/d is in lowest terms. After we have filled in the justification for these steps, we will have shown that $\sqrt{3}$ is irrational by adapting the proof that $\sqrt{2}$ is irrational. Note that this proof can be extended to show that $\sqrt{n}$ is irrational whenever n is a positive integer that is not a perfect square. We leave the details of this to Chapter 4. 

A good tip is to look for existing proofs that you might adapt when you are confronted with proving a new theorem, particularly when the new theorem seems similar to one you have already proved.

1.8.6 Looking for Counterexamples

In Section 1.7 we introduced the use of counterexamples to show that certain statements are false. When confronted with a conjecture, you might first try to prove this conjecture, and if your attempts are unsuccessful, you might try to find a counterexample, first by looking at the simplest, smallest examples. If you cannot find a counterexample, you might again try to prove the statement. In any case, looking for counterexamples is an extremely important pursuit, which often provides insights into problems. We will illustrate the role of counterexamples in Example 17.

EXAMPLE 17

Extra
Examples

In Example 15 in Section 1.7 we showed that the statement “Every positive integer is the sum of two squares of integers” is false by finding a counterexample. That is, there are positive integers that cannot be written as the sum of the squares of two integers. Although we cannot write every positive integer as the sum of the squares of two integers, maybe we can write every positive integer as the sum of the squares of three integers. That is, is the statement “Every positive integer is the sum of the squares of three integers” true or false?

Solution: Because we know that not every positive integer can be written as the sum of two squares of integers, we might initially be skeptical that every positive integer can be written as the sum of three squares of integers. So, we first look for a counterexample. That is, we can show that the statement “Every positive integer is the sum of three squares of integers” is false if we can find a particular integer that is not the sum of the squares of three integers. To look for a counterexample, we try to write successive positive integers as a sum of three squares. We find that $1 = 0^2 + 0^2 + 1^2$, $2 = 0^2 + 1^2 + 1^2$, $3 = 1^2 + 1^2 + 1^2$, $4 = 0^2 + 0^2 + 2^2$, $5 = 0^2 + 1^2 + 2^2$, $6 = 1^2 + 1^2 + 2^2$, but we cannot find a way to write 7 as the sum of three squares. To show that there are not three squares that add up to 7, we note that the only possible squares we can use are those not exceeding 7, namely, 0, 1, and 4. Because no three terms where each term is 0, 1, or 4 add up to 7, it follows that 7 is a counterexample. We conclude that the statement “Every positive integer is the sum of the squares of three integers” is false.

We have shown that not every positive integer is the sum of the squares of three integers. The next question to ask is whether every positive integer is the sum of the squares of four positive integers. Some experimentation provides evidence that the answer is yes. For example, $7 = 1^2 + 1^2 + 1^2 + 2^2$, $25 = 4^2 + 2^2 + 2^2 + 1^2$, and $87 = 9^2 + 2^2 + 1^2 + 1^2$. It turns out the conjecture “Every positive integer is the sum of the squares of four integers” is true. For a proof, see [Ro10].

1.8.7 Proof Strategy in Action

Mathematics is generally taught as if mathematical facts were carved in stone. Mathematics texts (including the bulk of this book) formally present theorems and their proofs. Such presentations do not convey the discovery process in mathematics. This process begins with exploring concepts and examples, asking questions, formulating conjectures, and attempting to settle these conjectures either by proof or by counterexample. These are the day-to-day activities of mathematicians. Believe it or not, the material taught in textbooks was originally developed in this way.

Extra
Examples

People formulate conjectures on the basis of many types of possible evidence. The examination of special cases can lead to a conjecture, as can the identification of possible patterns. Altering the hypotheses and conclusions of known theorems also can lead to plausible conjectures. At other times, conjectures are made based on intuition or a belief that a result holds. No matter how a conjecture was made, once it has been formulated, the goal is to prove or disprove it. When mathematicians believe that a conjecture may be true, they try to find a proof. If they cannot find a proof, they may look for a counterexample. When they cannot find a counterexample, they may switch gears and once again try to prove the conjecture. Although many

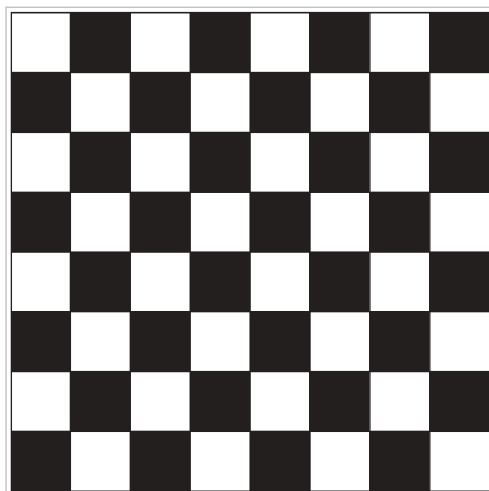


FIGURE 2 The standard checkerboard.

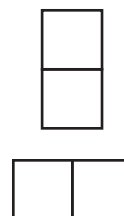


FIGURE 3 Two dominoes.

conjectures are quickly settled, a few conjectures resist attack for hundreds of years and lead to the development of new parts of mathematics. We will mention a few famous conjectures later in this section.

1.8.8 Tilings

Links

We can illustrate aspects of proof strategy through a brief study of tilings of checkerboards. Looking at tilings of checkerboards is a fruitful way to quickly discover many different results and construct their proofs using a variety of proof methods. There are almost an endless number of conjectures that can be made and studied in this area, too. To begin, we need to define some terms. A **checkerboard** is a rectangle divided into squares of the same size by horizontal and vertical lines. The game of checkers is played on a board with 8 rows and 8 columns; this board is called the **standard checkerboard** and is shown in Figure 2. In this section we use the term **board** to refer to a checkerboard of any rectangular size as well as parts of checkerboards obtained by removing one or more squares. A **domino** is a rectangular piece that is one square by two squares, as shown in Figure 3. We say that a board is **tiled** by dominoes when all its squares are covered with no overlapping dominoes and no dominoes overhanging the board. We now develop some results about tiling boards using dominoes.

EXAMPLE 18 Can we tile the standard checkerboard using dominoes?

Solution: We can find many ways to tile the standard checkerboard using dominoes. For example, we can tile it by placing 32 dominoes horizontally, as shown in Figure 4. The existence of one such tiling completes a constructive existence proof. There are a large number of other ways to do this tiling. We can place 32 dominoes vertically on the board or we can place some tiles vertically and some horizontally. But for a constructive existence proof we needed to find just one such tiling. ◀

EXAMPLE 19 Can we tile a board obtained by removing one of the four corner squares of a standard checkerboard?

Extra Examples

Solution: To answer this question, note that a standard checkerboard has 64 squares, so removing a square produces a board with 63 squares. Now suppose that we could tile a board obtained from the standard checkerboard by removing a corner square. The board has an even number of squares because each domino covers two squares and no two dominoes overlap and no dominoes

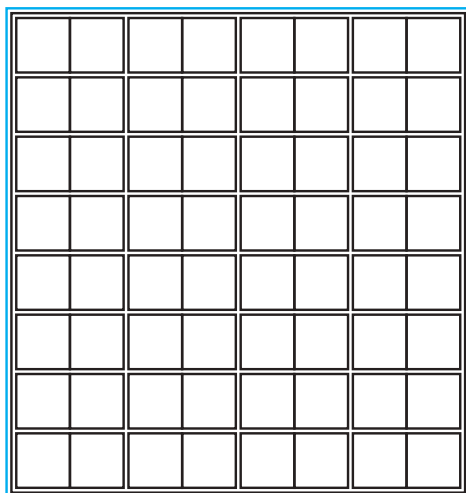


FIGURE 4 Tiling the standard checkerboard.

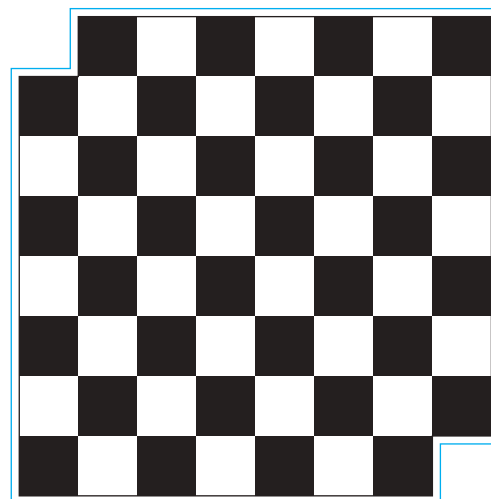


FIGURE 5 The standard checkerboard with the upper left and lower right squares removed.

overhang the board. Consequently, we can prove by contradiction that a standard checkerboard with one square removed cannot be tiled using dominoes because such a board has an odd number of squares. ◀

We now consider a trickier situation.

EXAMPLE 20 Can we tile the board obtained by deleting the upper left and lower right corner squares of a standard checkerboard, shown in Figure 5?

Solution: A board obtained by deleting two squares of a standard checkerboard contains $64 - 2 = 62$ squares. Because 62 is even, we cannot quickly rule out the existence of a tiling of the standard checkerboard with its upper left and lower right squares removed, unlike Example 19, where we ruled out the existence of a tiling of the standard checkerboard with one corner square removed. Trying to construct a tiling of this board by successively placing dominoes might be a first approach, as the reader should attempt. However, no matter how much we try, we cannot find such a tiling. Because our efforts do not produce a tiling, we are led to conjecture that no tiling exists.

We might try to prove that no tiling exists by showing that we reach a dead end however we successively place dominoes on the board. To construct such a proof, we would have to consider all possible cases that arise as we run through all possible choices of successively placing dominoes. For example, we have two choices for covering the square in the second column of the first row, next to the removed top left corner. We could cover it with a horizontally placed tile or a vertically placed tile. Each of these two choices leads to further choices, and so on. It does not take long to see that this is not a fruitful plan of attack for a person, although a computer could be used to complete such a proof by exhaustion. (Exercise 47 asks you to supply such a proof to show that a 4×4 checkerboard with opposite corners removed cannot be tiled.)

We need another approach. Perhaps there is an easier way to prove there is no tiling of a standard checkerboard with two opposite corners removed. As with many proofs, a key observation can help. We color the squares of this checkerboard using alternating white and black squares, as in Figure 2. Observe that a domino in a tiling of such a board covers one white square and one black square. Next, note that this board has unequal numbers of white squares and black

squares. We can use these observations to prove by contradiction that a standard checkerboard with opposite corners removed cannot be tiled using dominoes. We now present such a proof.

Proof: Suppose we can use dominoes to tile a standard checkerboard with opposite corners removed. Note that the standard checkerboard with opposite corners removed contains $64 - 2 = 62$ squares. The tiling would use $62/2 = 31$ dominoes. Note that each domino in this tiling covers one white and one black square. Consequently, the tiling covers 31 white squares and 31 black squares. However, when we remove two opposite corner squares, either 32 of the remaining squares are white and 30 are black or else 30 are white and 32 are black. This contradicts the assumption that we can use dominoes to cover a standard checkerboard with opposite corners removed, completing the proof. ◀

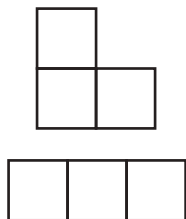


FIGURE 6 A right triomino and a straight triomino.

We can use other types of pieces besides dominoes in tilings. Instead of dominoes we can study tilings that use identically shaped pieces constructed from congruent squares that are connected along their edges. Such pieces are called **polyominoes**, a term coined in 1953 by the mathematician Solomon Golomb, the author of an entertaining book about them [Go94]. We will consider two polyominoes with the same number of squares the same if we can rotate and/or flip one of the polyominoes to get the other one. For example, there are two types of triominoes (see Figure 6), which are polyominoes made up of three squares connected by their sides. One type of triomino, the **straight triomino**, has three horizontally connected squares; the other type, **right triominoes**, resembles the letter L in shape, flipped and/or rotated, if necessary. We will study the tilings of a checkerboard by straight triominoes here; we will study tilings by right triominoes in Section 5.1.

EXAMPLE 21 Can you use straight triominoes to tile a standard checkerboard?

Solution: The standard checkerboard contains 64 squares and each triomino covers three squares. Consequently, if triominoes tile a board, the number of squares of the board must be a multiple of 3. Because 64 is not a multiple of 3, triominoes cannot be used to cover an 8×8 checkerboard. ◀

In Example 22, we consider the problem of using straight triominoes to tile a standard checkerboard with one corner missing.

EXAMPLE 22 Can we use straight triominoes to tile a standard checkerboard with one of its four corners removed? An 8×8 checkerboard with one corner removed contains $64 - 1 = 63$ squares. Any tiling by straight triominoes of one of these four boards uses $63/3 = 21$ triominoes. However, when we experiment, we cannot find a tiling of one of these boards using straight triominoes. A proof by exhaustion does not appear promising. Can we adapt our proof from Example 20 to prove that no such tiling exists?

Solution: We will color the squares of the checkerboard in an attempt to adapt the proof by contradiction we gave in Example 20 of the impossibility of using dominoes to tile a standard checkerboard with opposite corners removed. Because we are using straight triominoes rather than dominoes, we color the squares using three colors rather than two colors, as shown in Figure 7. Note that there are 21 blue squares, 21 black squares, and 22 white squares in this coloring. Next, we make the crucial observation that when a straight triomino covers three squares of the checkerboard, it covers one blue square, one black square, and one white square. Next, note that each of the three colors appears in a corner square. Thus, without loss of generality, we may assume that we have rotated the coloring so that the missing square is colored blue. Therefore, we assume that the remaining board contains 20 blue squares, 21 black squares, and 22 white squares.

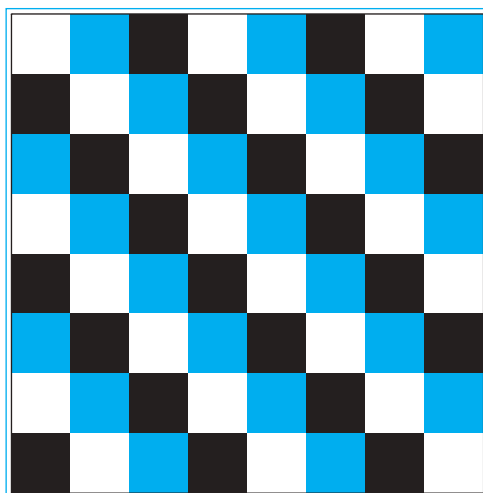


FIGURE 7 Coloring the squares of the standard checkerboard with three colors.

If we could tile this board using straight triominoes, then we would use $63/3 = 21$ straight triominoes. These triominoes would cover 21 blue squares, 21 black squares, and 21 white squares. This contradicts the fact that this board contains 20 blue squares, 21 black squares, and 22 white squares. Therefore, we cannot tile this board using straight triominoes. ◀

1.8.9 The Role of Open Problems

Many advances in mathematics have been made by people trying to solve famous unsolved problems. In the past 20 years, many unsolved problems have finally been resolved, such as the proof of a conjecture in number theory made more than 300 years ago. This conjecture asserts the truth of the statement known as **Fermat's last theorem**.

THEOREM 1 **FERMAT'S LAST THEOREM** The equation

$$x^n + y^n = z^n$$

has no solutions in integers x , y , and z with $xyz \neq 0$ whenever n is an integer with $n > 2$.

Links ▶ **Remark:** The equation $x^2 + y^2 = z^2$ has infinitely many solutions in integers x , y , and z ; these solutions are called Pythagorean triples and correspond to the lengths of the sides of right triangles with integer lengths. See Exercise 34.

This problem has a fascinating history. In the seventeenth century, Fermat jotted in the margin of his copy of the works of Diophantus that he had a “wondrous proof” that there are no integer solutions of $x^n + y^n = z^n$ when n is an integer greater than 2 with $xyz \neq 0$. However, he never published a proof (Fermat published almost nothing), and no proof could be found in the papers he left when he died. Mathematicians looked for a proof for three centuries without success, although many people were convinced that a relatively simple proof could be found. (Proofs of special cases were found, such as the proof of the case when $n = 3$ by Euler and the proof of the $n = 4$ case by Fermat himself.) Over the years, several established mathematicians thought that they had proved this theorem. In the nineteenth century, one of these failed

attempts led to the development of the part of number theory called algebraic number theory. A correct proof, requiring hundreds of pages of advanced mathematics, was not found until the 1990s, when Andrew Wiles used recently developed ideas from a sophisticated area of number theory called the theory of elliptic curves to prove Fermat's last theorem. Wiles's quest to find a proof of Fermat's last theorem using this powerful theory, described in a program in the *Nova* series on public television, took close to ten years! Moreover, his proof was based on major contributions of many mathematicians. (The interested reader should consult [Ro10] for more information about Fermat's last theorem and for additional references concerning this problem and its resolution.)

We now state an open problem that is simple to describe, but that seems quite difficult to resolve.

EXAMPLE 23

Links

The $3x + 1$ Conjecture Let T be the transformation that sends an even integer x to $x/2$ and an odd integer x to $3x + 1$. A famous conjecture, sometimes known as the **$3x + 1$ conjecture**, states that for all positive integers x , when we repeatedly apply the transformation T , we will eventually reach the integer 1. For example, starting with $x = 13$, we find $T(13) = 3 \cdot 13 + 1 = 40$, $T(40) = 40/2 = 20$, $T(20) = 20/2 = 10$, $T(10) = 10/2 = 5$, $T(5) = 3 \cdot 5 + 1 = 16$, $T(16) = 8$, $T(8) = 4$, $T(4) = 2$, and $T(2) = 1$. The $3x + 1$ conjecture has been verified using computers for all integers x up to $5.48 \cdot 10^{18}$.

The $3x + 1$ conjecture has an interesting history and has attracted the attention of mathematicians since the 1950s. The conjecture has been raised many times and goes by many other names, including the Collatz problem, Hasse's algorithm, Ulam's problem, the Syracuse problem, and Kakutani's problem. Many mathematicians have been diverted from their work to spend time attacking this conjecture. This led to the joke that this problem was part of a conspiracy to slow down American mathematical research. See the article by Jeffrey Lagarias [La10] for a fascinating discussion of this problem and the results that have been found by mathematicians attacking it.

Watch out! Working on the $3x + 1$ problem can be addictive.

There are a surprising number of important open problems throughout discrete mathematics. For instance, in Chapter 4 you will encounter many open questions about prime numbers. (Students already familiar with the basic notions about primes might want to explore Section



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ANDREW WILES (born 1953) Andrew Wiles was born in Cambridge, England. His father was a Professor of Divinity. Wiles attended the King's College School and the Leys School in Cambridge. Wiles became interested in Fermat's last theorem when at age ten he read a book stating the problem. He knew then that he would never let this problem go, as it looked simple but none of the great mathematicians could solve it. Wiles entered Merton College, Oxford in 1971. He received his B.A. in 1974, and then entered Clare College, Cambridge, for his graduate studies. He received his Ph.D. in 1980; his graduate research was on the theory of elliptic curves. He was a Benjamin Peirce Assistant Professor at Harvard University from 1977 until 1980. In 1981, he held a post at the Institute for Advanced Study in Princeton, and in 1982 he was appointed to a professorship at Princeton University. He was awarded a Guggenheim Fellowship in 1985 and spent a year at the Institut des Hautes Études Scientifiques and the École Normale Supérieure in Paris.

Ironically, he did not realize that during his years working on elliptic curves he was learning techniques that would later help him solve the problem that obsessed him.

In 1986 when Wiles learned of work that showed that Fermat's last theorem follows from a conjecture in the theory of elliptic curves, he realized that this led to a possible strategy for a proof. He abandoned his ongoing research and devoted himself entirely to working on Fermat's last theorem. It took him more than seven years to complete his proof and two more years for some parts of the proof to be corrected. During this time he spent time only on this problem and with his young daughters. In 1988 he took a position as a research professor at Oxford University, returning to Princeton in 1990, where he remained until 2011, when he rejoined Oxford University as the Royal Society Research Professor.

Not only did Wiles become famous when he proved Fermat's last theorem, he also won the Wolfskehl Prize, which was established in 1908 for the first correct proof. This prize included 100,000 German marks (in the currency of the day), which would have been worth over \$1,500,000 today. Although it was common wisdom that this prize had become worthless because of the two world wars, currency changes, and hyperinflation, Wiles received approximately \$50,000. Wiles has won many of the top awards in mathematics, including the Abel Prize, the Fermat Prize, and the Wolf Prize. In 2000, he was made a Knight Commander of the Order of the British Empire by the Queen of England, making him Sir Andrew Wiles.

4.3, where these open questions are discussed.) You will encounter many other open questions as you read this book. The study of such problems has played and continues to play an important role in the development of many parts of discrete mathematics.

Build up your arsenal of proof methods as you work through this book.

1.8.10 Additional Proof Methods

In this chapter we introduced the basic methods used in proofs. We also described how to leverage these methods to prove a variety of results. We will use these proof methods in all subsequent chapters. In particular, we will use them in Chapters 2, 3, and 4 to prove results about sets, functions, algorithms, and number theory and in Chapters 9, 10, and 11 to prove results in graph theory. Among the theorems we will prove is the famous halting theorem, which states that there is a problem that cannot be solved using any procedure. However, there are many important proof methods besides those we have covered. We will introduce some of these methods later in this book. In particular, in Section 5.1 we will discuss mathematical induction, which is an extremely useful method for proving statements of the form $\forall nP(n)$, where the domain consists of all positive integers. In Section 5.3 we will introduce structural induction, which can be used to prove results about recursively defined sets. We will use the Cantor diagonalization method, which can be used to prove results about the size of infinite sets, in Section 2.5. In Chapter 6 we will introduce the notion of combinatorial proofs, which can be used to prove results by counting arguments. The reader should note that entire books have been devoted to the activities discussed in this section, including many excellent works by George Pólya ([Po61], [Po71], [Po90]).

Finally, note that we have not given a procedure that can be used for proving theorems in mathematics. It is a deep theorem of mathematical logic that there is no such procedure.

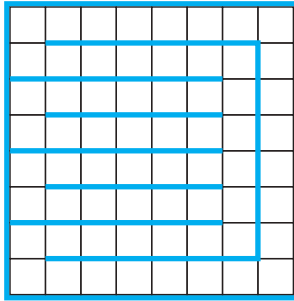
Exercises

1. Prove that $n^2 + 1 \geq 2^n$ when n is a positive integer with $1 \leq n \leq 4$.
2. Use a proof by cases to show that 10 is not the square of a positive integer. [Hint: Consider two cases: (i) $1 \leq x \leq 3$, (ii) $x \geq 4$.]
3. Use a proof by cases to show that 100 is not the cube of a positive integer. [Hint: Consider two cases: (i) $1 \leq x \leq 4$, (ii) $x \geq 5$.]
4. Prove that there are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers.
5. Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$. [Hint: Use a proof by cases, with the two cases corresponding to $x \geq y$ and $x < y$, respectively.]
6. Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a , b , and c are real numbers.
7. Prove using the notion of without loss of generality that $\min(x, y) = (x + y - |x - y|)/2$ and $\max(x, y) = (x + y + |x - y|)/2$ whenever x and y are real numbers.
8. Prove using the notion of without loss of generality that $5x + 5y$ is an odd integer when x and y are integers of opposite parity.
9. Prove the **triangle inequality**, which states that if x and y are real numbers, then $|x| + |y| \geq |x + y|$ (where $|x|$ represents the absolute value of x , which equals x if $x \geq 0$ and equals $-x$ if $x < 0$).
10. Prove that there is a positive integer that equals the sum of the positive integers not exceeding it. Is your proof constructive or nonconstructive?
11. Prove that there are 100 consecutive positive integers that are not perfect squares. Is your proof constructive or nonconstructive?
12. Prove that either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. Is your proof constructive or nonconstructive?
13. Prove that there exists a pair of consecutive integers such that one of these integers is a perfect square and the other is a perfect cube.
14. Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is nonnegative. Is your proof constructive or nonconstructive? [Hint: Do not try to evaluate these numbers!]
15. Prove or disprove that there is a rational number x and an irrational number y such that x^y is irrational.
16. Prove or disprove that if a and b are rational numbers, then a^b is also rational.

17. Show that each of these statements can be used to express the fact that there is a unique element x such that $P(x)$ is true. [Note that we can also write this statement as $\exists!xP(x)$.]
 - a) $\exists x\forall y(P(y) \leftrightarrow x = y)$
 - b) $\exists xP(x) \wedge \forall x\forall y(P(x) \wedge P(y) \rightarrow x = y)$
 - c) $\exists x(P(x) \wedge \forall y(P(y) \rightarrow x = y))$
18. Show that if a , b , and c are real numbers and $a \neq 0$, then there is a unique solution of the equation $ax + b = c$.
19. Suppose that a and b are odd integers with $a \neq b$. Show there is a unique integer c such that $|a - c| = |b - c|$.
20. Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than $1/2$.
21. Show that if n is an odd integer, then there is a unique integer k such that n is the sum of $k - 2$ and $k + 3$.
22. Prove that given a real number x there exist unique numbers n and ϵ such that $x = n + \epsilon$, n is an integer, and $0 \leq \epsilon < 1$.
23. Prove that given a real number x there exist unique numbers n and ϵ such that $x = n - \epsilon$, n is an integer, and $0 \leq \epsilon < 1$.
24. Use forward reasoning to show that if x is a nonzero real number, then $x^2 + 1/x^2 \geq 2$. [Hint: Start with the inequality $(x - 1/x)^2 \geq 0$, which holds for all nonzero real numbers x .]
25. The **harmonic mean** of two real numbers x and y equals $2xy/(x + y)$. By computing the harmonic and geometric means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.
26. The **quadratic mean** of two real numbers x and y equals $\sqrt{(x^2 + y^2)/2}$. By computing the arithmetic and quadratic means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.
- *27. Write the numbers $1, 2, \dots, 2n$ on a blackboard, where n is an odd integer. Pick any two of the numbers, j and k , write $|j - k|$ on the board and erase j and k . Continue this process until only one integer is written on the board. Prove that this integer must be odd.
- *28. Suppose that five ones and four zeros are arranged around a circle. Between any two equal bits you insert a 0 and between any two unequal bits you insert a 1 to produce nine new bits. Then you erase the nine original bits. Show that when you iterate this procedure, you can never get nine zeros. [Hint: Work backward, assuming that you did end up with nine zeros.]
29. Formulate a conjecture about the decimal digits that appear as the final decimal digit of the fourth power of an integer. Prove your conjecture using a proof by cases.
30. Formulate a conjecture about the final two decimal digits of the square of an integer. Prove your conjecture using a proof by cases.
31. Prove that there is no positive integer n such that $n^2 + n^3 = 100$.
32. Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.
33. Prove that there are no solutions in positive integers x and y to the equation $x^4 + y^4 = 625$.
34. Prove that there are infinitely many solutions in positive integers x , y , and z to the equation $x^2 + y^2 = z^2$. [Hint: Let $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$, where m and n are integers.]
35. Adapt the proof in Example 4 in Section 1.7 to prove that if $n = abc$, where a , b , and c are positive integers, then $a \leq \sqrt[3]{n}$, $b \leq \sqrt[3]{n}$, or $c \leq \sqrt[3]{n}$.
36. Prove that $\sqrt[3]{2}$ is irrational.
37. Prove that between every two rational numbers there is an irrational number.
38. Prove that between every rational number and every irrational number there is an irrational number.
- *39. Let $S = x_1y_1 + x_2y_2 + \dots + x_ny_n$, where $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ are orderings of two different sequences of positive real numbers, each containing n elements.
 - a) Show that S takes its maximum value over all orderings of the two sequences when both sequences are sorted (so that the elements in each sequence are in nondecreasing order).
 - b) Show that S takes its minimum value over all orderings of the two sequences when one sequence is sorted into nondecreasing order and the other is sorted into nonincreasing order.
40. Prove or disprove that if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons, respectively, then you can measure 4 gallons by successively pouring some of or all of the water in a jug into another jug.
41. Verify the $3x + 1$ conjecture for these integers.
 - a) 6 b) 7 c) 17 d) 21
42. Verify the $3x + 1$ conjecture for these integers.
 - a) 16 b) 11 c) 35 d) 113
43. Prove or disprove that you can use dominoes to tile the standard checkerboard with two adjacent corners removed (that is, corners that are not opposite).
44. Prove or disprove that you can use dominoes to tile a standard checkerboard with all four corners removed.
45. Prove that you can use dominoes to tile a rectangular checkerboard with an even number of squares.
46. Prove or disprove that you can use dominoes to tile a 5×5 checkerboard with three corners removed.
47. Use a proof by exhaustion to show that a tiling using dominoes of a 4×4 checkerboard with opposite corners removed does not exist. [Hint: First show that you can assume that the squares in the upper left and lower right corners are removed. Number the squares of the original checkerboard from 1 to 16, starting in the first row, moving right in this row, then starting in the leftmost square in the second row and moving right, and so on. Remove squares 1 and 16. To begin the proof, note that square 2 is covered either by a domino laid horizontally, which

covers squares 2 and 3, or vertically, which covers squares 2 and 6. Consider each of these cases separately, and work through all the subcases that arise.]

- *48. Prove that when a white square and a black square are removed from an 8×8 checkerboard (colored as in the text) you can tile the remaining squares of the checkerboard using dominoes. [Hint: Show that when one black and one white square are removed, each part of the partition of the remaining cells formed by inserting the barriers shown in the figure can be covered by dominoes.]



49. Show that by removing two white squares and two black squares from an 8×8 checkerboard (colored as in the text) you can make it impossible to tile the remaining squares using dominoes.

- *50. Find all squares, if they exist, on an 8×8 checkerboard such that the board obtained by removing one of these squares can be tiled using straight triominoes. [Hint: First use arguments based on coloring and rotations to eliminate as many squares as possible from consideration.]

- *51. a) Draw each of the five different tetrominoes, where a tetromino is a polyomino consisting of four squares.

- b) For each of the five different tetrominoes, prove or disprove that you can tile a standard checkerboard using these tetrominoes.

- *52. Prove or disprove that you can tile a 10×10 checkerboard using straight tetrominoes.

Key Terms and Results

TERMS

proposition: a statement that is true or false

propositional variable: a variable that represents a proposition

truth value: true or false

$\neg p$ (**negation of p**): the proposition with truth value opposite to the truth value of p

logical operators: operators used to combine propositions

compound proposition: a proposition constructed by combining propositions using logical operators

truth table: a table displaying all possible truth values of propositions

$p \vee q$ (**disjunction of p and q**): the proposition " p or q ," which is true if and only if at least one of p and q is true

$p \wedge q$ (**conjunction of p and q**): the proposition " p and q ," which is true if and only if both p and q are true

$p \oplus q$ (**exclusive or of p and q**): the proposition " p XOR q ," which is true when exactly one of p and q is true

$p \rightarrow q$ (**p implies q**): the proposition "if p , then q ," which is false if and only if p is true and q is false

converse of $p \rightarrow q$: the conditional statement $q \rightarrow p$

contrapositive of $p \rightarrow q$: the conditional statement $\neg q \rightarrow \neg p$

inverse of $p \rightarrow q$: the conditional statement $\neg p \rightarrow \neg q$

$p \leftrightarrow q$ (**biconditional**): the proposition " p if and only if q ," which is true if and only if p and q have the same truth value

bit: either a 0 or a 1

Boolean variable: a variable that has a value of 0 or 1

bit operation: an operation on a bit or bits

bit string: a list of bits

bitwise operations: operations on bit strings that operate on each bit in one string and the corresponding bit in the other string

logic gate: a logic element that performs a logical operation on one or more bits to produce an output bit

logic circuit: a switching circuit made up of logic gates that produces one or more output bits

tautology: a compound proposition that is always true

contradiction: a compound proposition that is always false

contingency: a compound proposition that is sometimes true and sometimes false

consistent compound propositions: compound propositions for which there is an assignment of truth values to the variables that makes all these propositions true

satisfiable compound proposition: a compound proposition for which there is an assignment of truth values to its variables that makes it true

logically equivalent compound propositions: compound propositions that always have the same truth values

predicate: part of a sentence that attributes a property to the subject

propositional function: a statement containing one or more variables that becomes a proposition when each of its variables is assigned a value or is bound by a quantifier

domain (or universe) of discourse: the values a variable in a propositional function may take

$\exists x P(x)$ (existential quantification of $P(x)$): the proposition that is true if and only if there exists an x in the domain such that $P(x)$ is true

$\forall x P(x)$ (universal quantification of $P(x)$): the proposition that is true if and only if $P(x)$ is true for every x in the domain

logically equivalent expressions: expressions that have the same truth value no matter which propositional functions and domains are used

free variable: a variable not bound in a propositional function

bound variable: a variable that is quantified

scope of a quantifier: portion of a statement where the quantifier binds its variable

argument: a sequence of statements

argument form: a sequence of compound propositions involving propositional variables

premise: a statement, in an argument, or argument form, other than the final one

conclusion: the final statement in an argument or argument form

valid argument form: a sequence of compound propositions involving propositional variables where the truth of all the premises implies the truth of the conclusion

valid argument: an argument with a valid argument form

rule of inference: a valid argument form that can be used in the demonstration that arguments are valid

fallacy: an invalid argument form often used incorrectly as a rule of inference (or sometimes, more generally, an incorrect argument)

circular reasoning or begging the question: reasoning where one or more steps are based on the truth of the statement being proved

theorem: a mathematical assertion that can be shown to be true

conjecture: a mathematical assertion proposed to be true, but that has not been proved

proof: a demonstration that a theorem is true

axiom: a statement that is assumed to be true and that can be used as a basis for proving theorems

lemma: a theorem used to prove other theorems

corollary: a proposition that can be proved as a consequence of a theorem that has just been proved

vacuous proof: a proof that $p \rightarrow q$ is true based on the fact that p is false

trivial proof: a proof that $p \rightarrow q$ is true based on the fact that q is true

direct proof: a proof that $p \rightarrow q$ is true that proceeds by showing that q must be true when p is true

proof by contraposition: a proof that $p \rightarrow q$ is true that proceeds by showing that p must be false when q is false

proof by contradiction: a proof that p is true based on the truth of the conditional statement $\neg p \rightarrow q$, where q is a contradiction

exhaustive proof: a proof that establishes a result by checking a list of all possible cases

proof by cases: a proof broken into separate cases, where these cases cover all possibilities

without loss of generality: an assumption in a proof that makes it possible to prove a theorem by reducing the number of cases to consider in the proof

counterexample: an element x such that $P(x)$ is false

constructive existence proof: a proof that an element with a specified property exists that explicitly finds such an element

nonconstructive existence proof: a proof that an element with a specified property exists that does not explicitly find such an element

rational number: a number that can be expressed as the ratio of two integers p and q such that $q \neq 0$

uniqueness proof: a proof that there is exactly one element satisfying a specified property

RESULTS

The logical equivalences given in Tables 6, 7, and 8 in Section 1.3.

De Morgan's laws for quantifiers.

Rules of inference for propositional calculus.

Rules of inference for quantified statements.

Review Questions

- Define the negation of a proposition.
 - What is the negation of "This is a boring course"?
- Define (using truth tables) the disjunction, conjunction, exclusive or, conditional, and biconditional of the propositions p and q .
 - What are the disjunction, conjunction, exclusive or, conditional, and biconditional of the propositions "I'll go to the movies tonight" and "I'll finish my discrete mathematics homework"?
- Describe at least five different ways to write the conditional statement $p \rightarrow q$ in English.
 - Define the converse and contrapositive of a conditional statement.
 - State the converse and the contrapositive of the conditional statement "If it is sunny tomorrow, then I will go for a walk in the woods."
- What does it mean for two propositions to be logically equivalent?
 - Describe the different ways to show that two compound propositions are logically equivalent.
 - Show in at least two different ways that the compound propositions $\neg p \vee (r \rightarrow \neg q)$ and $\neg p \vee \neg q \vee \neg r$ are equivalent.

5. (Depends on the Exercise Set in Section 1.3)
- Given a truth table, explain how to use disjunctive normal form to construct a compound proposition with this truth table.
 - Explain why part (a) shows that the operators $\wedge$, $\vee$, and $\neg$ are functionally complete.
 - Is there an operator such that the set containing just this operator is functionally complete?
6. What are the universal and existential quantifications of a predicate $P(x)$? What are their negations?
7. a) What is the difference between the quantification $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$, where $P(x, y)$ is a predicate?
- b) Give an example of a predicate $P(x, y)$ such that $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$ have different truth values.
8. Describe what is meant by a valid argument in propositional logic and show that the argument “If the earth is flat, then you can sail off the edge of the earth,” “You cannot sail off the edge of the earth,” therefore, “The earth is not flat” is a valid argument.
9. Use rules of inference to show that if the premises “All zebras have stripes” and “Mark is a zebra” are true, then the conclusion “Mark has stripes” is true.
10. a) Describe what is meant by a direct proof, a proof by contraposition, and a proof by contradiction of a conditional statement $p \rightarrow q$.
- b) Give a direct proof, a proof by contraposition, and a proof by contradiction of the statement: “If n is even, then $n + 4$ is even.”
11. a) Describe a way to prove the biconditional $p \leftrightarrow q$.
- b) Prove the statement: “The integer $3n + 2$ is odd if and only if the integer $9n + 5$ is even, where n is an integer.”
12. To prove that the statements p_1, p_2, p_3 , and p_4 are equivalent, is it sufficient to show that the conditional statements $p_4 \rightarrow p_2, p_3 \rightarrow p_1$, and $p_1 \rightarrow p_2$ are valid? If not, provide another collection of conditional statements that can be used to show that the four statements are equivalent.
13. a) Suppose that a statement of the form $\forall x P(x)$ is false. How can this be proved?
- b) Show that the statement “For every positive integer n , $n^2 \geq 2n$ ” is false.
14. What is the difference between a constructive and non-constructive existence proof? Give an example of each.
15. What are the elements of a proof that there is a unique element x such that $P(x)$, where $P(x)$ is a propositional function?
16. Explain how a proof by cases can be used to prove a result about absolute values, such as the fact that $|xy| = |x||y|$ for all real numbers x and y .

Supplementary Exercises

- Let p be the proposition “I will do every exercise in this book” and q be the proposition “I will get an A in this course.” Express each of these as a combination of p and q .
 - I will get an A in this course only if I do every exercise in this book.
 - I will get an A in this course and I will do every exercise in this book.
 - Either I will not get an A in this course or I will not do every exercise in this book.
 - For me to get an A in this course it is necessary and sufficient that I do every exercise in this book.
- Find the truth table of the compound proposition $(p \vee q) \rightarrow (p \wedge \neg r)$.
- Show that these compound propositions are tautologies.
 - $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
 - $((p \vee q) \wedge \neg p) \rightarrow q$
- Give the converse, the contrapositive, and the inverse of these conditional statements.
 - If it rains today, then I will drive to work.
 - If $|x| = x$, then $x \geq 0$.
 - If n is greater than 3, then n^2 is greater than 9.
- Given a conditional statement $p \rightarrow q$, find the converse of its inverse, the converse of its converse, and the converse of its contrapositive.
- Given a conditional statement $p \rightarrow q$, find the inverse of its inverse, the inverse of its converse, and the inverse of its contrapositive.
- Find a compound proposition involving the propositional variables p, q, r , and s that is true when exactly three of these propositional variables are true and is false otherwise.
- Show that these statements are inconsistent: “If Sergei takes the job offer, then he will get a signing bonus.” “If Sergei takes the job offer, then he will receive a higher salary.” “If Sergei gets a signing bonus, then he will not receive a higher salary.” “Sergei takes the job offer.”
- Show that these statements are inconsistent: “If Miranda does not take a course in discrete mathematics, then she will not graduate.” “If Miranda does not graduate, then she is not qualified for the job.” “If Miranda reads this book, then she is qualified for the job.” “Miranda does not take a course in discrete mathematics but she reads this book.”

Teachers in the Middle Ages supposedly tested the realtime propositional logic ability of a student via a technique known as an **obligato game**. In an obligato game, a number of rounds is set and in each round the teacher gives the student successive assertions that the student must either accept or reject as they are given. When the student accepts an assertion, it is

added as a commitment; when the student rejects an assertion its negation is added as a commitment. The student passes the test if the consistency of all commitments is maintained throughout the test.

10. Suppose that in a three-round obligato game, the teacher first gives the student the proposition $p \rightarrow q$, then the proposition $\neg(p \vee r) \vee q$, and finally, the proposition q . For which of the eight possible sequences of three answers will the student pass the test?
11. Suppose that in a four-round obligato game, the teacher first gives the student the proposition $\neg(p \rightarrow (q \wedge r))$, then the proposition $p \vee \neg q$, then the proposition $\neg r$, and finally, the proposition $(p \wedge r) \vee (q \rightarrow p)$. For which of the 16 possible sequences of four answers will the student pass the test?
12. Explain why every obligato game has a winning strategy. Exercises 13 and 14 are set on the island of knights and knaves described in Example 7 in Section 1.2.
13. Suppose that you meet three people, Aaron, Bohan, and Crystal. Can you determine what Aaron, Bohan, and Crystal are if Aaron says “All of us are knaves” and Bohan says “Exactly one of us is a knave”?
14. Suppose that you meet three people, Anita, Boris, and Carmen. What are Anita, Boris, and Carmen if Anita says “I am a knave and Boris is a knight” and Boris says “Exactly one of the three of us is a knight”?
15. (Adapted from [Sm78]) Suppose that on an island there are three types of people, knights, knaves, and normals (also known as spies). Knights always tell the truth, knaves always lie, and normals sometimes lie and sometimes tell the truth. Detectives questioned three inhabitants of the island—Amy, Brenda, and Claire—as part of the investigation of a crime. The detectives knew that one of the three committed the crime, but not which one. They also knew that the criminal was a knight, and that the other two were not. Additionally, the detectives recorded these statements: Amy: “I am innocent.” Brenda: “What Amy says is true.” Claire: “Brenda is not a normal.” After analyzing their information, the detectives positively identified the guilty party. Who was it?
16. Show that if S is a proposition, where S is the conditional statement “If S is true, then unicorns live,” then “Unicorns live” is true. Show that it follows that S cannot be a proposition. (This paradox is known as *Löb’s paradox*.)
17. Show that the argument with premises “The tooth fairy is a real person” and “The tooth fairy is not a real person” and conclusion “You can find gold at the end of the rainbow” is a valid argument. Does this show that the conclusion is true?
18. Suppose that the truth value of the proposition p_i is **T** whenever i is an odd positive integer and is **F** whenever i is an even positive integer. Find the truth values of $\bigvee_{i=1}^{100}(p_i \wedge p_{i+1})$ and $\bigwedge_{i=1}^{100}(p_i \vee p_{i+1})$.

- *19. Model 16×16 Sudoku puzzles (with 4×4 blocks) as satisfiability problems.
20. Let $P(x)$ be the statement “Student x knows calculus” and let $Q(y)$ be the statement “Class y contains a student who knows calculus.” Express each of these as quantifications of $P(x)$ and $Q(y)$.
 - a) Some students know calculus.
 - b) Not every student knows calculus.
 - c) Every class has a student in it who knows calculus.
 - d) Every student in every class knows calculus.
 - e) There is at least one class with no students who know calculus.
21. Let $P(m, n)$ be the statement “ m divides n ,” where the domain for both variables consists of all positive integers. (By “ m divides n ” we mean that $n = km$ for some integer k .) Determine the truth values of each of these statements.

a) $P(4, 5)$	b) $P(2, 4)$
c) $\forall m \forall n P(m, n)$	d) $\exists m \forall n P(m, n)$
e) $\exists n \forall m P(m, n)$	f) $\forall n P(1, n)$
22. Find a domain for the quantifiers in $\exists x \exists y (x \neq y \wedge \forall z ((z = x) \vee (z = y)))$ such that this statement is true.
23. Find a domain for the quantifiers in $\exists x \exists y (x \neq y \wedge \forall z ((z = x) \vee (z = y)))$ such that this statement is false.
24. Use existential and universal quantifiers to express the statement “No one has more than three grandmothers” using the propositional function $G(x, y)$, which represents “ x is the grandmother of y .”
25. Use existential and universal quantifiers to express the statement “Everyone has exactly two biological parents” using the propositional function $P(x, y)$, which represents “ x is the biological parent of y .”
26. The quantifier $\exists_n$ denotes “there exists exactly n ,” so that $\exists_n x P(x)$ means there exist exactly n values in the domain such that $P(x)$ is true. Determine the true value of these statements where the domain consists of all real numbers.

a) $\exists_0 x (x^2 = -1)$	b) $\exists_1 x (x = 0)$
c) $\exists_2 x (x^2 = 2)$	d) $\exists_3 x (x = x)$
27. Express each of these statements using existential and universal quantifiers and propositional logic, where $\exists_n$ is defined in Exercise 26.

a) $\exists_0 x P(x)$	b) $\exists_1 x P(x)$
c) $\exists_2 x P(x)$	d) $\exists_3 x P(x)$
28. Let $P(x, y)$ be a propositional function. Show that $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$ is a tautology.
29. Let $P(x)$ and $Q(x)$ be propositional functions. Show that $\exists x (P(x) \rightarrow Q(x))$ and $\forall x P(x) \rightarrow \exists x Q(x)$ always have the same truth value.
30. If $\forall y \exists x P(x, y)$ is true, does it necessarily follow that $\exists x \forall y P(x, y)$ is true?
31. If $\forall x \exists y P(x, y)$ is true, does it necessarily follow that $\exists x \forall y P(x, y)$ is true?

32. Find the negations of these statements.
 - a) If it snows today, then I will go skiing tomorrow.
 - b) Every person in this class understands mathematical induction.
 - c) Some students in this class do not like discrete mathematics.
 - d) In every mathematics class there is some student who falls asleep during lectures.
33. Express this statement using quantifiers: “Every student in this class has taken some course in every department in the school of mathematical sciences.”
34. Express this statement using quantifiers: “There is a building on the campus of some college in the United States in which every room is painted white.”
35. Express the statement “There is exactly one student in this class who has taken exactly one mathematics class at this school” using the uniqueness quantifier. Then express this statement using quantifiers, without using the uniqueness quantifier.
36. Describe a rule of inference that can be used to prove that there are exactly two elements x and y in a domain such that $P(x)$ and $P(y)$ are true. Express this rule of inference as a statement in English.
37. Use rules of inference to show that if the premises $\forall x(P(x) \rightarrow Q(x))$, $\forall x(Q(x) \rightarrow R(x))$, and $\neg R(a)$, where a is in the domain, are true, then the conclusion $\neg P(a)$ is true.
38. Prove that if x^3 is irrational, then x is irrational.
39. Prove or disprove that if x^2 is irrational, then x^3 is irrational.
40. Prove that given a nonnegative integer n , there is a unique nonnegative integer m such that $m^2 \leq n < (m+1)^2$.
41. Prove that there exists an integer m such that $m^2 > 10^{1000}$. Is your proof constructive or nonconstructive?
42. Prove that there is a positive integer that can be written as the sum of squares of positive integers in two different ways. (Use a computer or calculator to speed up your work.)
43. Disprove the statement that every positive integer is the sum of the cubes of eight nonnegative integers.
44. Disprove the statement that every positive integer is the sum of at most two squares and a cube of nonnegative integers.
45. Disprove the statement that every positive integer is the sum of 36 fifth powers of nonnegative integers.
46. Assuming the truth of the theorem that states that $\sqrt{n}$ is irrational whenever n is a positive integer that is not a perfect square, prove that $\sqrt{2} + \sqrt{3}$ is irrational.

Computer Projects

Write programs with the specified input and output.

1. Given the truth values of the propositions p and q , find the truth values of the conjunction, disjunction, exclusive or, conditional statement, and biconditional of these propositions.
2. Given two bit strings of length n , find the bitwise *AND*, bitwise *OR*, and bitwise *XOR* of these strings.
- *3. Give a compound proposition, determine whether it is satisfiable by checking its truth value for all positive assignments of truth values to its propositional variables.
4. Given the truth values of the propositions p and q in fuzzy logic, find the truth value of the disjunction and the conjunction of p and q (see Exercises 50 and 51 of Section 1.1).
- *5. Given positive integers m and n , interactively play the game of Chomp.
- *6. Given a portion of a checkerboard, look for tilings of this checkerboard with various types of polyominoes, including dominoes, the two types of triominoes, and larger polyominoes.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

1. Look for positive integers that are not the sum of the cubes of nine different positive integers.
2. Look for positive integers greater than 79 that are not the sum of the fourth powers of 18 positive integers.
3. Find as many positive integers as you can that can be written as the sum of cubes of positive integers, in two different ways, sharing this property with 1729.
- *4. Try to find winning strategies for the game of Chomp for different initial configurations of cookies.
5. Construct the 12 different pentominoes, where a pentomino is a polyomino consisting of five squares.
6. Find all the rectangles of 60 squares that can be tiled using every one of the 12 different pentominoes.

Writing Projects

Respond to these with essays using outside sources.

1. Discuss logical paradoxes, including the paradox of Epimenides the Cretan, Jourdain's card paradox, and the barber paradox, and how they are resolved.
2. Describe how fuzzy logic is being applied to practical applications. Consult one or more of the recent books on fuzzy logic written for general audiences.
3. Describe some of the practical problems that can be modeled as satisfiability problems.
4. Explain how satisfiability can be used to model a round robin tournament.
5. Describe some of the techniques that have been devised to help people solve Sudoku puzzles without the use of a computer.
6. Describe the basic rules of *WFF'N PROOF*, *The Game of Modern Logic*, developed by Layman Allen. Give examples of some of the games included in *WFF'N PROOF*.
7. Read some of the writings of Lewis Carroll on symbolic logic. Describe in detail some of the models he used to represent logical arguments and the rules of inference he used in these arguments.
8. Extend the discussion of Prolog given in Section 1.4, explaining in more depth how Prolog employs resolution.
9. Discuss some of the techniques used in computational logic, including Skolem's rule.
10. "Automated theorem proving" is the task of using computers to mechanically prove theorems. Discuss the goals and applications of automated theorem proving and the progress made in developing automated theorem provers.
11. Describe how DNA computing has been used to solve instances of the satisfiability problem.
12. Look up some of the incorrect proofs of famous open questions and open questions that were solved since 1970 and describe the type of error made in each proof.
13. Discuss what is known about winning strategies in the game of Chomp.
14. Describe various aspects of proof strategy discussed by George Pólya in his writings on reasoning, including [Po62], [Po71], and [Po90].
15. Describe a few problems and results about tilings with polyominoes, as described in [Go94] and [Ma91], for example.